Notes on Grimmett and Welsh's Probability

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This document provides a summary of the results presented in *Grimmett, G. & Welsh, D. (2014), Probability: an introduction (2nd ed.).*

The content is highly summarised and most of the examples, exercises, and problems are omitted.

The purpose of this document is mostly for my own understanding, as I find it easier to learn from a book when taking notes. It may also be used as a summary of the most important results.

Very occasionally, I have included some additional results and explanations where they have aided my understanding. The explanations in particular are introduced with a question written in italics, which I present a usually intuitive rather than rigorous answer to.

This document should not be treated as a substitution or sole learning resource. All errors, whether mathematical or formatting, are almost surely my own.

Last updated: October 2026. Currently almost complete with Chapter 12.

1 Events and Probabilities

An experiment is a course of action whose consequence is not predetermined.

We model experiments using probability spaces.

If an experiment is denoted by ε, its sample space is the set of all possible outcomes of the experiment and is denoted Ω. Each member of Ω is called an elementary event.

The collection of subsets of Ω which describe events we are interested in is called the event space, and is denoted by F.

Why set theory? Set theoretic operations are quite natural here, because the set $A \cup B$ corresponds to "either A or B", $A \cap B$ corresponds to "both A and B", and A^c corresponds to "A does not occur". These are all events we would likely wish to study if we are studying the event A, and set theoretic operations make this natural.

We say that a subset C of Ω occurs whenever the outcome of ε lies in C.

These natural nations of which events we may be interested in given a collection of events we are already interested in give rise to a natural definition of F.

Definition 1.1

The collection of subsets of Ω F is an event space if:

- F is non-empty.

- If $A \in F$ then $A^c \in F$.
- If $A_1, A_2, \ldots \in F$ then $\bigcup_i A_i \in F$.

F is therefore closed under complements and countable unions.

Simple consequences of this definition include:

- F contains the empty set.
- F contains Ω.
- F is closed under finite unions.
- F is closed under finite intersections.

Definition 1.13

A mapping $P : F \to \mathbb{R}$ is a probability measure on (Ω, F) if:

- $P(A) \geq 0$ for $A \in F$.
- $P(\Omega) = 1$ and $P(\varnothing) = 0$.
- If $A_1, A_2, \ldots$ are disjoint events in F, then

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i).$$

Note that P is defined only on subsets of Ω in F, and not necessarily all subsets of Ω.

We can omit the second part of the second condition, as taking $A_1 = \Omega$, $A_i = \varnothing$ in the third condition results in $P(\varnothing) = 0$.

Definition 1.18

A probability space is a triple (Ω, F, P) such that:

- Ω is a non-empty set.
- F is an event space of subsets of Ω.
- P is a probability measure on (Ω, F).

The conditions of the previous definitions allow for some easy general results regarding probability spaces. For instance:

- If $A, B \in F$ then $A \setminus B \in F$.
- If $A_1, A_2, \ldots \in F$ then $\bigcap_i A_i \in F$.
- If $A \in F$ then $P(A) + P(A^c) = 1$.
- If $A, B \in F$ then $P(A \cap B) + P(A \cup B) = P(A) + P(B)$.

- If $A, B \in F$ and A is a subset of B then $P(A) \leq P(B)$.

The cardinality of Ω is highly significant to the structure of probability spaces including Ω. If Ω is countable, we normally take $F = \mathcal{P}(\Omega)$. This is because if we include each elementary event in F, then by closure under countable unions, we require $F = \mathcal{P}(\Omega)$. This doesn't necessary hold if we don't include each elementary event in F, but this is unlikely to occur naturally.

In this case, we can express

$$P(A) = P\Big(\bigcup(\omega_1, \omega_2, \ldots, \omega_n)\Big)$$

for some ω_i elementary events, which are necessarily disjoint. Consequently, we have

$$P(A) = \sum_i P(\omega_i).$$

When we have Ω such that $|\Omega|$ is finite, we can define a probability measure such that $P(\omega_i) = P(\omega_j)$ for all i, j. Then $P(A) = |A|/|\Omega|$. Problems involving such probability measures are often combinatorial and their difficulty amounts to calculating $|A|$ and $|\Omega|$. This probability measure is not always available for countable Ω; try to construct a probability measure P on $(\mathbb{Z}_+, \mathcal{P}(\mathbb{Z}_+))$ such that $P(m) = P(n)$ for all $m, n \in \mathbb{Z}_+$.

Suppose we are given some information about ε, say that B occurs. Now A occurs if and only if $A \cap B$ occurs. Hence:

Definition 1.31

If $A, B \in F$ and $P(B) > 0$, then the probability of A given B is defined

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

Why this definition? It allows for a new probability space (Ω, F, Q) with $Q(A) = P(A \mid B)$, which can be easily verified.

Definition 1.38

Events A and B are independent if $P(A \cap B) = P(A)P(B)$. Otherwise, A and B are dependent.

Noting $P(A \cap B) = P(B)P(A \mid B)$, we have, as long as $P(A), P(B) > 0$,

$$P(B)P(A \mid B) = P(A)P(B),$$

and hence $P(A \mid B) = P(A)$.

Symmetrically, $P(B \mid A) = P(B)$ for independent events A and B with $P(A), P(B) > 0$.

This can be easily generalised. Events A_i are independent when

$$P\left(\bigcap_{i \in J} A_i\right) = \prod_{i \in J} P(A_i)$$

for all finite subsets J of I. The events A_i are called pairwise independent if this equation holds whenever $|J| = 2$. It is possible for groups of events to be pairwise independent but not independent.

Example 1.41

Throw a tetrahedral die. Take $\Omega = \{1, 2, 3, 4\}$. Then $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{1, 4\}$ are:

- pairwise independent, as $P(A \cap B) = P(A)P(B) = 1/4$, and likewise for the other sets of size two.
- not independent, as $P(A \cap B \cap C) = 1/4$ but $P(A)P(B)P(C) = 1/8$.

How should this result be interpreted?

Pairwise independence intuitively means if you know A occurs, this doesn't affect the chance of B occurring (and similarly for other pairs). This still holds here. But if instead we are told "A occurred and B occurred", then we do have more information about whether C occurred (in fact, it did occur). Hence we do not have independence.

Theorem 1.48: partition theorem, law of total probability

If $\{B_1, B_2, \ldots\}$ is a partition of Ω with $P(B_i) > 0$ for each i, then

$$P(A) = \sum_i P(A \mid B_i)P(B_i).$$

Proof.

$$P(A) = P\left(A \cap \left(\bigcup_i B_i\right)\right)$$

by B_i partition.

$$\Rightarrow P(A) = P\left(\bigcup_i (A \cap B_i)\right),$$

which can be seen intuitively on a Venn diagram.

$$\Rightarrow P(A) = \sum_i P(A \cap B_i),$$

as the $(A \cap B_i)$ are disjoint by B_i being disjoint, and then applying this condition of the definition of a probability measure.

$$\Rightarrow P(A) = \sum_i P(A \mid B_i)P(B_i)$$

by the definition of conditional probability.

We require $P(B_i) > 0$ here only because $P(A \mid B_i)$ is undefined in this case.

Theorem 1.50: Bayes' theorem

If $\{B_1, B_2, \ldots\}$ is a partition of Ω with $P(B_i) > 0$ for each i, then

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{\sum_i P(A \mid B_i)P(B_i)}.$$

Proof.

$$P(B_j \mid A) = \frac{P(A \mid B_j)P(B_j)}{P(A)}$$

from the definition of conditional probability.

The result follows expressing $P(A)$ using the partition theorem.

Note that the intermediate result with denominator $P(A)$ is sometimes also given as Bayes' theorem.

Example 1.51

A classic result goes something like this. A rare disease occurs with probability $1/10^5$. A test is positive with probability $9/10$ on a true positive, and is positive with probability $1/20$ on a true negative. Given a test is positive, what is the probability of a true positive?

$$\begin{aligned} P(TP \mid P) &= \frac{P(P \mid TP)P(TP)}{P(P \mid TP)P(TP) + P(P \mid TN)P(TN)} \\ &\Rightarrow P(TP \mid P) = \frac{\frac{9}{10} \cdot \frac{1}{10^5}}{\frac{9}{10} \cdot \frac{1}{10^5} + \frac{1}{20} \cdot \frac{10^5 - 1}{10^5}} \\ &\Rightarrow P(TP \mid P) = 0.0002. \end{aligned}$$

This is to 4dp upon calculation, which is much lower than one might expect.

Theorem 1.54: continuity of probability measures

If $A_1, A_2, \ldots$ is an increasing sequence of events – this means $A_1 \subseteq A_2 \subseteq A_3 \subseteq \cdots$ – with limit $A = \bigcup_i A_i$ then

$$P(A) = \lim_{n \to \infty} P(A_n).$$

Proof.

$$\begin{aligned} A &= A_1 \cup (A_2 \setminus A_1) \cup (A_3 \setminus A_2) \cup \cdots, \quad \text{which are disjoint events.} \\ &\Rightarrow P(A) = P(A_1) + P(A_2 \setminus A_1) + P(A_3 \setminus A_2) + \cdots. \end{aligned}$$

But each A_{i-1}, $(A_i \setminus A_{i-1})$ are disjoint events with union A_i, so

$$P(A_{i-1}) + P(A_i \setminus A_{i-1}) = P(A_i)$$

by the definition of a probability measure. Hence:

$$\begin{aligned} P(A) &= P(A_1) + P(A_2) - P(A_1) + P(A_3) - P(A_2) + \cdots \\ &\Rightarrow P(A) = \lim_{n \to \infty} P(A_n) \quad \text{when } \{A_i\} \text{ converges.} \end{aligned}$$

This theorem holds for decreasing sequences of events, replacing the union with intersection.

Either way, these results are most often applied to rigorously prove intuitively "obvious" statements.

2 Discrete Random Variables

A discrete random variable X on the probability space (Ω, F, P) is defined to be a mapping

$$X : \Omega \to \mathbb{R}$$

such that the image $X(\Omega)$ is a countable subset of $\mathbb{R}$ and

$$\{\omega \in \Omega : X(\omega) = x\} \in F$$

for $x \in \mathbb{R}$.

The first condition is what makes this a discrete random variable. The second condition is required so that we can measure the probability that X takes a given value x. This happens if and only if the result of ε lies in the subset of Ω mapped to x. Consequently, we require this subset to belong to F.

We abbreviate $\{\omega \in \Omega : X(\omega) = x\}$ to $\{X = x\}$.

Definition 2.3

The probability mass function of a discrete random variable X is the function

$$p_X : \mathbb{R} \to [0, 1]$$

with $p_X(x) = P(X = x)$.

Theorem 2.7

Let $S = \{s_i\}$ be a countable set of distinct real numbers and let $\{\pi_i\}$ be a collection of non-negative real numbers with sum 1. Then there exists a probability space (Ω, F, P) and a discrete random variable X on (Ω, F, P) such that the probability mass function of X is given by

$$P(X = s_i) = \pi_i$$

and $P(X = s) = 0$ for $s \notin S$.

This theorem can be thought of similarly to things like the chain rule in analysis, which we may rigorously prove, and then forget about the details in its application. Here, we use probability spaces to prove that we can treat discrete random variables rigorously, and we do not need to think much about probability spaces further.

Some examples of discrete random variables are given.

Bernoulli distribution.

$$P(X = 0) = 1 - p, \qquad P(X = 1) = p.$$

p is the parameter.

Binomial distribution.

$$P(X = k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n.$$

n and p are the parameters.

Poisson distribution.

$$P(X = k) = \frac{1}{k!} \lambda^k e^{-\lambda}, \qquad k = 0, 1, 2, \ldots$$

λ is the parameter.

Geometric distribution.

$$P(X = k) = pq^{k-1}, \qquad k = 1, 2, 3, \ldots$$

p is the parameter.

Negative binomial distribution.

$$P(X = k) = \binom{k-1}{n-1} p^n q^{k-n}, \qquad k = n, n+1, n+2, \ldots$$

n and p are the parameters.

For $g : \mathbb{R} \to \mathbb{R}$, we have $Y = g(X)$ is a discrete random variable on the same probability space as X. Recall that for a discrete random variable, we require its image to be countable and it must be F-measurable. It is intuitively clear, and can be shown rigorously quite easily as well, that these conditions applying to X also apply to Y.

If $Y = g(X)$, then the mass function of Y is given by

$$P_Y(y) = \sum_{x:g(x)=y} P(X = x),$$

which is intuitively quite clear.

Definition 2.27

For a discrete random variable X, the expectation of X is defined as

$$E(X) = \sum_{x \in \mathrm{Im}(X)} xP(X = x)$$

whenever the sum converges absolutely.

Absolute convergence is required to avoid problems with rearrangement when dealing with X such that $\mathrm{Im}(X)$ is infinite.

The indexing of $x \in \mathrm{Im}(X)$ is redundant; we can just say $x \in \mathbb{R}$, as the contributions from $x \notin \mathrm{Im}(X)$ are then zero.

Theorem 2.29: LOTUS

If X is a discrete random variable and $g : \mathbb{R} \to \mathbb{R}$, then

$$E(g(X)) = \sum_{x \in \mathrm{Im}(X)} g(x)P(X = x)$$

whenever the sum converges absolutely.

Proof. Let $Y = g(X)$.

$$\begin{aligned}
E(Y) &= \sum_{y \in \mathrm{Im}(Y)} yP(Y = y) && \text{by definition,} \\
E(Y) &= \sum_{y \in \mathrm{Im}(Y)} y \sum_{x : g(x) = y} P(X = x), \\
E(Y) &= \sum_{y \in \mathrm{Im}(Y)} \sum_{x : g(x) = y} yP(X = x), \\
E(Y) &= \sum_{x \in \mathrm{Im}(X)} yP(X = x), \\
E(g(X)) &= \sum_{x \in \mathrm{Im}(X)} g(x)P(X = x) && \text{back-substituting.}
\end{aligned}$$

Some useful properties of expectation include:

- If $P(X \geq 0) = 1$ and $E(X) = 0$ then $P(X = 0) = 1$. Or, if X is a non-negative discrete random variable with mean zero, then X is identically zero.
- $E(aX + b) = aE(X) + b$. This can be shown using LOTUS with $g(X) = aX + b$.

Definition 2.32

The variance of a discrete random variable X is defined by

$$\mathrm{Var}(X) = E\big((X - E(X))^2\big).$$

Writing $E(X) = \mu$, we have:

$$\begin{aligned}
\mathrm{Var}(X) &= E(X^2 - 2X\mu + \mu^2), \\
\mathrm{Var}(X) &= E(X^2) - 2\mu E(X) + \mu^2 && \text{by the linearity of expectation above,} \\
\mathrm{Var}(X) &= E(X^2) - 2\mu^2 + \mu^2, \\
\mathrm{Var}(X) &= E(X^2) - E(X)^2,
\end{aligned}$$

the other familiar definition of expectation.

Definition 2.40

If X is a discrete random variable and $P(B) > 0$, then the conditional expectation of X given B is defined as

$$E(X \mid B) = \sum_{x \in \mathrm{Im}(X)} xP(X = x \mid B)$$

whenever the sum converges absolutely.

Theorem 2.42: partition theorem for expectation

If X is a discrete random variable and $\{B_1, B_2, \ldots\}$ is a partition of the sample space with $P(B_i) > 0$, then

$$E(X) = \sum_i E(X \mid B_i)P(B_i)$$

whenever the sum converges absolutely.

Example 2.44

A biased coin is tossed repeatedly with probability of heads p. Find the expected length of the initial run (of either heads or tails). Let H be the event that the first toss is heads and H^c the complement. Then if X is the length of the initial run, we have:

$$P(X = k \mid H) = p^{k-1}q,$$
$$P(X = k \mid H^c) = pq^{k-1}.$$

Then:

$$E(X \mid H) = \sum_{k=1}^{\infty} kp^{k-1}q = \frac{1}{q} \qquad \text{as this is a geometric series,}$$
$$E(X \mid H^c) = \frac{1}{p} \qquad \text{similarly.}$$

Then by the partition theorem:

$$E(X) = pE(X \mid H) + qE(X \mid H^c),$$
$$E(X) = \frac{p}{q} + \frac{q}{p},$$
$$E(X) = \frac{p^2 + q^2}{pq},$$
$$E(X) = \frac{2p^2 - 2p + 1}{p - p^2},$$
$$E(X) = -2 + \frac{1}{p(1-p)},$$
$$E(X) = \frac{1}{p(1-p)} - 2.$$

Note that the last few lines are just algebraic manipulation, which can be quite difficult when using p and $q = 1 - p$ in problems.

3 Multivariate Discrete Distributions and Independence

If X and Y are discrete random variables on a probability space, we may regard the pair (X, Y) as a random vector in $\mathbb{R}^2$.

Definition 3.1

The joint probability mass function $p_{X,Y}$ of X and Y is the function

$$p_{X,Y} : \mathbb{R}^2 \to [0, 1]$$

defined by

$$p_{X,Y}(x, y) = P(X = x, Y = y).$$

We have

$$\sum_x \sum_y p_{X,Y}(x, y) = 1.$$

The individual mass functions p_X and p_Y can be easily found from $p_{X,Y}$.

$$P(X = x) = \sum_y P(X = x, Y = y),$$
$$P(Y = y) = \sum_x P(X = x, Y = y).$$

These mass functions are called the marginal mass functions of X and Y.

Similarly, the joint mass function of $\mathbf{X} = (X_1, X_2, \ldots, X_n)$ for discrete random variables X_i is given by

$$P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n).$$

These higher-order families of discrete random variables are rarely encountered.

Similarly to before, we can prove that for any $g : \mathbb{R}^2 \to \mathbb{R}$, $Z = g(X, Y)$ is a discrete random variable on the probability space on which X and Y are discrete random variables. We then have

$$E(g(X, Y)) = \sum_x \sum_y g(x, y)P(X = x, Y = y).$$

Using this theorem with $g(x, y) = ax + by$, we have

$$E(aX + bY) = aE(X) + bE(Y),$$

which is a very useful result.

Definition 3.13

Discrete random variables X and Y are independent if the events $\{X = x\}$ and $\{Y = y\}$ are independent for all $x, y \in \mathbb{R}$. This is true if and only if

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

for $x, y \in \mathbb{R}$. Random variables which are not independent are dependent.

We can express this condition as

$$P(X = x, Y = y) = \left(\sum_y P(X = x, Y = y)\right)\left(\sum_x P(X = x, Y = y)\right).$$

From this, we obtain the important result below.

Theorem 3.16

Discrete random variables X and Y are independent if and only if there exist functions $f, g : \mathbb{R} \to \mathbb{R}$ such that:

$$P(X = x, Y = y) = f(x)g(y).$$

Proof. The condition is already necessary by definition 3.13. We need only prove sufficiency, so assume the condition holds. Then

$$P(X = x) = f(x) \sum_y g(y),$$
$$P(Y = y) = g(y) \sum_x f(x).$$

And

$$\sum_{x,y} f(x)g(y) = \sum_x f(x) \sum_y g(y) = 1.$$

So we can write

$$P(X = x, Y = y) = f(x)g(y) = f(x)g(y) \sum_x f(x) \sum_y g(y),$$

multiplying by 1. Then

$$P(X = x, Y = y) = P(X = x)P(Y = y),$$

which is our independence criterion.

Theorem 3.19

If X and Y are independent discrete random variables with expectations $E(X)$ and $E(Y)$, then

$$E(XY) = E(X)E(Y).$$

The proof follows from the definition of expectation in the multivariate case, independence, and rearrangement.

Note that the converse $E(XY) = E(X)E(Y)$ given does not imply independence.

Theorem 3.20

If X and Y are discrete random variables such that

$$E(g(X)h(Y)) = E(g(X))E(h(Y))$$

for all functions $g, h : \mathbb{R} \to \mathbb{R}$ such that the expectations exist, then X and Y are independent. This is a necessary and sufficient condition for independence.

Proof. The condition is already necessary by theorem 3.19. We need only prove sufficiency, so assume the condition holds. Define families of functions g, h such that g is the indicator of $x = a$ and h is the indicator of $y = b$ for some $a, b \in \mathbb{R}$. Then

$$E(g(X)h(Y)) = P(X = a, Y = b).$$

With

$$E(g(X)) = P(X = a), \qquad E(h(Y)) = P(Y = b).$$

Since we have assumed the condition holds, we have

$$P(X = a, Y = b) = P(X = a)P(Y = b)$$

for all $a, b \in \mathbb{R}$. Consequently, X and Y are independent.

Example 3.22

For an easy case where X and Y are dependent but satisfy $E(XY) = E(X)E(Y)$, use: X with distribution

$$p(-1) = p(0) = p(1) = \frac{1}{3}$$

and Y as the indicator variable of $X \neq 0$. Then it is easy to verify

$$P(X = 0, Y = 1) \neq P(X = 0)P(Y = 1),$$

and hence X and Y are dependent. Nevertheless, we do have $E(XY) = E(X)E(Y)$.

If $Z = X + Y$, then we have $Z = z$ if and only if $X = x$ and $Y = z - x$ for some x. Consequently,

$$P(Z = z) = \sum_x P(X = x, Y = z - x).$$

If X and Y are independent, we can use the factorisation of their joint mass function to obtain

Theorem 3.27: convolution for discrete random variables

$$P(Z = z) = \sum_x P(X = x)P(Y = z - x).$$

The indicator function of an event A is the random variable often denoted $\mathbf{1}_A$ or I_A and defined by:

$$I_A(\omega) = \begin{cases} 1, & \omega \in A, \\ 0, & \omega \notin A. \end{cases}$$

Recall here ω is an elementary event.

Indicator functions have many useful properties:

$$\begin{aligned} E(I_A) &= P(A), \\ I_{A\cap B} &= I_A I_B, \\ I_A + I_{A^c} &= 1, \\ I_{A\cup B} &= I_A + I_B - I_{A\cap B}. \end{aligned}$$

Indicator functions are one way to derive the inclusion-exclusion formula.

Example 3.36: inclusion-exclusion formula

Let $A_1, A_2, \ldots, A_n$ be events. The union

$$A = A_1 \cup A_2 \cup \cdots \cup A_n$$

has indicator function

$$I_A = 1 - \prod_{i=1}^{n} (1 - I_{A_i}).$$

Expanding,

$$I_A = \sum_i I_{A_i} - \sum_{i<j} I_{A_i} I_{A_j} + \cdots + (-1)^{n+1} I_{A_1} I_{A_2} \cdots I_{A_n}.$$

On taking expectations,

$$P\left(\bigcup_i A_i\right) = \sum_i P(A_i) - \sum_{i<j} P(A_i \cap A_j) + \cdots + (-1)^{n+1} P\left(\bigcap_i A_i\right).$$

This result is quite clunky when written out, but is intuitively more clear and is widely applicable.

4 Probability Generating Functions

A generating function is a function which somehow encodes the sequence. For example, a generating function a sequence $\{u_k\}$ might be the power series

$$u_0 + u_1 s + u_2 s^2 + \cdots .$$

Generating functions are useful ways of studying sequences, especially when they specify a sequence uniquely.

Other types of generating functions exist, like the exponential generating function

$$u_0 + u_1 s + \frac{1}{2!} u_2 s^2 + \cdots .$$

We do not normally explicitly state the radius of convergence of a generating sequence.

Many interesting random variables are non-negative integers. We can associate a sequence with these, namely the sequence $\{p_k\}$ defined as $p_k = P(X = k)$. Recall we have $p_k \geq 0$ for all k and

$$\sum_{k=0}^{\infty} p_k = 1$$

following from some of our earliest definitions.

Definition 4.7

The probability generating function of X is the function $G_X(s)$ defined by

$$G_X(s) = p_0 + p_1 s + p_2 s^2 + \cdots$$

for all s for which the right side converges absolutely.

We have some immediate properties:

$$G_X(0) = p_0, \qquad G_X(1) = 1.$$

Less obvious is

$$G_X(s) = E(s^X),$$

but this follows immediately from LOTUS. We have the restriction that this expectation has to exist.

The radius of convergence must be at least one about the origin, as

$$\sum_{k=0}^{\infty} |p_k s^k| \leq \sum_{k=0}^{\infty} p_k s^k \leq 1$$

for all $s \leq 1$.

Theorem 4.13: uniqueness of probability generating functions

$$G_X(s) = G_Y(s) \quad \text{for all } s$$

if and only if

$$P(X = k) = P(Y = k) \quad \text{for all } k \geq 0.$$

Proof. We need only prove the forward implication, and this follows from the uniqueness of power series expansions around the origin given our radius of convergence.

We will now give some common probability generating functions. These can almost all be easily proven by substituting the probability mass function of a distribution into the general formula for a probability generating function.

Bernoulli distribution.

$$G_X(s) = q + ps.$$

Binomial distribution.

$$G_X(s) = (q + ps)^n.$$

Poisson distribution.

$$G_X(s) = e^{\lambda(s-1)}.$$

Geometric distribution.

$$G_X(s) = \frac{ps}{1 - qs}.$$

Negative binomial distribution.

$$G_X(s) = \left(\frac{ps}{1 - qs}\right)^n.$$

The mean of a distribution $E(X)$ encodes some information about the distribution. We have further similar quantities called moments.

Definition 4.20

The kth moment of the random variable X is defined as $E(X^k)$.

If X is a non-negative integer random variable, then the moments of X are found by calculating the derivatives of the probability generating function at $s = 1$.

In particular, we have:

Theorem 4.23

We have for a non-negative integer random variable X, the rth derivative of its probability generating function is:

$$G_X^{(r)}(1) = E\big(X(X-1)\cdots(X-r+1)\big).$$

This comes from differentiating term-by-term. This is somewhat difficult to justify, but we will not concern ourselves with that.

For instance, we may apply theorem 4.23 to calculate $E(X^2)$ as follows. Write

$$X^2 = X(X-1) + X.$$

Then

$$E(X^2) = E(X(X-1)) + E(X).$$

Then

$$E(X^2) = G''_X(1) + G'_X(1).$$

Probability generating functions are most useful for considering sums of independent random variables. We have the useful result:

Theorem 4.33

For independent non-negative integer random variables X and Y, we have:

$$G_{X+Y}(s) = G_X(s)G_Y(s).$$

Proof.

$$\begin{aligned} &G_{X+Y}(s) = E(s^{X+Y}) = E(s^X s^Y), \\ &G_{X+Y}(s) = E(s^X)E(s^Y) && \text{by independence,} \\ &G_{X+Y}(s) = G_X(s)G_Y(s). \end{aligned}$$

The result can also be obtained by using the convolution formula.

Theorem 4.36: random sum formula

Let N and $X_1, X_2, \ldots$ be independent non-negative integer random variables. If the X_i are identically distributed, then the sum

$$S = X_1 + X_2 + \cdots + X_N$$

has probability generating function

$$G_S(s) = G_N(G_X(s)).$$

Proof.

$$\begin{aligned} G_S(s) &= E\left(s^{X_1+\cdots+X_N}\right), \\ G_S(s) &= \sum_{n=0}^{\infty} E\left(s^{X_1+\cdots+X_N} \mid N=n\right) P(N=n), \end{aligned}$$

using the partition theorem for expectation.

$$\begin{aligned} G_S(s) &= \sum_{n=0}^{\infty} E\left(s^{X_1+\cdots+X_n}\right) P(N=n), \\ G_S(s) &= \sum_{n=0}^{\infty} G_X(s)^n P(N=n), \end{aligned}$$

by independence and theorem 4.33.

$$G_S(s) = G_N(G_X(s))$$

by the definition of a probability generating function.

Note the distinction here; the number of variables we are adding up is itself a random variable.

We can now easily obtain $E(S)$ by differentiation. In particular,

$$E(S) = E(N)E(X),$$

a nice (and quite intuitive) result.

5 Distribution Functions and Density Functions

We now move onto continuous random variables. We must briefly return to rigorous discussion of probability spaces to define a broader definition of a random variable X.

Specifically, we say that we require

$$\{\omega \in \Omega : X(\omega) \leq x\} \in F$$

for all $x \in \mathbb{R}$.

We will abbreviate this to $\{X \leq x\}$.

Why do we do this? We will often be interested in the probability that a random variable lies in an interval (a, b). If the event space only necessarily contained the events $\{X = x\}$, we cannot investigate the interval (a, b), as we cannot express it as a union of countably many events in the event space. Whereas if the event space contains the events $\{X \leq x\}$, we can investigate these intervals. This can be rigorously expressed with some yap about sigma-algebras.

All discrete random variables are trivially random variables under this definition.

Definition 5.2

If X is a random variable on (Ω, F, P), then the distribution function of X is the function

$$F_X : \mathbb{R} \to [0, 1]$$

defined by

$$F_X(x) = P(X \leq x).$$

This is also often referred to as the cumulative distribution function.

This definition allows for some immediate properties:

- If $x \leq y$, then $F_X(x) \leq F_X(y)$ because $\{X \leq x\}$ is a subset of $\{X \leq y\}$.
- $F_X(x) \to 0$ as $x \to -\infty$.
- $F_X(x) \to 1$ as $x \to \infty$.

These last two properties rely on the continuity of a probability measure (theorem 1.54) to formally verify.

Curiously, F_X is continuous from the right, meaning

$$F_X(x+\varepsilon) \to F_X(x) \quad \text{as } \varepsilon \to 0^+.$$

However, F_X is not necessarily continuous from the left. A counterexample can be easily demonstrated by sketching the cumulative distribution function of a discrete random variable.

It turns out that the three bullet-pointed properties and continuity from the right are sufficient for any function f satisfying them to be a cumulative density function.

Note finally that

$$P(a < X \leq b) = P(X \leq b) - P(X \leq a).$$

We now give some examples of continuous distributions and their cumulative distribution functions.

Example 5.15: uniform distribution

It is characterised by parameters a, b with $a < b$.

$$F(x) = \begin{cases} 0, & x < a, \\ \dfrac{x-a}{b-a}, & a \leq x \leq b, \\ 1, & x > b. \end{cases}$$

Example 5.16: exponential distribution

It is characterised by a positive parameter λ.

$$F(x) = \begin{cases} 0, & x \leq 0, \\ 1 - e^{\lambda x}, & x > 0. \end{cases}$$

Definition 5.20

A random variable X is continuous if its cumulative distribution function F_X may be written in the form

$$F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(u)\, du$$

for some non-negative function f_X. We say that X has probability density function f_X.

Why this definition? A more natural definition might be a random variable with F_X continuous. However, it turns out that this is not strong enough to give rise to many of the useful properties we require for working with continuous random variables. This results in some variables which are neither continuous nor discrete, like those following a Cantor distribution. For most purposes, these are beyond our scope.

By the fundamental theorem of calculus, we have

$$f_X(x) = \frac{d}{dx} F_X(x)$$

if this derivative exists at x, and 0 else.

There are very many analogues between discrete random variables and continuous random variables. One example of this is the pdf having many properties of the probability mass function of a discrete random variable. For instance,

$$f_X(x) \geq 0 \quad \text{for } x \in \mathbb{R},$$

and

$$\int_{-\infty}^{\infty} f_X(x)\, dx = 1.$$

One place where this does not hold is f_X may exceed 1.

There is some measure-theoretic sense in which f_X is a probability measure. In particular,

$$P(x < X \leq x + \delta x) \sim f_X(x)\, \delta x.$$

However, this is beyond what we need to consider.

Theorem 5.27

For a continuous random variable X with density function f_X, we have:

$$P(X = x) = 0,$$

and

$$P(a \leq X \leq b) = \int_a^b f_X(u)\, du.$$

From our discussion of pdfs so far, we can say any function f satisfying:

$$f(x) \geq 0 \quad \text{for } x \in \mathbb{R},$$

$$\int_{-\infty}^{\infty} f(x)\, dx = 1$$

is in fact a pdf. This is easy to verify by defining the corresponding cumulative distribution function and verifying that its required properties hold.

We will now give some examples of common pdfs.

Uniform distribution.

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a < x < b, \\ 0, & \text{else.} \end{cases}$$

a and b are the parameters.

Exponential distribution.

$$f(x) = \begin{cases} \lambda e^{-\lambda x}, & x > 0, \\ 0, & \text{else.} \end{cases}$$

λ is the parameter.

Normal distribution.

$$f(x) = \frac{1}{\sqrt{2\mu\sigma^2}} \exp\left(-\frac{1}{2\sigma^2}(x-\mu)^2\right), \qquad x \in \mathbb{R}.$$

μ and σ^2 are the parameters.

Cauchy distribution.

$$f(x) = \frac{1}{\pi(1+x^2)}, \qquad x \in \mathbb{R}.$$

Gamma distribution.

$$f(x) = \begin{cases} \dfrac{1}{\Gamma(w)}\lambda^w x^{w-1} e^{-\lambda x}, & x > 0, \\ 0, & \text{else.} \end{cases}$$

w and λ are the parameters.

Beta distribution.

$$f(x) = \begin{cases} \dfrac{1}{B(s,t)} x^{s-1}(1-x)^{t-1}, & 0 \le x \le 1, \\ 0, & \text{else.} \end{cases}$$

s and t are the parameters.

Chi-squared distribution with n degrees of freedom.

$$f(x) = \begin{cases} \dfrac{1}{2\Gamma(\frac{1}{2}n)} \left(\dfrac{x}{2}\right)^{\frac{1}{2}n-1} e^{-\frac{1}{2}nx}, & x > 0, \\ 0, & \text{else.} \end{cases}$$

If we have a random variable X on (Ω, F, P), then $Y = g(X)$ is a random variable on (Ω, F, P) for sufficiently nice g. We will not go into detail about which specific nice properties we need here.

We already know how to find the mass function of Y if X is discrete. If X is continuous, we need another method to find the pdf. Specifically, we have

Theorem 5.50

If X is a continuous random variable with pdf f_X and $g : \mathbb{R} \to \mathbb{R}$ is an increasing differentiable function, we have

$$f_Y(y) = f_X(g^{-1}(y)) \frac{d}{dy} g^{-1}(y).$$

Where g^{-1} is the inverse of g.

The specific formula does not generally need to be memorised, and the derivation allows for more understanding. Essentially, we find the distribution of Y by writing

$$P(Y \le y) = P(X \le g^{-1}(y))$$

and use this to find f_Y. Note that this rewriting of $Y \le y$ is why we must specify g as an increasing function.

We can derive f_Y similarly when g is strictly decreasing.

If g is not monotonic, we break up our process into branches. For instance,

Example 5.53

If X has pdf f_X and $g(x) = x^2$, find the pdf of $Y = g(X)$. We have

$$P(Y \leq y) = P(X^2 \leq y).$$

Then

$$X^2 \leq y \iff -\sqrt{y} \leq X \leq \sqrt{y}.$$

So

$$P(Y \leq y) = \begin{cases} 0, & y < 0, \\ P(-\sqrt{y} \leq X \leq \sqrt{y}), & y \geq 0. \end{cases}$$

Hence

$$\begin{aligned} f_Y(y) &= \frac{d}{dy} P(-\sqrt{y} \leq X \leq \sqrt{y}) && \text{for } y > 0, \\ f_Y(y) &= \frac{d}{dy} \left(F_X(\sqrt{y}) - F_X(-\sqrt{y}) \right), \\ f_Y(y) &= \frac{1}{2\sqrt{y}} \left(f_X(\sqrt{y}) - f_X(-\sqrt{y}) \right). \end{aligned}$$

Definition 5.56

For a continuous random variable X with pdf f_X, we have

$$E(X) = \int_{-\infty}^{\infty} x f_X(x)\, dx$$

whenever the integral converges absolutely.

This is analogous to the definition of expectation for discrete random variables.

Theorem 5.58: LOTUS

For a continuous random variable X with pdf f_X and $g : \mathbb{R} \to \mathbb{R}$, we have

$$E(g(X)) = \int_{-\infty}^{\infty} g(x) f_X(x)\, dx$$

whenever the integral converges absolutely.

This turns out to be somewhat difficult to prove, but the analogy to the discrete case is highly obvious.

Again, we have

$$\text{Var}(X) = E(X^2) - E(X)^2$$

for continuous random variables.

The case of expectations not existing turns up in practice anecdotally more often for the continuous case than for the discrete case, though of course both can occur. For instance, the Cauchy distribution is a common continuous distribution, but

$$\int_{-\infty}^{\infty} \frac{x}{\pi(1+x^2)}\,dx$$

diverges, and so its expectation does not exist.

We will briefly discuss probability in geometrical calculations using our knowledge so far. This is somewhat removed from the rest of this chapter.

The paradox of Joseph Louis Francois Bertrand

A chord of the unit circle is picked at random. Several questions can then be asked about this chord, which result in different answers resulting from choosing a chord "at random" in different ways, which give rise to different distributions.

Let D be the perpendicular between the centre and the chord. Let D be uniformly distributed on the interval $(0, 1)$.

Let A be the acute angle between the chord and its tangent. Let A be uniformly distributed on the interval $(0, \pi/2)$.

Let C be the centre of the chord. Let C be uniformly distributed on the interior of the circle.

If, for instance, we ask for the probability that an equilateral triangle with the chord as its base fits inside the circle, we obtain the answers:

$$\frac{1}{2}, \qquad \frac{2}{3}, \qquad \frac{3}{4}.$$

The needle of Georges Louis Leclerc, Comte de Buffon

A plane is ruled by straight lines separated by unit distances. What is the probability that a unit needle, dropped at random, intersects a line?

If (X, Y) are the coordinates of the midpoint of the needle and θ its angle with the horizontal, we have:

- $Z = Y - \lfloor Y \rfloor$ uniformly distributed on $[0, 1)$,
- θ uniformly distributed on $[0, \pi)$,
- Z, θ independent.

Then

$$f_{Z,\theta}(z, \theta) = \frac{1}{\pi}.$$

We have an intersection when

$$z \leq \frac{1}{2}\sin\theta \quad \text{or} \quad 1 - z \leq \frac{1}{2}\sin\theta,$$

which is more obvious with a sketch. Hence,

$$\begin{aligned}
P(\text{intersection}) &= \iint_{z\leq \frac{1}{2}\sin\theta \text{ or } 1-z\leq \frac{1}{2}\sin\theta} \frac{1}{\pi}\, dz\, d\theta, \\
P(\text{intersection}) &= \iint_{z\leq \frac{1}{2}\sin\theta \text{ or } z\geq 1-\frac{1}{2}\sin\theta} \frac{1}{\pi}\, dz\, d\theta, \\
P(\text{intersection}) &= \frac{1}{\pi}\int_0^{\pi} \left(\int_0^{\frac{1}{2}\sin\theta} dz + \int_{1-\frac{1}{2}\sin\theta}^{1} dz \right) d\theta, \\
P(\text{intersection}) &= \frac{1}{\pi}\int_0^{\pi} \left(\frac{1}{2}\sin\theta + \frac{1}{2}\sin\theta \right) d\theta, \\
P(\text{intersection}) &= \frac{1}{\pi}\int_0^{\pi} \sin\theta\, d\theta, \\
P(\text{intersection}) &= \frac{2}{\pi}.
\end{aligned}$$

We can use this result to develop a Monte Carlo method to approximate the value of π.

Stick breaking

A stick of unit length is broken at two places X and Y uniformly chosen along the stick. What is the probability that three pieces can be used to make a triangle?

We express the lengths of the three new sticks in terms of X and Y, apply the triangle inequality, and sketch the resulting inequalities on the unit square. This is a common application of a technique for problems involving two uniform random variables, where the probability can be found by considering the area of the sample space which satisfies some condition corresponding to an event.

6 Multivariate distributions and independence

We established previously that it can be useful to think of them acting as a random vector (X, Y) taking values in $\mathbb{R}^2$. If X and Y are discrete, we may use the joint mass function of X and Y to study this vector. For continuous random variables, we instead consider their joint distribution function.

Definition 6.1

The joint distribution function of the pair X, Y of random variables is defined as a mapping

$$F_{X,Y} : \mathbb{R}^2 \to [0, 1]$$

given by

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y).$$

We again have some obvious properties:

$$\begin{aligned}
\lim_{(x,y)\to(-\infty,-\infty)} F_{X,Y}(x, y) &= 0, \\
\lim_{(x,y)\to(\infty,\infty)} F_{X,Y}(x, y) &= 1,
\end{aligned}$$

and if $x_1 \leq x_2$ and $y_1 \leq y_2$ then

$$F_{X,Y}(x_1, y_1) \leq F_{X,Y}(x_2, y_2).$$

We can again recover the cumulative distribution functions of X and Y, called the marginal distribution functions, from the joint distribution:

$$F_X(x) = \lim_{y \to \infty} F_{X,Y}(x, y),$$
$$F_Y(y) = \lim_{x \to \infty} F_{X,Y}(x, y).$$

Definition 6.7

We say X and Y are independent if for all $x, y \in \mathbb{R}$, the events $\{X \leq x\}$ and $\{Y \leq y\}$ are independent. Again, this is more often expressed as X and Y are independent if and only if

$$P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y).$$

Which can also be expressed as saying the joint distribution factorises as the product of the marginal distributions.

Again, we have the same notion of independence, and the added notion of pairwise independence, for larger families of random variables.

Definition 6.16

The pair X, Y of random variables on (Ω, F, P) is called jointly continuous if its joint distribution function is expressible in the form

$$F_{X,Y}(x, y) = P(X \leq x, Y \leq y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u, v)\, du\, dv$$

for some function

$$f : \mathbb{R}^2 \to [0, \infty).$$

We say that X and Y have the joint pdf f, usually denoted $f_{X,Y}$.

The joint pdf can be obtained from the joint distribution function using:

$$f_{X,Y}(x, y) = \frac{\partial^2}{\partial x\, \partial y} F_{X,Y}(x, y)$$

if the derivative exists at (x, y), and 0 otherwise.

And once more we have:

$$f_{X,Y}(x, y) \geq 0,$$
$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y)\, dx\, dy = 1.$$

With these properties being necessary and sufficient for f to be the joint pdf of some random variables X and Y.

Theorem 6.22

If A is any regular subset of $\mathbb{R}^2$ and X and Y are jointly continuous random variables with joint pdf $f_{X,Y}$, then

$$P((X,Y) \in A) = \iint_{(x,y)\in A} f_{X,Y}(x,y)\,dx\,dy.$$

Example 6.24

If X and Y have the joint pdf

$$f(x,y) = \begin{cases} \dfrac{1}{ab}, & 0 < x < a \text{ and } 0 < y < b, \\ 0, & \text{else,} \end{cases}$$

then define the rectangle

$$B = (0,a) \times (0,b).$$

Then

$$\begin{aligned}
P((X,Y) \in A) &= \iint_A f(x,y)\,dx\,dy, \\
P((X,Y) \in A) &= \iint_{A\cap B} f(x,y)\,dx\,dy + \iint_{A\setminus B} f(x,y)\,dx\,dy, \\
P((X,Y) \in A) &= \iint_{A\cap B} \frac{1}{ab}\,dx\,dy + 0, \\
P((X,Y) \in A) &= \frac{\text{Area}(A\cap B)}{ab}, \\
P((X,Y) \in A) &= \frac{\text{Area}(A\cap B)}{\text{Area}(B)}.
\end{aligned}$$

This result is the most common interpretation of a uniform distribution of a vector (X,Y) over a rectangle B. Problems will often state something like "(X,Y) is uniformly distributed over a rectangle B such that the probability of (X,Y) being in a region is proportional to its area".

From the criterion of independence we derived, namely

$$F_{X,Y}(x,y) = F_X(x)F_Y(y),$$

we may differentiate with respect to x and y to obtain the further criterion

$$f_{X,Y}(x,y) = f_X(x)f_Y(y)$$

for $x, y \in \mathbb{R}$.

This result can be generalised to give

Theorem 6.31

Jointly continuous random variables X and Y are independent if and only if their joint pdf can be expressed as

$$f_{X,Y}(x,y) = g(x)h(y)$$

for $x, y \in \mathbb{R}$.

Proof idea. Adapt the proof of theorem 3.16 using integrals rather than sums.

Note that the marginal pdf is given by

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,dy,$$

and symmetrically for $f_Y(y)$.

To find the pdf of the sum $Z = X + Y$ for jointly continuous random variables X and Y, we again find the cumulative distribution function of Z first:

$$P(Z \leq z) = P(X + Y \leq z).$$

Also,

$$P(Z \leq z) = \iint_A f_{X,Y}(x,y)\,dx\,dy$$

where A is the region such that $x + y \leq z$. In particular,

$$P(Z \leq z) = \int_{-\infty}^{\infty} \int_{-\infty}^{z-x} f_{X,Y}(x,y)\,dx\,dy.$$

Substituting $u = x$, $v = x + y^2$, we obtain

$$P(Z \leq z) = \int_{-\infty}^{z} \int_{-\infty}^{\infty} f_{X,Y}(u, v-u)\,du\,dv.$$

And differentiating we obtain

$$f_Z(z) = \int_{-\infty}^{\infty} f_{X,Y}(u, z-u)\,du,$$

which is again a relatively intuitive result.

When X and Y are independent, we can factorise $f_{X,Y}$ to obtain:

Theorem 6.38: convolution for continuous random variables

$$f_Z(z) = \int_{-\infty}^{\infty} f_X(x) f_Y(z-x)\,dx.$$

If we have random variables X, Y and function $u, v : \mathbb{R}^2 \to \mathbb{R}$, what can we say about the joint distribution of the pair (U, V) of random variables defined by $U = u(X, Y)$ and $V = v(X, Y)$? There are some conditions which may make the following argument invalid, but we may say:

Define

$$T : \mathbb{R}^2 \to \mathbb{R}^2$$

as $T(x, y) = (u(x, y), v(x, y))$. Suppose that T is a bijection between its domain D and range S. Then T may be inverted to obtain

$$T^{-1} : S \to D,$$

which is itself a bijection. Hence, for each $(u(x, y), v(x, y))$ we have a point (x, y) in D. Write

$$x = x(u(x,y), v(x,y)), \qquad y = y(u(x,y), v(x,y)).$$

Then we may find the Jacobian of T^{-1} and use standard calculus results to obtain theorem 6.50 below.

Recall that the Jacobian is given by

$$J(u,v) = \det \begin{pmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{pmatrix}.$$

Theorem 6.50: Jacobian formula

$$f_{U,V}(u,v) = \begin{cases} f_{X,Y}(x(u,v), y(u,v))|J(u,v)|, & (u,v) \in S, \\ 0, & \text{else.} \end{cases}$$

The details of this argument are beyond what we seek to cover here, however, it may be useful to apply this result.

Example 6.53

Let X and Y follow

$$f(x,y) = \begin{cases} e^{-x-y}, & x,y > 0, \\ 0, & \text{else.} \end{cases}$$

Let

$$U = X + Y, \qquad V = \frac{X}{X+Y}.$$

Find the joint pdf of U and V and the marginal pdf of V.

Solution. Define $T(x,y) = (u,v)$, where

$$u = x + y, \qquad v = \frac{x}{x+y}.$$

T is a bijection from

$$D = \{(x,y) : x,y > 0\}$$

to

$$S = \{(u,v) : 0 < u < \infty,\ 0 < v < 1\}.$$

It has inverse $T^{-1}(u,v) = (x,y)$ where

$$x = uv, \qquad y = u(1-v).$$

Then the Jacobian of T^{-1} is

$$\det \begin{pmatrix} v & u \\ 1-v & -u \end{pmatrix} = -u.$$

So U and V have joint density function

$$f_{U,V}(u,v) = f_{X,Y}(uv, u(1-v))u$$

for $0 < u < \infty$ and $0 < v < 1$. Simplifying, we have

$$f_{U,V}(u,v) = e^{-uv-u+uv}u.$$

And thus

$$f_{U,V}(u,v) = ue^{-u}.$$

The marginal pdf of V is given by

$$f_V(v) = \int_0^\infty ue^{-u}\,du = 1$$

for $0 < v < 1$ and 0 else. Hence V is uniformly distributed on the interval $(0,1)$.

Note that in this case, the most efficient way to show T is a bijection is just to solve explicitly for x and y in terms of u and v, which results in just one possible inverse.

Calculating conditional distributions of Y given $X = x$ requires something other than a direct analogue of the method for discrete variables when X and Y are continuous. Recall we have

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}.$$

But for continuous X and Y, we would have

$$P(Y \leq y \mid X = x) = \frac{P(Y \leq y \cap X = x)}{P(X = x)},$$

which results in division by zero.

Instead, we use the following approach:

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \frac{P(Y \leq y,\ x \leq X \leq x + \delta x)}{P(x \leq X \leq x + \delta x)}.$$

Also,

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \frac{\displaystyle\int_x^{x+\delta x} \int_{-\infty}^{y} f_{X,Y}(u, v)\, du\, dv}{\displaystyle\int_x^{x+\delta x} f_X(u)\, du}.$$

Define the numerator as $A(t)$, where the upper limit of the outer integral is t. Define similarly $B(t)$ for the denominator. Then:

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = A(x + \delta x) = B(x + \delta x).$$

Notice $A(x) = 0$ and $B(x) = 0$, so:

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \frac{\dfrac{A(x + \delta x) - A(x)}{\delta x}}{\dfrac{B(x + \delta x) - B(x)}{\delta x}}.$$

Taking the limit as $\delta x \to 0$,

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \frac{A'(x)}{B'(x)}$$

by the definition of a derivative,

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \frac{\displaystyle\int_{-\infty}^{y} f_{X,Y}(x, v)\, dv}{f_X(x)},$$

and

$$P(Y \leq y \mid x \leq X \leq x + \delta x) = \int_{-\infty}^{y} \frac{f_{X,Y}(x, v)}{f_X(x)}\, dv.$$

Now this right-hand side is a cumulative distribution function, so we can differentiate both sides to obtain

Definition 6.57

The conditional density function of Y given that $X = x$ is defined as

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)}.$$

Compare this with the discrete analogue,

$$P(Y = y \mid X = x) = \frac{P(X = x, Y = y)}{P(X = x)},$$

and we see the similar structure.

And if X and Y are independent, we obtain

$$f_{Y|X}(y \mid x) = \frac{f_X(x) f_Y(y)}{f_X(x)} = f_Y(y)$$

as expected.

Example 6.59

Let X and Y have joint pdf

$$f(x, y) = \begin{cases} 2e^{-x-y}, & 0 < x < y < \infty, \\ 0, & \text{else.} \end{cases}$$

The marginals are

$$f_X(x) = 2e^{-2x} \quad \text{for } x > 0$$

and

$$f_Y(y) = 2e^{-y}(1 - e^{-y}) \quad \text{for } y > 0.$$

Note the restrictions on the ranges of x and y affecting the domain over which we integrate.

The conditional pdf of Y given $X = x$ is therefore:

$$f_{Y|X}(y \mid x) = \begin{cases} \dfrac{2e^{-x-y}}{2e^{-2x}}, & y > x, \\ 0, & \text{else,} \end{cases}$$

so

$$f_{Y|X}(y \mid x) = e^{x-y} \quad \text{for } y > x.$$

We also have:

$$f_{X|Y}(x \mid y) = \begin{cases} \dfrac{2e^{-x-y}}{2e^{-y}(1 - e^{-y})}, & 0 < x < y, \\ 0, & \text{else,} \end{cases}$$

so

$$f_{X|Y}(x \mid y) = \frac{e^{-x}}{1 - e^{-y}} \quad \text{for } 0 < x < y.$$

Theorem 6.62

For continuous random variables X and Y and a function $g : \mathbb{R}^2 \to \mathbb{R}$, we have the random variable $Z = g(X, Y)$. To calculate the expectation of Z, we use:

$$E(Z) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y)\, dx\, dy.$$

And again, we may use this to obtain

$$E(aX + bY) = aE(X) + bE(Y),$$

this time for continuous random variables.

Recall we had for discrete random variables $E(XY) = E(X)E(Y)$ if X and Y are independent (though not necessarily the converse). This also holds for continuous random variables.

Theorem 6.66

If X and Y are continuous random variables such that

$$E(g(X)h(Y)) = E(g(X))E(h(Y))$$

for all functions $g, h : \mathbb{R} \to \mathbb{R}$ such that the expectations exist, then X and Y are independent. This is a necessary and sufficient condition for independence.

Proof idea. Use indicator functions, identically to 3.19 with the necessary changes for continuous rather than discrete random variables.

Definition 6.68

The conditional expectation of Y given $X = x$ is defined as

$$E(Y \mid X = x) = \int_{-\infty}^{\infty} y f_{Y|X}(y \mid x)\, dy = \int_{-\infty}^{\infty} y \frac{f_{X,Y}(x, y)}{f_X(x)}\, dy.$$

Theorem 6.69: analogue of the partition theorem for continuous random variables

If X and Y are jointly continuous random variables then

$$E(Y) = \int_{x : f_X(x) > 0} E(Y \mid X = x) f_X(x)\, dx.$$

Proof. This may be proved more straightforwardly, but we may also apply the tower property of expectation, which has not been touched on yet. The tower property is given below. We have

$$E(E(Y \mid X = x)) = E(Y).$$

Then simply applying LOTUS results in:

$$E(Y) = \int_{-\infty}^{\infty} f_X(x) E(Y \mid X = x)\, dx.$$

This is not exactly rigorous as $E(Y \mid X = x)$ is undefined when $f_X(x) = 0$, but removing these points does not change the value of our integral, so

$$E(Y) = \int_{x : f_X(x) > 0} E(Y \mid X = x) f_X(x)\, dx.$$

Tower property of expectation

Define

$$g(x) = E(Y \mid X = x).$$

Then

$$E(g(X)) = E(Y).$$

Proof in the discrete case.

$$\begin{aligned}
E(g(X)) &= \sum_x E(Y \mid X = x)P(X = x), \\
E(g(X)) &= \sum_x \left(\sum_y yP(Y = y \mid X = x)P(X = x) \right), \\
E(g(X)) &= \sum_x \left(\sum_y y\frac{P(X = x, Y = y)}{P(X = x)} \right) P(X = x), \\
E(g(X)) &= \sum_x \left(\sum_y yP(X = x, Y = y) \right), \\
E(g(X)) &= \sum_y \left(y\sum_x P(X = x, Y = y) \right), \\
E(g(X)) &= \sum_y yP(Y = y), \\
E(g(X)) &= E(Y).
\end{aligned}$$

Exercise 6.70

If

$$f_{X,Y}(x, y) = \begin{cases} \frac{1}{\pi}, & x^2 + y^2 \leq 1, \\ 0, & \text{else}, \end{cases}$$

find $E(\sqrt{X^2 + Y^2})$.

Solution. By LOTUS,

$$\begin{aligned}
E(\sqrt{X^2 + Y^2}) &= \iint_{x^2+y^2\leq 1} \sqrt{x^2 + y^2}\frac{1}{\pi}\, dx\, dy, \\
E(\sqrt{X^2 + Y^2}) &= \frac{1}{\pi}\int_0^{2\pi}\int_0^1 r \cdot r\, dr\, d\theta \qquad \text{switching to polar coordinates,} \\
E(\sqrt{X^2 + Y^2}) &= \frac{1}{\pi}\int_0^{2\pi}\frac{1}{3}\, d\theta, \\
E(\sqrt{X^2 + Y^2}) &= \frac{1}{\pi}\cdot\frac{2\pi}{3}, \\
E(\sqrt{X^2 + Y^2}) &= \frac{2}{3}.
\end{aligned}$$

The bivariate normal distribution has the following joint pdf:

$$f(x, y) = \frac{1}{2\pi\sqrt{1 - \rho^2}} \exp\left(-\frac{1}{2(1 - \rho^2)}(x^2 - 2\rho xy + y^2) \right)$$

for $x, y \in \mathbb{R}$ and $\rho \in (-1, 1)$.

Notably:

- X and Y each are normally distributed with mean 0 and variance 1.
- The conditional density function of Y given $X = x$ is the normal distribution with mean ρx and variance $1 - \rho^2$.
- The conditional expectation of Y given $X = x$ is ρx.
- X and Y are independent if and only if $\rho = 0$.
- $E(XY) = \rho$, resulting in $\rho = E(XY) - E(X)E(Y)$. So X and Y are independent if and only if $E(XY) = E(X)E(Y)$.

Why is this useful? It might seem strange to use the criterion $E(XY) = E(X)E(Y)$ for independence, when we can check the simpler criterion $\rho = 0$, since ρ must be specified to give the distribution anyway. However, it can be shown that if the sum of two random variables $aX + bY$ follows a univariate normal distribution, then (X, Y) follows a bivariate normal distribution. It may be hard to calculate ρ for the distribution of (X, Y), but finding the relevant expectations is easy, and hence we can test easily for the independence of X and Y. Supposing for instance $X = 3A + 4B$ and $Y = 4A - 3B$, where A and B are independently distributed on $\mathcal{N}(0, 1)$, we have:

- $aX + bY$ follows a univariate normal distribution as this is a sum of normally distributed variables.
- $E(X) = E(Y) = 0$.
- $E(XY) = E(12A^2 + 7AB - 12B^2) = 12 + 0 - 12 = 0$.

Hence $E(XY) = E(X)E(Y)$, and thus X and Y are independent, despite not knowing ρ for their bivariate normal distribution.

7 Moments and Moment Generating Functions

We previously defined the kth moment of X as $E(X^k)$ whenever this expectation exists. We will study moments in more detail.

Example 7.6

Recall the Cauchy distribution with

$$f(x) = \frac{1}{\pi(1 + x^2)}.$$

It is easy to verify that the integral

$$\int_{-\infty}^{\infty} \left| \frac{x^k}{\pi(1 + x^2)} \right| dx$$

diverges for all $k \geq 1$, and therefore the Cauchy distribution has no moments. Recall that for a continuous expectation to exist, the corresponding integral must converge absolutely.

Theorem 7.7: uniqueness theorem for moments

Suppose that all moments of the random variable X exist and that the series

$$\sum_{k=0}^{\infty} \frac{1}{k!} t^k E(X^k)$$

is absolutely convergent for some $t > 0$. Then the sequence of moments uniquely determines the distribution of X.

The condition of absolute convergence is sufficient but not necessary for uniqueness.

The series in this theorem is the exponential generating function of the sequence of moments.

Example 7.9

If $f(x)$ is the pdf of the log-normal distribution, which we will not go into detail with but which has a well-defined pdf, then

$$f_a(x) = (1 + a \sin(2\pi \ln x)) f(x)$$

is also a density function for $-1 \leq a \leq 1$. It turns out that f and f_a have equal finite moments, despite describing different distributions. By the contrapositive, we have that the sum in theorem 7.7 does not absolutely converge for these series. Intuitively, the sin term in f_a adds some waves to the pdf f, which when calculating moments cancel out to leave the moments the same as for f. The details of this are of course somewhat more involved.

For any random variable Y,

$$E(Y^2) = 0$$

if and only if

$$P(Y = 0) = 1.$$

Hence

$$\mathrm{Var}(X) = E((X - \mu)^2) = 0$$

if and only if $X = \mu$ always, or in other words $P(X = \mu) = 1$.

It is also useful to note the inequality

$$\mathrm{Var}(X) \geq 0,$$

and thus

$$E(X^2) \geq E(X)^2$$

by the alternate definition of $\mathrm{Var}(X)$.

We have also

$$\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X).$$

This is somewhat undesirable; we might prefer out measure of dispersion to be linear. Therefore, it is sometimes preferable to use the standard deviation of x, $\sigma(X)$, with the property

$$\sigma(aX + b) = a\sigma(X).$$

We have also

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + 2E((X - E(X))(Y - E(Y))) + \mathrm{Var}(Y)$$

upon applying the definition of variance.

Definition 7.19

We define the covariance of the random variables X and Y as

$$\mathrm{Cov}(X,Y) = E((X - E(X))(Y - E(Y)))$$

whenever the expectations exist.

Hence

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y).$$

Expanding, we may also obtain

$$\begin{aligned}
\mathrm{Cov}(X,Y) &= E(XY - XE(Y) - YE(X) + E(X)E(Y)),\\
\mathrm{Cov}(X,Y) &= E(XY) - E(X)E(Y) - E(X)E(Y) + E(X)E(Y),\\
\mathrm{Cov}(X,Y) &= E(XY) - E(X)E(Y).
\end{aligned}$$

If X and Y are independent, $E(XY) = E(X)E(Y)$ and so $\mathrm{Cov}(X,Y) = 0$. Thus, for independent X and Y, we have:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

The converse is not necessarily true.

One disadvantage of covariance is that it is not scale-invariant. For instance, if X and Y are random measurements in inches while U and V are the same random measurements in centimetres, we do not have $\mathrm{Cov}(X,Y) = \mathrm{Cov}(U,V)$, despite the dependence between the values necessarily being the same. This motivates a definition of the correlation coefficient below.

Definition 7.25

The correlation coefficient of the random variables X and Y is defined as

$$\rho(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}$$

whenever the quantities on the right hand side exist and

$$\mathrm{Var}(X)\,\mathrm{Var}(Y) \neq 0.$$

It can be routinely shown that

$$\rho(aX+b, cY+d) = \rho(X,Y)$$

as desired (for $ac \neq 0$).

Theorem 7.28

We have

$$-1 \leq \rho(X,Y) \leq 1$$

whenever this correlation exists.

Proof. We use theorem 7.30 given below. Setting $U = X - E(X)$ and $V = Y - E(Y)$, we obtain

$$\mathrm{Cov}(X,Y)^2 \leq \mathrm{Var}(X)\,\mathrm{Var}(Y).$$

This is equivalent to

$$\rho(X,Y)^2 \leq 1,$$

which is equivalent to the desired inequality.

Theorem 7.30: Cauchy-Schwartz inequality for random variables

For random variables U and V, we have

$$(E(UV))^2 \leq E(U^2)E(V^2)$$

whenever the expectations exist.

Proof. Let $s \in \mathbb{R}$ and consider the variable $sU + V$. We have $(sU + V)^2 \geq 0$ always, so

$$\begin{aligned} 0 &\leq E(s^2U^2 + 2sUV + V^2), \\ 0 &\leq s^2E(U^2) + 2sE(UV) + E(V^2). \end{aligned}$$

Now consider the right hand side as a quadratic in s. We have $E(U^2) \geq 0$. If $E(U^2) = 0$, then $P(U = 0) = 1$, and the inequality holds trivially. So we need only consider $E(U^2) > 0$. Then we have a non-degenerate quadratic which we can apply the discriminant inequality to, as we know it has at most one root. Consequently,

$$4E(UV)^2 - 4E(U^2)E(V^2) \leq 0.$$

Dividing by 4 obtains the desired result.

Let us return to the Cauchy-Schwartz inequality with $U = X - E(X)$ and $V = Y - E(Y)$. Suppose we have

$$\rho(X, Y) = \pm 1.$$

Then the discriminant involving s is

$$\begin{aligned} \Delta &= 4\operatorname{Cov}(X, Y)^2 - 4\operatorname{Var}(X)\operatorname{Var}(Y), \\ \Delta &= 4\operatorname{Var}(X)\operatorname{Var}(Y)(\rho(X, Y)^2 - 1), \\ \Delta &= 0. \end{aligned}$$

So the quadratic has one real root, say at $s = \alpha$. Hence, we have

$$E(\alpha^2U^2 + 2\alpha UV + V^2) = 0$$

for $U = X - E(X)$ and $V = Y - E(Y)$. So the random variable $\alpha^2U^2 + 2\alpha UV + V^2$ has

$$P(\alpha^2U^2 + 2\alpha UV + V^2 = 0) = 1.$$

So

$$\alpha^2U^2 + 2\alpha UV + V^2 = 0.$$

Upon back-substituting and rearranging, we obtain

$$Y = -\alpha X + \beta$$

for $\beta = \alpha E(X) + E(Y)$. More carefully, we can also obtain $\rho(X, Y) = 1$ iff. $Y = \alpha X + \beta$ for $\alpha > 0$ and $\rho(X, Y) = 1$ iff. $Y = \alpha X + \beta$ for $\alpha < 0$.

In summary, we may use $\rho(X, Y)$ as a measure of the dependence of X and Y, so long as they have non-zero variances. We have:

- If X and Y are independent, then $\rho(X, Y) = 0$. The converse need not be true, but we can then say that X and Y are uncorrelated.
- Y is a linear increasing function of X if and only if $\rho(X, Y) = 1$ and Y is a linear decreasing function of X if and only if $\rho(X, Y) = -1$.

Definition 7.39

The moment generating function of the random variable X is defined as

$$M_X(t) = E(e^{tX})$$

for all $t \in \mathbb{R}$ for which the expectation exists.

Consequently,

$$M_X(t) = \sum_x e^{tx} P(X = x)$$

for discrete X, and

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} f_X(x)\, dx$$

for continuous X, whenever the sum or integral converges absolutely.

When X is a non-negative integer random variable, we have

$$M_X(t) = G_X(e^t).$$

It is important that $E(e^{tX})$ exists in some neighborhood $t \in (-\delta, \delta)$. Otherwise, moment generating functions have less utility.

Example 7.43

The moment generating function of the normal distribution with mean 0 and variance 1 exists for all $t \in \mathbb{R}$.

Example 7.45

The moment generating function of the exponential distribution with parameter λ exists for $t < \lambda$.

Example 7.47

The moment generating function of the Cauchy distribution exists only for $t = 0$.

$$M_X(t) = E\left(1 + tX + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \cdots\right),$$
$$M_X(t) = 1 + tE(X) + \frac{t^2}{2!}E(X^2) + \frac{t^3}{3!}E(X^3) + \cdots.$$

Hence, $M_X(t)$ is the exponential generating function of the moments of X. Note that there is some lack of rigour in the interchanging of expectation and summation here, but this is okay for our purposes.

Theorem 7.49

If $M_X(t)$ exists in a neighbourhood of 0, then

$$E(X^k) = M_K^{(k)}(0),$$

the kth derivative of $M_X(t)$ at $t = 0$.

Proof sketch. Interchanging expectation and differentiation in the definition of $M_X(t)$ immediately gives the result.

We shall again examine how this generating function interacts with sums of random variables. We have

$$\begin{aligned} M_{aX+b}(t) &= E(e^{t(aX+b)}), \\ M_{aX+b}(t) &= E(e^{atX}e^{bt}), \\ M_{aX+b}(t) &= e^{bt}M_X(at). \end{aligned}$$

Theorem 7.52

If X and Y are independent random variables, then $X + Y$ has moment generating function

$$M_{X+Y}(t) = M_X(t)M_Y(t).$$

Proof.

$$\begin{aligned} M_{X+Y}(t) &= E(e^{tX}e^{tY}), \\ M_{X+Y}(t) &= E(e^{tX})E(e^{tY}) \qquad \text{by theorem 6.66.} \end{aligned}$$

The result follows.

The expected result

$$M_S(t) = M_{X_1}(t) + M_{X_2}(t) + \cdots + M_{X_N}(t)$$

holds also for $S = X_1 + X_2 + \cdots + X_N$.

Theorem 7.55: uniqueness theorem for moment generating functions

If $M_X(t)$ is finite for all $t \in (-\delta, \delta)$ for some $\delta > 0$, there is a unique distribution with moment generating function M_X. We also have each moment being finite under this condition. Consequently, we can legitimately expand $M_X(t)$ as a power series around the origin with radius of convergence δ if we know $M_X(t)$ is the moment generating function of a distribution.

This requires some stronger machinery to prove, involving Laplace transforms.

Example 7.57

If X and Y follow independent normal distributions with parameters μ_1, σ_1^2 and μ_2, σ_2^2, then we can use theorem 7.55 to show that $Z = X + Y$ follows a normal distribution with parameters $(\mu_1 + \mu_2)$, $(\sigma_1^2 + \sigma_2^2)$.

We have some inequalities associated with the moments of a random variable.

Markov's inequality tells us something about the distribution of X given that its mean is finite. Specifically, it places a bound on the right tail.

Theorem 7.63: Markov's inequality

For any non-negative random variable X,

$$P(X \geq t) \leq \frac{E(X)}{t}$$

for $t > 0$.

Proof. We can use the definition of expectation as a sum or integral. However, we may also use an indicator function approach as follows. We have the trivial inequality $X \geq t$ for $X \geq t$ and $X \geq 0$ for $X < t$ (recall t is positive). This implies

$$X \geq tI_A,$$

where I_A is the indicator function of the event $\{X \geq t\}$. Taking expectations, we have

$$E(X) \geq tP(X \geq t).$$

Hence

$$P(X \geq t) \leq \frac{E(X)}{t}$$

(again, using the condition that t is positive).

Example 7.65

The real number m is called a median of a random variable X if

$$P(X < m) \leq \frac{1}{2} \leq P(X \leq m).$$

Every random variable has at least one median m. By Markov's inequality, we have

$$\frac{1}{2} \leq P(X \geq m) \leq \frac{E(X)}{m},$$

and thus

$$m \leq 2E(X)$$

taking the extreme left and right sides.

Theorem 7.67: Jensen's inequality

Let X be a random variable taking values in the possibly infinite interval (a, b) such that $E(X)$ exists. Let $g : (a, b) \to \mathbb{R}$ be a convex function such that

$$E|g(X)| < \infty.$$

Then

$$E(g(X)) \geq g(E(X)).$$

Here, we will define a convex function as one where every chord of the curve $y = g(x)$ lies above the curve, and we will not go into more rigour. This definition allows for the following intuition: taking a mean of two points of the curve will result in a greater value than the value of the curve on the midpoint, as the first value lies on the chord line, which is vertically above the curve. Note that this statement is very similar to that of Jensen's inequality, but with a two-point average rather than the expectation operation.

Theorem 7.68: supporting tangent theorem

Let $g : (a, b) \to \mathbb{R}$ be convex. Then for $a < w < b$, there exists a straight line that touches the curve at $(w, g(w))$ such that the curve never passes below the line. If the line has equation

$$y = \alpha(x - w) = g(w)$$

for some α, then our statement is equivalent to

$$g(x) \geq g(w) + \alpha(x - w)$$

for $x \in (a, b)$.

Intuition. One can be convinced of this theorem through a diagram.

Proof of theorem 7.67. Let X take values in (a, b) with mean $\mu = E(X)$. Let $g : (a, b) \to \mathbb{R}$ be a convex function such that $E|g(X)| < \infty$. Then by theorem 7.68, we have

$$g(x) \geq g(\mu) + \alpha(x - \mu)$$

for some α and for $x \in (a, b)$. So

$$g(X) \geq g(\mu) + \alpha(X - \mu)$$

in the general case. Taking expectations,

$$E(g(X)) \geq g(E(X)) + \alpha(E(X)) - \alpha(E(X))$$

replacing μ with $E(X)$. So

$$E(g(X)) \geq g(E(X))$$

as required.

Example 7.70: AM-GM inequality

Note that

$$g(x) = -\log x$$

is convex on $(0, \infty)$. Hence,

$$E(\log X) \leq \log(E(X))$$

for any positive random variable X with finite mean. Now let X be a discrete random variable which is equally likely to take any of the positive values $x_1, x_2, \ldots, x_n$. We have

$$E(\log X) = \frac{1}{n} \sum_{i=1}^{n} \log(x_i) = \log \left(\left(\prod_{i=1}^{n} x_i \right)^{1/n} \right).$$

And

$$\log(E(X)) = \log \left(\frac{x_1 + x_2 + \cdots + x_n}{n} \right).$$

Taking exponentials, we have

$$\left(\prod_{i=1}^{n} x_i \right)^{1/n} \leq \frac{x_1 + x_2 + \cdots + x_n}{n}.$$

Which is the familiar AM-GM inequality.

We return to moment generating functions. We can make a change to ensure that another type of generating function with broadly the same properties always exists.

Definition 7.76

The characteristic function of the random variable X is defined as

$$\phi_X(t) = E(e^{itX})$$

for $t \in \mathbb{R}$.

We can also write

$$\phi_X(t) = E(\cos(tX)) + iE(\sin(tX)).$$

We have, crucially,

$$|\phi_X(t)| \leq 1$$

for all $t \in \mathbb{R}$, guaranteeing finiteness.

Some examples of characteristic functions are given below.

Example 7.78

The random variable X taking the value a with probably p or the value b with probability $1 - p$ has characteristic function

$$\phi_X(t) = E(e^{itX}) = pe^{ita} + (1-p)e^{itb}.$$

Example 7.79

The random variable X with exponential distribution with parameter λ has characteristic function

$$\begin{aligned}
\phi_X(t) &= \int_0^\infty e^{itx}\lambda e^{-\lambda x}\,dx, \\
\phi_X(t) &= \lambda\int_0^\infty e^{x(it-\lambda)}\,dx, \\
\phi_X(t) &= \frac{\lambda}{it-\lambda}\left[e^{itx}e^{-\lambda x}\right]_0^\infty, \\
\phi_X(t) &= \frac{\lambda}{it-\lambda}[0-1], \\
\phi_X(t) &= \frac{\lambda}{\lambda - it}.
\end{aligned}$$

Example 7.80

The random variable X with Cauchy distribution has characteristic function

$$\begin{aligned}
\phi_X(t) &= \int_{-\infty}^\infty \frac{e^{itx}}{\pi(1+x^2)}\,dx, \\
\phi_X(t) &= e^{-|t|},
\end{aligned}$$

obtained using contour integration.

Sometimes, contour integration may be avoided using

$$\phi_X(t) = E(e^{itX}) = M_X(it).$$

This sometimes works, but it is not rigorous and sometimes fails, as with the Cauchy distribution. However, it does in fact always work if the moment generating function M_X is finite in a neighbourhood of the origin.

Example 7.82

If X has the normal distribution with mean 0 and variance 1, we have

$$M_X(s) = e^{\frac{1}{2}s^2}$$

for $t \in \mathbb{R}$. Hence,

$$\phi_X(t) = e^{-\frac{1}{2}t^2}.$$

Theorem 7.85

If

$$E|X^N| < \infty$$

for some $N \in \mathbb{Z}_+$, then

$$\phi_X(t) = \sum_{k=0}^{N} \frac{1}{k!}(it)^k E(X^k) + o(t^N)$$

as $t \to 0$.

Here, $o(h)$ denotes a function of h which satisfies $f(h)/h \to 0$ as $h \to 0$, meaning $o(h)$ is of very small magnitude close to $h = 0$. Thus, the remainder $o(t^N)$ is negligible compared to the t^k terms. A function of $o(t^N)$ is smaller near zero than a function of $o(t^k)$ for $k < N$.

Some properties of characteristic functions are given below.

Theorem 7.87

Let X and Y be independent random variables with characteristic functions ϕ_X and ϕ_Y. Then:

- if $Z = aX + b$, then
$$\phi_Z(t) = e^{itb}\phi_X(at),$$
analogous to the result for moment generating functions.
- if $Z = X + Y$, then
$$\phi_Z(t) = \phi_X(t)\phi_Y(t),$$
also analogous.

These can be proven using standard expectation results for independent random variables.

Theorem 7.88: uniqueness theorem for characteristic functions

$$\phi_X(t) = \phi_Y(t)$$

is necessary and sufficient for X and Y to be identically distributed.

Consequently, any characteristic function ϕ corresponds to a unique distribution function. However, it may be hard to find this distribution. For distributions with density functions, we have a relatively simple form.

Theorem 7.89: inversion theorem

Let X have characteristic function ϕ. Then the pdf of X is given by

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t)\, dt.$$

Proof. We start with the right hand side. Let

$$I = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \phi(t)\, dt,$$
$$I = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-itx} \left(\int_{-\infty}^{\infty} e^{ity} f(y)\, dy \right) dt,$$

by definition of $\phi(t)$ for continuous X.

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-itx} e^{ity} f(y)\, dy\, dt,$$
$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-itx} e^{ity} f(y)\, dt\, dy,$$
$$I = \int_{-\infty}^{\infty} f(y) \left(\int_{-\infty}^{\infty} \frac{1}{2\pi} e^{-it(y-x)}\, dt \right) dy,$$
$$I = \int_{-\infty}^{\infty} f(y) \delta(y-x)\, dy.$$

This is because the inner integral is the definition of the Dirac delta function at $y - x$.

$$I = f(x),$$

as $\delta(y - x) \neq 0$ only at $y = x$, then applying this fundamental property of the Dirac delta function.

Unfortunately, we may only apply theorem 7.89 once we know that a given ϕ comes from a continuous random variable. However, there is no (nice) necessary and sufficient test. One sufficient condition is that

$$\int_{-\infty}^{\infty} |\phi(t)|\, dt$$

is finite. This holds for the normal distribution, but fails for the exponential as just two examples.

Characteristic functions are closely related to Fourier analysis. This is more concrete when dealing with non-negative integer random variables, where we have

$$\phi(t) = \sum_{k=0}^{\infty} e^{itk} P(X = x).$$

Suppose we wish to find $P(X = j)$. Multiplying by e^{-itj}, we may obtain

$$e^{-itj} \phi(t) = \sum_{k=0}^{\infty} e^{it(k-j)} P(X = x).$$

For integers m, we have the result

$$\int_0^{2\pi} e^{imt}\, dt = \begin{cases} 2\pi, & m = 0, \\ 0, & \text{otherwise.} \end{cases}$$

So integrating the previous equation, we have

$$\int_0^{2\pi} e^{-itj}\phi(t)\,dt = 2\pi P(X = j).$$

Rearranging,

$$P(X = j) = \frac{1}{2\pi}\int_0^{2\pi} e^{-itj}\phi(t)\,dt.$$

The standard formula for a Fourier coefficient is

$$c_j = \frac{1}{2\pi}\int_0^{2\pi} e^{-itj}g(t)\,dt,$$

and so for non-negative integer variables X, we have the probabilities of X corresponding to Fourier coefficients.

8 The Main Limit Theorems

We aim to develop the two main limit theorems, namely the law of large numbers and the central limit theorem.

Suppose we throw a million dice and record the results. We note the average, 3.500867. This is unsurprising as the expected value of each throw is 3.5. However, we would like to show that the average

$$\frac{1}{n}(x_1 + x_2 + \cdots + x_n)$$

really does converge to the underlying mean value μ as $n \to \infty$.

There are different ways in which random variables can be said to converge.

Definition 8.3

We say that the sequence $Z_1, Z_2, \ldots$ of random variables converges in mean square to the random variable Z if

$$E((Z_n - Z)^2) \to 0$$

as $n \to \infty$. We write $Z_n \to Z$ in mean square as $n \to \infty$.

Example 8.5

Define a random variable Z_n with

$$P(Z_n = 1) = \frac{1}{n}, \qquad P(Z_n = 2) = 1 - \frac{1}{n}.$$

Then

$$E((Z_n - 2)^2) = \frac{1}{n}\cdot 1 + \left(1 - \frac{1}{n}\right)\cdot 0.$$

So

$$E((Z_n - 2)^2) = \frac{1}{n}.$$

And thus $Z_n \to 2$ in mean square as $n \to \infty$ (as the expectation goes to zero).

Theorem 8.6: mean-square law of large numbers

Let $X_1, X_2, \ldots$ be a sequence of independent random variables, each with mean μ and variance σ^2. The average of the first n of the X_i converges to the constant variable μ in mean square as $n \to \infty$.

Proof. Write S_n as the nth partial sum. Then the average of the first n of the X_i is $E(\frac{1}{n}S_n)$.

$$E\left(\frac{1}{n}S_n\right) = \frac{1}{n}(E(X_1) + E(X_2) + \cdots + E(X_n)),$$
$$E\left(\frac{1}{n}S_n\right) = \frac{1}{n}(\mu + \mu + \cdots + \mu),$$
$$E\left(\frac{1}{n}S_n\right) = \mu.$$

Now note

$$\mathrm{Var}\left(\frac{1}{n}S_n\right) = E\left(\left(\frac{1}{n}S_n - \mu\right)^2\right) - E\left(\frac{1}{n}S_n - \mu\right)^2.$$

So

$$\mathrm{Var}\left(\frac{1}{n}S_n\right) = E\left(\left(\frac{1}{n}S_n - \mu\right)^2\right).$$

But

$$\mathrm{Var}\left(\frac{1}{n}S_n\right) = \frac{1}{n^2}(\mathrm{Var}(X_1) + \mathrm{Var}(X_2) + \cdots + \mathrm{Var}(X_n)).$$

So

$$E\left(\left(\frac{1}{n}S_n - \mu\right)^2\right) = \frac{1}{n^2}(\sigma^2 + \sigma^2 + \cdots + \sigma^2).$$

So

$$E\left(\left(\frac{1}{n}S_n - \mu\right)^2\right) = \frac{\sigma^2}{n} \to 0$$

as $n \to \infty$. So the average of the first n of the X_i converges to the constant variable μ in mean square as $n \to \infty$.

The next definition of convergence we will use is in some sense more intuitive.

Definition 8.12

We say that the sequence $Z_1, Z_2, \ldots$ of random variables converges in probability to Z as $n \to \infty$ if, for all $\varepsilon > 0$,

$$P(|Z_n - Z| > \varepsilon) \to 0$$

as $n \to \infty$. We write $Z_n \to Z$ in probability as $n \to \infty$.

We can rewrite this criterion as requiring

$$P(|Z_n - Z| \leq \varepsilon) \geq 1 - \delta$$

for all small positive δ and ε for sufficiently large n.

Although not immediately clear, it turns out that convergence in mean square is stronger than convergence in probability.

Theorem 8.14

If $Z_1, Z_2, \ldots$ is a sequence of random variables and $Z_n \to Z$ in mean square as $n \to \infty$, then $Z_n \to Z$ in probability also.

Proof. We apply Chebyshev's inequality, given as theorem 8.15 below, to the random variable $Z_n - Z$. Then

$$P(|Z_n - Z| \geq \varepsilon) \leq \frac{E((Z_n - Z)^2)}{\varepsilon^2}$$

for $\varepsilon > 0$. If we have convergence in mean square, the right hand side goes to zero and thus the left hand side must also go to zero. The left hand side going to zero as $n \to \infty$ is the definition of convergence in probability.

Theorem 8.15: Chebyshev's inequality

If Y is a random variable and $E(Y^2) < \infty$, then

$$P(|Y| \geq t) \leq \frac{E(Y^2)}{t^2}$$

for $t > 0$.

Proof. Recall theorem 7.63, Markov's inequality, states that

$$P(X \geq s) \leq \frac{E(X)}{s}$$

for $s > 0$ for non-negative random variables X. We apply Markov's inequality to the non-negative random variable Y^2, giving

$$P(Y^2 \geq s) \leq \frac{E(Y^2)}{s}$$

for $s > 0$. Let $s = t^2$ and note that

$$P(Y^2 \geq t^2) = P(|Y| \geq t).$$

Then we obtain the result.

Note that the converse of theorem 8.14 is false; there exist sequences of random variables which converge in probability but not in mean square. This is why we say that convergence in mean square is stronger than convergence in probability.

Combining the mean-square law of large numbers with theorem 8.14 gives rise to the weak law of large numbers.

Theorem 8.17: weak law of large numbers

Let $X_1, X_2, \ldots$ be a sequence of independent random variables, each with mean μ and variance σ^2. The average of the first n of the X_i converges to the constant variable μ in probability as $n \to \infty$.

One advantage of the weak law of large numbers over the mean-square law of large numbers is that the former does not require the X_i to have finite variance. Our proof has not yet allowed for this assumption to be dropped, but the conclusion nonetheless applies.

There are many other laws of large numbers which work with different assumptions. Some may have weak assumptions and strong conclusions; this makes them more powerful but harder to prove.

For instance, the strong law of large numbers assumes only a finite mean (less assumptions than mean-square) but assumes independence (more assumptions than mean-square, which only requires pairwise uncorrelated variables). It states that

$$P\left(\lim_{n\to\infty} \bar{X}_n = \mu\right) = 1,$$

which is subtly different to the mean-square law, which we can write in limit notation as

$$\lim_{n\to\infty} E((\bar{X}_n - \mu)^2) = 0.$$

It can be very difficult to intuitively understand the difference between these laws, and one would not be blamed for forgoing this understanding in favour of at least being able to complete problems involving these laws.

Example 8.19

Suppose Z_n is a random variable with

$$P(Z_n = 0) = 1 - \frac{1}{n}, \qquad P(Z_n = n) = \frac{1}{n}.$$

Then

$$P(|Z_n| > \varepsilon) = P(Z_n = n) = \frac{1}{n} \to 0$$

as $n \to \infty$ (assuming $\varepsilon > 0$). Hence Z_n converges in probability. But

$$E((Z_n - 0)^2) = E(Z_n^2) = n^2 \cdot \frac{1}{n} = n \to \infty$$

as $n \to \infty$. Hence Z_n does not converge (to zero) in mean square.

Moving on from laws of large numbers, we aim to show the central limit theorem, a result which concerns sums of independent variables.

Let $X_1, X_2, \ldots$ be independent identically distributed random variables each with mean μ and non-zero variance σ^2. By the weak law of large numbers, we know that

$$S_n \sim n\mu$$

for large n. We wish to investigate the error $S_n - n\mu$, which it turns out has order $\sqrt{n}$.

We will work with Z_n, the standardised version of S_n. Define

$$Z_n = \frac{S_n - E(S_n)}{\sqrt{\mathrm{Var}(S_n)}}.$$

Note that Z_n is a linear function of S_n with $E(Z_n) = 0$ and $\mathrm{Var}(Z_n) = 1$.

We recall $E(S_n) = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$, so in fact

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}.$$

Theorem 8.25: central limit theorem

Let $X_1, X_2, \ldots$ be a sequence of independent identically distributed random variables, each with mean μ and non-zero variance σ^2. Then

$$Z_n = Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}}$$

of the sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

satisfies, as $n \to \infty$,

$$P(Z_n \leq x) \to \int_{-\infty}^{x} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}u^2}\, du$$

for $x \in \mathbb{R}$.

The right hand side of this limit is just the distribution function of the standard normal distribution, so that we may say

$$P(Z_n \leq x) \to P(Z \leq x)$$

where Z is a random variable with the standard normal distribution.

Theorem 8.27: continuity theorem

If the moment generating function of Z_n converges to the moment generating function of the normal distribution, then the distribution function of Z_n converges to the distribution function of the normal distribution.

Proof of the central limit theorem in the case where the moment generating function of the X_i exists. Let

$$U_i = X_i - \mu.$$

Then $U_1, U_2, \ldots$ are independent identically distributed random variables with $E(U_i) = 0$ and $\mathrm{Var}(U_i) = \sigma^2$ and

$$M_U(t) = M_X(t)e^{-\mu t}.$$

Now

$$Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} = \frac{1}{\sigma\sqrt{n}} \sum_{i=1}^{n} U_i.$$

So

$$M_{Z_n}(t) = E\left(\exp\left(\sum_{i=1}^{n} \frac{t}{\sigma\sqrt{n}} U_i\right)\right).$$

Recall

$$M_{aX+b}(t) = e^{bt} M_X(at)$$

and

$$M_{X+Y}(t) = M_X(t) M_Y(t)$$

for independent X, Y. So

$$M_{Z_n}(t) = \left(M_U\left(\frac{t}{\sigma\sqrt{n}}\right)\right)^n.$$

By theorem 7.55, we can expand M_U as a power series about the origin. We have

$$M_{Z_n}(t) = \left(1 + \frac{t}{\sigma\sqrt{n}} E(U_1) + \frac{t}{\sigma\sqrt{n}} E(U_1^2) + o\left(\left(\frac{t}{\sigma\sqrt{n}}\right)^2\right)\right)^n.$$

So

$$M_{Z_n}(t) = \left(1 + \frac{1}{2}\sigma^2 \left(\frac{t}{\sigma\sqrt{n}}\right)^2 + o\left(\frac{1}{n}\right)\right)^n.$$

Taking the limit of the right hand side as $n \to \infty$, we have the definition of e^x with $x = \frac{1}{2}t^2$. So

$$M_{Z_n}(t) \to e^{\frac{1}{2}t^2}$$

as $n \to \infty$. This is the moment generating function of the standard normal distribution. So by theorem 8.27, the distribution function of Z_n converges to the distribution function of the normal distribution.

Example 4.30

An unknown fraction of a population p are jedi knights. It is desired to estimate p with error less than 0.005. How large of a sample is needed?

Solution. Denoting the sample mean

$$\hat{p} = \frac{1}{n}S_n,$$

we wish to choose n large enough so that

$$|\hat{p} - p| \leq 0.005.$$

However, this cannot be done, as $|\hat{p} - p|$ is a random variable and so may take a larger value than 0.005 for any n. We instead seet n such that

$$P(|\hat{p} - p| \leq 0.005) \geq 0.95,$$

following convention of taking a 5% probability of error as a maximum.

We have, by the central limit theorem,

$$\hat{p} \sim \mathcal{N}\left(p, \frac{p(1-p)}{n}\right)$$

for large n. Standardising $\hat{p}$ gives

$$\frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim \mathcal{N}(0, 1).$$

Note

$$P(|\hat{p} - p| \leq 0.005) = P\left(\frac{|\hat{p} - p|}{\sqrt{p(1-p)/n}} \leq \frac{0.005}{\sqrt{p(1-p)/n}}\right).$$

Hence

$$P(|\hat{p} - p| \leq 0.005) \sim P\left(|Z| \leq \frac{0.005}{\sqrt{p(1-p)/n}}\right)$$

for Z following a standard normal distribution. So

$$P(|\hat{p} - p| \leq 0.005) \geq 0.95 \Rightarrow P\left(|Z| \leq \frac{0.005}{\sqrt{p(1-p)/n}}\right) \geq 0.95$$

under this approximation.

$$P\left(|Z| \leq \frac{0.005}{\sqrt{p(1-p)/n}}\right) \geq 0.95 \Rightarrow -1.960 \geq Z \geq 1.960$$

using statistical tables. So we must have

$$\frac{0.005}{\sqrt{p(1-p)/n}} \geq 1.960,$$

or upon rearranging

$$n \geq 153664p(1-p).$$

Now

$$p(1-p) \leq \frac{1}{4}$$

for all p, so our worst case is requiring

$$n \geq 38416.$$

This is our minimum sample size. Note that until the last line, this is a standard A-Level FS1 problem.

A consequence of the central limit theorem is that the deviations of S_n are unlikely to exceed the mean $n\mu$ by more than order $\sqrt{n}$. This is demonstrated below.

$$\begin{aligned} D &= P(|S_n - n\mu| > cn^a), \\ D &= P\left(\frac{|S_n - n\mu|}{\sigma\sqrt{n}} > \frac{cn^a}{\sigma\sqrt{n}}\right), \\ D &= P\left(|Z_n| > \frac{cn^a}{\sigma\sqrt{n}}\right). \end{aligned}$$

For large n, we have

$$D \sim P\left(|Z| > \frac{cn^a}{\sigma\sqrt{n}}\right)$$

for $Z \sim \mathcal{N}(0,1)$. Now the RHS goes to ∞ for any $a > \frac{1}{2}$, meaning $D \to 0$ in this case.

We have now found the approximate order of the mean and deviations of S_n. It proves fruitful to next study the unlikely large deviations of S_n.

In dealing with this subject, we shall make a few assumptions:

- $X_1, X_2, \ldots$ are independent identically distributed random variables with mean 0.
- If this does not hold, replace X_i with $X_i - \mu$.
- The common moment generating function of the X_i, $M(t)$, is finite for values of $t \in (-\delta, \delta)$.
- As usual, $S_n = X_1 + X_2 + \cdots + X_n$.

8.34: Chernoff bound

Let $t > 0$. We have $S_n > an$ iff. $g(S_n) > g(an)$, where

$$g(x) = e^{tx}.$$

By theorem 7.63, Markov's inequality, we have

$$P(g(S_n) > g(an)) \leq \frac{E(g(S_n))}{g(an)}.$$

So

$$P(S_n > an) \leq \frac{E(e^{tS_n})}{e^{tan}},$$

using our fact from the first line to simplify the left hand side. By theorem 7.52,

$$E(e^{tS_n}) = M(t)^n,$$

so

$$P(S_n > an) \leq \left(\frac{M(t)}{e^{at}}\right)^n$$

for $t > 0$. Note that if $t > \delta$, then the inequality above still technically applies, but is trivial as we have the right hand side infinite. Minimising the right hand side over t, we obtain

$$P(S_n > an) \leq \left(\inf\{e^{-at}M(t) : t > 0\}\right)^n,$$

which is an exponentially decaying bound for the probability of a large deviation.

What do we mean precisely by exponentially decaying here? $P(S_n > an) = c^n$ for a constant c, dependent on a. It remains to be shown $c < 1$ for this to be exponential decay. This seems somewhat difficult to show, but it can be done rigorously, the details of which we will omit here. If $a \leq 0$, then in fact we do not have $c < 1$, and this corresponds to these no longer being deviations.

It turns out that, neglecting sub-exponential corrections, this bound is an equality, and this is the content of Cramer's theorem.

Theorem 8.36: Cramer's theorem, large deviation theorem

Let

$$\lambda(t) = \log(M(t)).$$

Then, define the Fenchel-Legendre transform of λ by

$$\lambda^*(A) = \sup\{at - \lambda(t) : t \in \mathbb{R}\}.$$

Now, let $X_1, X_2, \ldots$ be independent identically distributed random variables with mean 0 whose common moment generating function

$$M(t) = E(e^{tX})$$

is finite for $t \in (-\delta, \delta)$. Let $a > 0$ be such that $P(X > a) > 0$. Then $\lambda^*(a) > 0$ and we have

$$\frac{1}{n}\log(P(S_n > an)) \to -\lambda^*(a)$$

as $n \to \infty$.

Proof idea.

1. We show $\lambda(t)$ is convex.
2. We show that $\lambda'(0) = 0$ and hence $\sup\{at - \lambda(t) : t \in \mathbb{R}\}$ may be taken over $t > 0$.
3. We show that $\lambda^*(a) > 0$, when the supremum is taken over sufficiently small t.
4. We rewrite the Chernoff bound as

   $$P(S_n > an) \leq e^{-n\lambda^*(a)}$$

   using our equation from (2).

5. We take logarithms and divide by n to obtain the result.

We may show that this limit is asymptotically sharp rather than just an upper bound, but this requires a much harder argument.

The theorem applies to deviations above the mean. For deviations below the mean, the theorem may be applied to the sequence of variables $-X_i$.

What is the benefit of this bound over the previous one? This bound is asymptotically equivalent to the previous one, but is only valid for $n \to \infty$ rather than for all finite n. However, this bound provides an exact rate rather than just an upper bound. According to the previous bound, we could have had $P(S_n > an)$ decaying like e^{-x^2} or even just identically zero, whereas now we know that (for very large n) we do have this exponential decay precisely.

This theorem is tedious to apply, but is often not too difficult.

Example 8.40

Let X be a random variable with

$$P(X = -1) = P(X = 1) = \frac{1}{2}.$$

Let X have the moment generating function $M(t)$.

$$M(t) = E(e^{tX}) = \frac{1}{2}(e^t + e^{-t}).$$

The Fenchel-Legendre transformation is given by

$$\lambda^*(A) = \sup\{at - \lambda(t) : t \in \mathbb{R}\}.$$

Now

$$\frac{d}{dt}(at - \lambda(t)) = a - \frac{M'(t)}{M(t)},$$

recalling the definition of $\lambda(t)$. Chugging through the calculus, we have

$$\frac{d}{dt}(at - \lambda(t)) = 0 \Rightarrow t = \log\left(\sqrt{\frac{1+a}{1-a}}\right).$$

Then

$$\lambda^*(A) = \log\left(\sqrt{(1-a)^{1-a}(1+a)^{1+a}}\right)$$

upon doing some algebraic manipulation. So

$$\frac{1}{n}\log(P(S_n > an)) \to -\log\left(\sqrt{(1-a)^{1-a}(1+a)^{1+a}}\right)$$

by Cramer's theorem, theorem 8.36. So

$$P(S_n > an) \to n^{1/\sqrt{(1-a)^{1-a}(1+a)^{1+a}}}$$

as $n \to \infty$ upon exponentiation.

We have so far seen convergence in mean square and convergence in probability (where the former implies the latter). We will now also convergence in distribution, which we have already seen in the central limit theorem.

Definition 8.45

The sequence $Z_1, Z_2, \ldots$ is said to converge in distribution or converge weakly to Z as $n \to \infty$ if

$$P(Z_n \leq x) \to P(Z \leq x)$$

for $x \in C$. Where C is the subset of $\mathbb{R}$ where the cumulative distribution function $P(Z \leq z)$ is continuous. If this holds, we write

$$Z_n \Rightarrow Z.$$

Convergence in distribution is a property of the distributions of random variables rather than a property of the random variables themselves, and for this reason, explicit reference to the limit random variable Z is often omitted. For example, the conclusion of the central limit theorem may be expressed as "Z_n converges in distribution to the normal distribution with mean 0 and variance 1."

Convergence in distribution is in fact weaker than both convergence in mean square and convergence in probability.

Theorem 8.46

If $Z_1, Z_2, \ldots$ is a sequence of random variables with $Z_n \to Z$ in probability, we have

$$Z_n \Rightarrow Z.$$

A trivial corollary is convergence in mean square implying convergence in probability.

There does however exist a partial converse.

Theorem 8.47

If $Z_1, Z_2, \ldots$ is a sequence of random variables with $Z_n \Rightarrow c$ for a constant variable c, we have $Z_n \to c$ in probability.

Proof. If $Z_n \Rightarrow c$, we must have the cumulative distribution function F_n of Z_n satisfying:

$$F_n(z) \to \begin{cases} 0, & z < c, \\ 1, & z > c. \end{cases}$$

Then considering $P(|Z_n - c| > \varepsilon)$ for $\varepsilon > 0$, we have

$$\begin{aligned} P(|Z_n - c| > \varepsilon) &= P(Z_n < c - \varepsilon) + P(Z_n > c + \varepsilon), \\ P(|Z_n - c| > \varepsilon) &\leq F_n(c - \varepsilon) + 1 - F_n(c + \varepsilon). \end{aligned}$$

Note that we have an inequality in the line above due to the possibility of $Z_n = c \pm \varepsilon$ exactly. The right hand side converges to 0, so $P(|Z_n - c| > \varepsilon)$ converges to 0. Hence Z_n converges to c in probability.

Example 8.49

Let U be a random variable with

$$P(U = 1) = P(U = -1) = \frac{1}{2}.$$

Define $Z_1, Z_2, \ldots$ by

$$Z_n = \begin{cases} U, & n \text{ is odd,} \\ -U, & n \text{ is even.} \end{cases}$$

Then each Z_n has the same distribution, so $Z_n \Rightarrow U$. However,

$$Z_n - U = \begin{cases} 0, & n \text{ is odd,} \\ -2U, & n \text{ is even.} \end{cases}$$

So

$$P(|Z_{2m} - U| > 1) = P(|-2U| > 1) = P(|U| > 1/2) = 1.$$

Hence,

$$P(|Z_{2m} - U| < \varepsilon) = 0$$

for all $\varepsilon < 1$, and convergence in probability fails.

Finally, we return to characteristic functions. In proving the central limit theorem we employed a result, theorem 8.27, linking the convergence of moment generating functions to convergence in distribution. This result is a weak form of the continuity theorem.

Theorem 8.50: continuity theorem

Let $Z, Z_1, Z_2, \ldots$ be random variables with characteristic functions $\phi, \phi_1, \phi_2, \ldots$. Then

$$Z_n \Rightarrow Z$$

if and only if

$$\phi_n(t) \to \phi(t)$$

for $t \in \mathbb{R}$.

The proof of this is difficult.

Example 8.51

If $Z_n \Rightarrow Z$, then we wish to show

$$aZ_n + b \Rightarrow aZ + b$$

for $a, b \in \mathbb{R}$.

$\phi_{aZ_n+b}(t) = e^{itb}\phi_n(at)$	by theorem 7.87,
$\phi_{aZ_n+b}(t) \to e^{itb}\phi(at)$	as $n \to \infty$ by theorem 8.50,
$\phi_{aZ_n+b}(t) \to \phi_{aZ+b}(t)$	by theorem 7.87,
$aZ_n + b \Rightarrow aZ + b$	by theorem 8.50.

Example 8.52: alternate law of weak large numbers

Let $X_1, X_2, \ldots$ be independent identically distributed random variables with mean μ, and let

$$U_n = \frac{1}{n}(X_1 + X_2 + \cdots + X_n).$$

Then the characteristic function ϕ_n of U_n is given by

$$\phi_n(t) = \phi_X(t/n)^n$$

by theorem 7.87 where ϕ_X is the common characteristic function of the X_i. By theorem 7.85, together with the assumption $E(X_i) = \mu$, we can write

$$\phi_n(t) = \left(1 + \frac{it\mu}{n} + o(t/n)\right)^n.$$

Using the limit definition of e^x, we have

$$\phi_n(t) \to e^{it\mu}$$

as $n \to \infty$. Now for the constant variable $X = \mu$, we have

$$\phi_X(t) = e^{it\mu},$$

so $U_n \to \mu$ by theorem 8.50.

This version of the law of weak large numbers is slightly different than theorem 8.17; we assume identical distribution but not finite variance, while theorem 8.17 did assume finite variance but not identical distribution.

Example 8.54: central limit theorem

The central limit theorem (theorem 8.23) may be proven for random variables without finite moment generating functions, which we previously assumed, using similar arguments but replacing moment generating functions with characteristic functions.

9 Branching Processes

We will now turn our attention to applications of the probability theory we have developed so far. Many such applications involve quantities which develop randomly as time passes. These are called random processes or stochastic processes. Some processes can be modelled accurately with just the tools we have developed. For instance:

- branching processes: modelling the growth of a self-reproducing population.
- random walks: modelling the movement of a particle moving erratically within a medium.
- Poisson processes: modelling processes such as the emission of radioactive particles.

The theory of stochastic processes is divided into concrete processes like those above, and an abstract theory of general stochastic processes, which is beyond our scope. Between these extremes, we have families of processes which share key properties but are sufficiently specific for the development of useful theory; one such property is the Markov property, discussed in Chapter 12.

We define the term nomad to be a hypothetical object which is able to reproduce itself according to the following rules:

- At time $n = 0$, there exists a single nomad. It lives for a unit of time and then, at time $n = 1$, dies in the act of childbirth and is replaced by a family of offspring nomads.
- These nomads have similar biographies, each surviving only until time $n = 2$ and then dying and being replaced by a family of offspring.

- This process continues at each time point.

We assume the following:

- The family sizes of a nomad's offspring are independent random variables taking values in $\{0, 1, 2, \ldots\}$.
- The family sizes are identically distributed with known mass function p such that the number C of children of a typical nomad has mass function

$$P(C = k) = p_k.$$

Such a process is called a branching process and may be used as a simple model for bacterial growth or the spread of a family name in western cultures. The basic objective is to say something about how the development of the process depends on the function p. We shall suppose that

$$P(C = k) \neq 1$$

for any k to avoid trivial processes.

We must also define some notation. The set of nomads born at time n is called the nth generation, and we write Z_n for the number of such nomads. The process is described by the sequence $Z_0, Z_1, \ldots$ of random variables.

We will wish to answer some questions which help with our goal of understanding the development of the process based on the function p.

- What is the mean and variance of Z_n?
- What is the mass function of Z_n?
- What is the probability that the population is extinct by time n?
- What is the probability that the population ever goes extinct?

Clearly,

$$Z_0 = 1, \qquad P(Z_1 = k) = p_k.$$

The mass function of Z_2 is more difficult to obtain, and requires an appeal to some previous results. Writing C_i for the number of the children of the ith nomad in the first generation, we have

$$Z_2 = C_1 + C_2 + \cdots + C_{Z_1}.$$

And more generally,

$$Z_n = C_1 + C_2 + \cdots + C_{Z_{n-1}}$$

if the C_i are defined with reference to the $(n-1)$th generation.

Recall that the sum of a random number of random variables has some nice properties regarding its probability generating function. Write

$$G_n(s) = \sum_{k=0}^{\infty} s^k P(Z_n = k)$$

so that G_n is the probability generating function of Z_n and

$$G(s) = \sum_{k=0}^{\infty} s^k p_k$$

for the probability generating function of a typical family size.

Theorem 9.4

The functions $G_0, G_1, \ldots$ satisfy

$$G_0(s) = s, \qquad G_n(s) = G_{n-1}(G(s))$$

for $n \in \mathbb{Z}_+$. Hence,

$$\begin{aligned} G_1(s) &= G(s), \\ G_2(s) &= G(G(s)), \\ &\vdots \\ G_n(s) &= G(G(\cdots(G(s))\cdots)) \quad \text{with } n \text{ } G\text{'s.} \end{aligned}$$

Proof. We have $Z_0 = 1$ so $G_0(s) = s$. Then in theorem 4.36, which is reproduced below, take $N = Z_{n-1}$ and $X_i = C_i$ to obtain

$$G_n(s) = G_{n-1}(G(s)),$$

remembering that in our notation we have written G_k for the probability generating function of Z_k and G for the probability generating function of the C_i.

Theorem 4.36: random sum formula

Let N and $X_1, X_2, \ldots$ be independent non-negative integer random variables. If the X_i are identically distributed, then the sum

$$S = X_1 + X_2 + \cdots + X_N$$

has probability generating function

$$G_S(s) = G_N(G_X(s)).$$

Theorem 9.8

The mean of Z_n is

$$E(Z_n) = \mu^n,$$

where μ is the mean of the family-size distribution.

Proof.

$E(Z_n) = G_n'(1)$ by G_n being a probability generating function,

$E(Z_n) = \dfrac{d}{ds}(G_{n-1}(G(s)))\Big|_{s=1}$,

$E(Z_n) = G'(1)G_{n-1}'(G(1))$,

$E(Z_n) = G'(1)G_{n-1}'(1)$ as $G(1)$ is the sum of probabilities, which is 1 by definition,

$E(Z_n) = \mu E(Z_{n-1})$.

We now have a recurrence relation for $E(Z_n)$ with $E(Z_0) = 1$. So

$$E(Z_n) = \mu^n.$$

It follows that $E(Z_n) \to 0$ if $\mu < 1$, $E(Z_n) \to 1$ if $\mu = 1$, and $E(Z_n) \to \infty$ if $\mu > 1$. We will show that the condition $\mu \leq 1$ results in the nomad population going extinct, and if $\mu > 1$, then there is a positive probability that the nomad population will never go extinct.

We may similarly derive the variance of Z_n.

Exercise 9.12

If μ and σ^2 are the mean and variance of the family-size distribution, we have:

$$\mathrm{Var}(Z_n) = n\sigma^2 \quad \text{if } \mu = 1,$$

and

$$\mathrm{Var}(Z_n) = \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1} \quad \text{if } \mu \neq 1.$$

This functional equation for the probability generating function of Z_n is key to the analysis of branching process. Here is an example where it can be solved for a closed form.

Suppose that the mass function of each family size is given by

$$p_k = \left(\frac{1}{2}\right)^{k+1}.$$

Then

$$G(s) = \sum_{k=0}^{\infty} s^k \left(\frac{1}{2}\right)^{k+1} = \frac{1/2}{1 - (s/2)} = \frac{1}{2-s}.$$

So

$$G_1(s) = \frac{1}{2-s}.$$

Hence,

$$G_2(s) = \frac{2-s}{3-2s}, \qquad G_3(s) = \frac{3-2s}{4-3s}, \qquad \ldots$$

where these results are obtained using the functional equation. We might guess that

$$G_n(s) = \frac{n - (n-1)s}{n + 1 - ns},$$

and this may be proved by induction or by considering matrix transformations.

By expanding the right hand side of this result as a power series in s, we can also obtain the mass function of Z_n.

$$P(Z_n = k) = \begin{cases} \dfrac{n}{n+1}, & k = 0, \\ \dfrac{n^{k-1}}{(n+1)^{k+1}}, & k \in \mathbb{Z}_+. \end{cases}$$

In particular,

$$P(Z_n = 0) = \frac{n}{n+1} \to 1$$

as $n \to \infty$, verifying our previous claim that this population will go extinct.

The general case with mass function of each family size given by $p_k = pq^k$ can be solved similarly. The solution is

$$G_n(s) = \frac{p\big((q^n - p^n) - qs(q^{n-1} - p^{n-1})\big)}{(q^{n+1} - p^{n+1}) - qs(q^n - p^n)}.$$

The probability of extinction

$$P(Z_n = 0) = \frac{\mu^n - 1}{\mu^{n+1} - 1}$$

for $\mu = q/p$, the mean family size.

We can strengthen our previous result to obtain $P(Z_n) = 0 \to 1$ if $\mu < 1$ and $P(Z_n = 0) = 1/\mu$ if $\mu > 1$. Additionally, we have solved the special case $\mu = 1$ (where $p = q = 1/2$) and found that

this leads to extinction, culminating in the result $P(Z_n) = 0 \to 1$ if $\mu \leq 1$ and $P(Z_n = 0) = 1/\mu$ if $\mu > 1$.

In fact, we also find that this conclusion — extinction if and only if $\mu \leq 1$ — is valid for all non-trivial branching processes, regardless of the form of their associated mass function p_k.

Define the extinction probability

$$e = P(Z_n = 0 \text{ for some } n \geq 0).$$

Let

$$E_n = \{Z_n = 0\}$$

be the event that the process is extinct by the nth generation and let $e_n = P(E_n)$. Then

$$\{Z_n = 0 \text{ for some } n \geq 0\} = \bigcup_{n=0}^{\infty} E_n.$$

We note E_n is a subset of E_{n+q}, so $e_n \leq e_{n+1}$ and the sequence $\{E_n\}$ is increasing. By theorem 1.54, we have

$$e = \lim_{n \to \infty} e_n.$$

Now, we must calculate e. If $p_0 = 0$, $e = 0$ clearly. In the general case, we use the following theorem.

Theorem 9.19: extinction probability theorem

The probability e of extinction is the smallest non-negative root of the equation

$$x = G(x).$$

Proof. Since $e_n = P(Z_n = 0)$, we have that

$$e_n = G_n(0).$$

Recall we have

$$G_n(s) = G(G(\cdots(G(s))\cdots))$$

so that

$$G_n(s) = G(G_{n-1}(s)).$$

Hence,

$$G_n(0) = G(G_{n-1}(0)).$$

So

$$e_n = G(e_{n-1}).$$

Taking limits as $n \to \infty$, and using the result we derived based on the continuity of probability measures, we obtain:

$$e = G(e),$$

so that e is a root of the equation $x = G(x)$.

It remains to show e is specifically the smallest non-negative root. Suppose that η is any non-negative root of $x = G(x)$. Then as G is a probability generating function, it has non-negative coefficients, and so is non-decreasing on the interval $(0, 1)$. Hence,

$$\begin{aligned} e_0 = 0 \leq \eta &\Rightarrow e_1 = G(e_0) \leq G(\eta) = \eta, \\ e_1 \leq \eta &\Rightarrow e_2 = G(e_1) \leq G(\eta) = \eta, \\ &\vdots \\ e_{n-1} \leq \eta &\Rightarrow e_n = G(e_{n-1}) \leq G(\eta) = \eta, \end{aligned}$$

by the property $e_n \leq e_{n+1}$ we demonstrated by considering the events E_n. As for any non-negative root η, we have $e \leq \eta$, e must be the smallest non-negative root of $x = G(x)$.

Theorem 9.22: extinction theorem, survival theorem

For $p_1 \neq 1$, the probability e of ultimate extinction satisfies $e = 1$ if and only if the mean family size $\mu \geq 1$.

Proof. The $p_0 = 0$ case is trivial; we consider the other cases. We know by theorem 9.19 that e is the smallest non-negative root of the equation $x = G(x)$. We will now investigate G. We find that on the interval $[0, 1]$:

- G is continuous with radius of convergence at least 1 (by the power series structure of G).
- G is non-decreasing since $G'(x)$ is another power series with positive coefficients.
- G is convex since $G''(x)$ is another power series with positive coefficients.

These facts allow us to justify the statement that $x = 1$ is the unique root of the equation $x = G(x)$ on this interval if and only if $G'(1) \leq 1$, either by a non-rigorous sketch or using ideas from analysis. Now $x = 1$ is always a root, as $G(1) = 1$ is a property of probability generating functions. So $x = 1$ is the smallest root if and only if $G'(1) \leq 1$. So $x = 1$ is the smallest root if and only if $\mu \leq 1$ by the property $G'(1) = \mu$ for probability generating functions. So $e = 1$ if and only if $\mu \leq 1$.

10 Random Walks

We will consider the simplest type of random walk, where a hypothetical particle inhabits a one-dimensional set (say, a line) and moves randomly within this set as time passes.

For simplicity, we assure that both space and time are discrete. Therefore, we need only observe the particle's position at each of the discrete time points $0, 1, 2, \ldots$ and at each time point we allow the particle to be located at an integer position.

We now specify how the particle moves. Let p be a real number with $0 < p < 1$ and $q = 1 - p$. During each unit of time, the particle moves its location either one unit left or one unit right with probabilities q and p respectively. Thus, if S_n describes the location of the particle at time n, then

$$S_{n+1} = \begin{cases} S_n + 1, & \text{with probability } p, \\ S_n - 1, & \text{with probability } q. \end{cases}$$

We suppose that the direction of each jump is independent of other jumps, giving rise to the equation

$$S_n = S_0 + X_1 + X_2 + \cdots + X_n$$

for a starting position S_0 and $X_1, X_2, \ldots$ being independent identically distributed random variables taking the value 1 with probability p and -1 with probability q.

We call the process $S_0, S_1, \ldots$ a simple random walk. It is called symmetric if $p = q = 1/2$ and asymmetric otherwise.

We will see that random walks are examples of Markov chains. In particular, we shall later extend these one-dimensional calculations to d dimensions in the general case.

Example 10.2: gambling

A gambler enters a casino with £1000 and plays a game where they can either gain or lose £1 based on a (possibly biased) coinflip. We have all the conditions for a simple random walk with $S_0 = 1000$. We might be interested in the conditions for the gambler to go bankrupt. It turns out that

$$P(S_n = 0 \text{ for some } n \geq 1) < 1$$

if $p > 1/2$ and

$$P(S_n = 0 \text{ for some } n \geq 1) = 1$$

is $p \leq 1/2$.

We now consider a simple random walk with $S_0 = 0$. What is the probability that $S_n = k$? We calculate these probabilities using counting arguments.

Theorem 10.6

Let

$$u_n = P(S_n = S_0)$$

be the probability that a simple random walk revisits its starting point at time n. Then

$$u_{2m+1} = 0$$

and

$$u_{2m} = \binom{2m}{m} p^m q^m.$$

Proof. Suppose without loss of generality that $S_0 = 0$. If $S_0 = S_n = 0$ then the walk is made up of equal numbers of left and right steps. Clearly, this is only possible in the case where n is even.

$$S_{2m} = X_1 + X_2 + \cdots + X_{2m}.$$

We require exactly m of the X_i to be 1 and m of the X_i to be -1. There are

$$\binom{2m}{m}$$

ways of choosing a set of m of the X_i. Then, we have that this set has all of the X_i equal to 1 is $p^m q^m$.

More general transition probabilities may be calculated similarly. For instance, the random variable

$$\frac{1}{2}(X_i + 1)$$

has Bernoulli distribution with parameter p, giving that

$$B_n = \frac{1}{2}(S_n + n)$$

has the binomial distribution with parameters n and p. Hence

$$P(S_n = k \mid S_0 = 0) = P\left(B_n = \frac{1}{2}(k+n)\right).$$

Thus

$$P(S_n = k \mid S_0 = 0) = \binom{n}{\frac{1}{2}(k+n)} p^{\frac{1}{2}(k+n)} q^{\frac{1}{2}(k+n)}.$$

This is valid for k such that $\frac{1}{2}(n+k)$ is an integer between 0 and n.

We will introduce some terminology. Consider a simple random walk starting with $S_0 = 0$. In the subsequent motion, the random walk is recurrent if it revisits its starting point with probability 1 and is called transient otherwise. We will show that a simple random walk is recurrent if and only if it is symmetric.

The position at time n is the sum

$$S_n = X_1 + X_2 + \cdots + X_n$$

of independent random variables with mean $E(X) = p - q$ and finite variance $\mathrm{Var}(X)$. Hence,

$$\frac{1}{n} S_n \to p - q$$

as $n \to \infty$ in mean square by theorem 8.6. Thus, if $p > q$, the walk tends to drift linearly towards ∞, whilst if $p < q$, the drift is linear towards $-\infty$. If $p = q$, then $\frac{1}{n} S_n \to 0$ in mean square. This provides an intuitive basis for our claim below.

Theorem 10.12

The probability that a simple random walk ever revisits its starting point is given by

$$P(S_n = 0 \text{ for some } n \mid S_0 = 0) = 1 - |p - q|,$$

and hence the walk is recurrent if and only if $p = q = 1/2$.

Proof. Write

$$A_n = \{S_n = 0\}$$

and

$$B_n = \{S_n = 0,\ S_k \neq 0 \text{ for } k < n\}.$$

If A_n occurs, exactly one of $B_1, B_2, \ldots, B_n$ occurs. Now the B_i are disjoint, and A_n is a subset of whichever B_i it is that occurs. So

$$A_n = (A_n \cap B_1) \cup (A_n \cap B_2) \cup \cdots \cup (A_n \cap B_n).$$

Hence by a property of probability measures,

$$P(A_n) = \sum_{k=1}^{n} P(A_n \cap B_k).$$

Now $\{A_n \cap B_k\}$ is the event that the walk returns for the first time at k and then returns again after another $n-k$ time units. Hence,

$$P(A_n \cap B_k) = P(B_k)P(A_{n-k})$$

by independence. Let us now write

$$f_n = P(B_n), \qquad u_n = P(A_n).$$

We have

$$u_n = \sum_{k=1}^{n} f_k u_{n-k}.$$

Now by theorem 10.6, we may find expressions for the u_i. We therefore want to find the f_k. We introduce:

$$U(s) = \sum_{n=0}^{\infty} u_n s^n,$$

the generating function of u_i, and

$$F(s) = \sum_{n=0}^{\infty} f_n s^n,$$

the generating function of f_k. Note that $u_0 = 0$, $f_0 = 0$ trivially. Now multiplying our previous sum by s^n and summing over n, we have

$$\begin{aligned}
\sum_{n=1}^{\infty} u_n s^n &= \sum_{n=1}^{\infty} \left(\sum_{k=1}^{n} f_k u_{n-k} s^n \right), \\
\sum_{n=1}^{\infty} u_n s^n &= \sum_{k=1}^{\infty} \left(\sum_{n=k}^{\infty} f_k u_{n-k} s^k s^{n-k} \right) && \text{interchanging sums,} \\
\sum_{n=1}^{\infty} u_n s^n &= \sum_{k=1}^{\infty} \left(f_k s^k \sum_{m=0}^{\infty} u_m s^m \right) && \text{writing } m = n-k, \\
\sum_{n=1}^{\infty} u_n s^n &= F(s)U(s), \\
U(s) - u_0 &= F(s)U(s), \\
U(s) &= F(s)U(s) + 1, \\
F(s) &= 1 - \frac{1}{U(s)}.
\end{aligned}$$

Now we return to calculating $U(s)$ explicitly.

$$\begin{aligned}
U(s) &= \sum_{n=0}^{\infty} u_n s^n, \\
U(s) &= \sum_{m=0}^{\infty} u_{2m} s^{2m} && \text{by theorem 10.6,} \\
U(s) &= \sum_{m=0}^{\infty} \binom{2m}{m} (pqs^2)^m, \\
U(s) &= (1 - 4pqs^2)^{-1/2}.
\end{aligned}$$

Hence,

$$F(s) = 1 - \sqrt{1 - 4pqs^2}.$$

Now

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = P(B_1 \cup B_2 \cup \cdots).$$

So

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = P(B_1) + P(B_2) + \cdots$$

as the B_i are disjoint and using definition 1.13. Hence

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = f_1 + f_2 + \cdots .$$

Now we may use Abel's lemma to justify that this right hand side is equal to

$$\lim_{s \to 1^-} \sum_{n=0}^{\infty} f_n s^n = F(1).$$

Thus,

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = 1 - \sqrt{1 - 4pq}.$$

Now $p + q = 1$ so

$$1 - 4pq = (p+q)^2 - 4pq = (p-q)^2,$$

giving rise to

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = 1 - |p - q|.$$

We can say some more about the symmetric, recurrent case where $p = q$. Although we have

$$P(S_n = 0 \text{ for some } n \geq 1 \mid S_0 = 0) = 1$$

here, if we set

$$T = \min\{n \geq 1 : S_n = 0\}$$

so that T is the time of the first return, we have

$$P(T < \infty) = 1.$$

However, $E(T)$ is infinite. We have

$$E(T) = \lim_{s \to 1^-} F'(s),$$

again justified by Abel's lemma. Now

$$F'(s) = \frac{s}{\sqrt{1 - s^2}}$$

in the case where $p = q = 1/2$, and this diverges to ∞ as $s \to 1^-$.

Hence, although a symmetric random walk is certain to return to its starting point, the expected time before this happens is infinite.

We turn our attention to the gambler's ruin problem. It concerns a game between two players, A and B. Initially A possesses £a and B possesses £$(N - a)$, so that their total capital is £N, where $N \geq 1$. A coin is flipped repeatedly and comes up either heads with probability p or tails with probability q. Each time heads turns up, player B gives £1 to player A, while each time tails turns up, player A gives £1 to player B. This game continues until either A or B runs out of money.

We may record the state of the game by noting A's capital after each flip. This sequence is a random walk on the set $\{0, 1, \ldots, N\}$; it starts at the point a and stops when it reaches either 0 or N. We refer to 0 and N as absorbing barriers. If the random walk stops at N, we say A wins and vice versa. The most natural question to ask is the probability that A wins the game.

Theorem 10.23: gambler's ruin

Consider a simple random walk on $\{0, 1, \ldots, N\}$ with absorbing barriers at 0 and N. If the walk begins at a, then the probability $v(a)$ that the walk is absorbed at N is given by

$$v(a) = \begin{cases} \dfrac{(q/p)^a - 1}{(q/p)^N - 1}, & p \neq q, \\ \dfrac{a}{N}, & p = q = 1/2. \end{cases}$$

Proof. We set up a difference equation for $v(0), v(1), \ldots, v(N)$. Let H be the event that the first flip is heads. We have, by the partition theorem, that

$$P(\text{A wins}) = P(\text{A wins} \mid H)P(H) + P(\text{A wins} \mid H^c)P(H^c).$$

Now if H occurs, we have A's capital increasing to $(a+1)$, and if H^c occurs, we have A's capital decreasing to $(a-1)$. So

$$v(a) = pv(a+1) + qv(a-1).$$

This is a second-order difference equation with boundary conditions

$$v(0) = 0, \qquad v(N) = 1.$$

Solving, we obtain

$$v(a) = \alpha + \beta(q/p)^a \quad \text{if } p \neq q,$$
$$v(a) = \alpha + \beta a \quad \text{if } p = q,$$

where α and β are constants to be obtained from the boundary conditions.

Another natural question to ask about a gambler's ruin problem is the expected length of the game.

Theorem 10.28: recurrence/transience of a random walk

Consider a simple random walk on $\{0, 1, \ldots, N\}$ with absorbing barriers at 0 and N. If the walk begins at a, then the expected number of steps before the walk is absorbed is given by

$$e(a) = \begin{cases} \dfrac{1}{p-q}\left(N\dfrac{(q/p)^a - 1}{(q/p)^N - 1} - a\right), & p \neq q, \\ a(N-a), & p = q = 1/2. \end{cases}$$

Proof. We proceed similarly. Let F be the number of flips of the coin before the game ends. We have

$$E(F) = E(F \mid H)P(H) + E(F \mid H^c)P(H^c).$$

So

$$e(a) = (1 + e(a+1))p + (1 + e(a-1))q.$$

Rearranging,

$$pe(a+1) - e(a) + qe(a-1) = -1$$

with boundary conditions

$$e(0) = 0, \qquad e(N) = 0.$$

We then solve in the standard way.

Finally, we investigate a player's chances when their opponent has infinite capital. Of course, the player with infinite wealth cannot lose, but we may find probabilities relating to whether the first player goes bankrupt.

Theorem 10.32

Consider a simple random walk on $\{0, 1, 2, \ldots\}$ with an absorbing barrier at 0. If the walk begins at a, then the probability that the walk is ultimately absorbed at 0 given by

$$\pi(a) = \begin{cases} (q/p)^a, & p > q, \\ 1, & p \leq q. \end{cases}$$

The player a, who wishes to move right with probability p, may be able to play forever if and only if they have better odds.

Proof sketch.

$$\pi(a) = \lim_{N \to \infty} (1 - v(a))$$

where this v function comes from theorem 10.23. However, this is not exactly rigorous, and we may instead wish to proceed with a standard difference equation approach.

11 Random Processes in Continuous Time

The random processes we have covered so far — branching processes and random walks — are called discrete time processes since they involve observations at discrete times. Many other processes involve observations made continuously as time passes and are called continuous time processes.

Rather than being a family $\{Z_n\}$ of random variables indexed by non-negative integer n, a continuous time process is a family

$$Z = (Z_t : t \geq 0)$$

of random variables. We will restrict our attention to processes which take integer values only to simplify our study, so that Z_t is a random integer for each t.

The main difference is moving from difference equations to differential equations in in problems involving continuous time processes.

Our first example goes as follows. Bill has to answer incoming telephone calls. He cannot predict with certainty when the telephone will ring. We assume that Bill deals with every call instantaneously — though there are models which drop this assumption — and we assume calls arrive "homogenously" in time, in the sense that the chance the telephone rings during any period of time depends only upon the length of the period.

We describe time by a parameter $t \in \mathbb{R}_+$ and let N_t represent the number of calls which have arrived in the interval $[0, t]$. We suppose that the random process

$$N = (N_t : t \geq 0)$$

evolves such that:

- N_t is a non-negative integer random variable.
- $N_0 = 0$.
- If $s \leq t$, then $N_s \leq N_t$.
- If $0 \leq s < t$ then the number of calls in the interval $(s, t]$ is independent of the arrivals prior to time s.

- There exists $\lambda > 0$ such that for small positive h,

$$P(N_{t+h} = n + 1 \mid N_t = n) = \lambda h + o(h),$$

$$P(N_{t+h} = n \mid N_t = n) = 1 - \lambda h + o(h).$$

The final condition proposes that the probability of a call arriving in a short interval of length h is approximately a linear function of h and that this approximation improves for smaller h. We also have that the probability of two calls in this short interval is of $o(h)$, an insignificant probability.

We call the random process $N = (N_t : t \geq 0)$ a Poisson process with rate λ. Poisson processes may be used to model phenomena like:

- the arrival of customers in a shop.
- the clicks of a Geiger counter.
- the incidence of deaths in a small town with a stable population.

It is useful to represent the outcomes of a Poisson process by a graph of N_t against t. We may define

$$T_i = \inf\{t : N_t = i\},$$

giving rise to a sequence

$$T_0 = 0, \qquad T_0 \leq T_1 \leq T_2 \leq \cdots,$$

which may be thought of as the inverse process of N. We have

$$N_t = \max\{n : T_n \leq t\}.$$

Based on the conditions we have defined for a Poisson process (1–5), we seek to answer similar questions to those which we investigated for previous random processes. In particular, what is the mass function of N_t and what can be said about the distribution of the sequence $\{T_i\}$?

Theorem 11.6

For each $t > 0$, the random variable N_t has the Poisson distribution with parameter λt, so that

$$P(N_t = k) = \frac{1}{k!}(\lambda t)^k e^{-\lambda t}$$

and

$$E(N_t) = \mathrm{Var}(N_t) = \lambda t.$$

Proof. Let

$$p_k(t) = P(N_t = k).$$

We have

$$P(N_{t+h} = k) = \sum_{i=0}^{k} P(N_{t+h} = k \mid N_t = i) P(N_t = i)$$

using the partition theorem.

$$\begin{aligned} p_k(t+h) &= P(N_{t+h} = k \mid N_t = k-1)P(N_t = k-1) \\ &\quad + P(N_{t+h} = k \mid N_t = k)P(N_t = k) + o(h), \\ p_k(t+h) &= (\lambda h + o(h))P(N_t = k-1) + (1 - \lambda h + o(h))P(N_t = k) + o(h), \\ p_k(t+h) &= \lambda h P(N_t = k-1) + (1 - \lambda h)P(N_t = k) + o(h), \\ p_k(t+h) - p_k(t) &= \lambda h(p_{k-1}(t) - p_k(t)) + o(h). \end{aligned}$$

Dividing by h and using the limit definition of the derivative and functions of order $o(h)$, we have

$$p_k'(t) = \lambda p_{k-1}(t) - \lambda p_k(t).$$

We also have

$$P(N_{t+h} = 0) = P(N_{t+h} = 0 \mid N_t = 0)P(N_t = 0)$$

as this is the only case when $N_{t+h} = 0$. Hence

$$p_0(t+h) = (1 - \lambda h)p_0(t) + o(h).$$

And by a similar method,

$$p_0'(t) = -\lambda p_0(t).$$

We have a peculiar system of equations

$$p_k'(t) = \lambda p_{k-1}(t) - \lambda p_k(t), \qquad p_0'(t) = -\lambda p_0(t),$$

and the boundary conditions

$$p_k(0) = \begin{cases} 1, & k = 0, \\ 0, & \text{else.} \end{cases}$$

We may solve this by induction with a guess or by generating functions.

The probability generating function of N_t is

$$G(s,t) = \sum_{k=0}^{\infty} p_k(t)s^k.$$

Now we have from our first equation, multiplying by s^k and summing over k,

$$\sum_{k=1}^{\infty} p_k'(t)s^k = \lambda \sum_{k=1}^{\infty} p_{k-1}(t)s^k - \lambda \sum_{k=1}^{\infty} p_k(t)s^k.$$

$$\sum_{k=0}^{\infty} p_k'(t)s^k = \lambda \sum_{k=1}^{\infty} p_{k-1}(t)s^k - \lambda \sum_{k=0}^{\infty} p_k(t)s^k,$$

adding the second equation to the line above. Now

$$\sum_{k=0}^{\infty} p_k'(t)s^k = \frac{\partial G}{\partial t},$$

which needs some more work to rigorously justify. And

$$\sum_{k=1}^{\infty} p_{k-1}(t)s^k = sG(s,t).$$

So we have

$$\frac{\partial G}{\partial t} = \lambda s G(s,t) - \lambda G(s,t).$$

Then

$$\frac{1}{G}\frac{\partial G}{\partial t} = \lambda(s-1).$$

Integrating with respect to t, we have

$$\log G = \lambda t(s-1) + A(s)$$

for an arbitrary function $A(s)$. We use our boundary condition to find that $A(s) = 0$ for all s, so

$$G(s,t) = e^{-\lambda t(s-1)}.$$

Expanding this exponential and taking the coefficient of s^k, we have

$$P(N_t = k) = \frac{1}{k!}(\lambda t)^k e^{-\lambda t}.$$

Now let N be a process with rate λ. The arrival times $T_0, T_1, \ldots$ of N are defined as before, with $T_0 = 0$ and

$$T_i = \inf\{t : N_t = i\}.$$

The inter-arrival times $X_1, X_2, \ldots$ are the times between successive arrivals so that

$$X_i = T_i - T_{i-1}.$$

Theorem 11.23

In a Poisson process with rate λ, the inter-arrival times X_i are independent random variables each with the exponential distribution with the parameter λ.

We recall condition (4) for a Poisson process: $0 \leq s < t$ then the number of calls in the interval $(s,t]$ is independent of the arrivals prior to time s. Theorem 11.23 is motivated by the lack-of-memory property of the exponential distribution.

Definition 11.24

A positive random variable X is said to have the lack-of-memory property if

$$P(X > u + v \mid X > u) = P(X > v).$$

Thinking about X as time which elapses before some event A, then this definition requires that if A has not occurred by time u then the time which subsequently elapses does not depend on u.

Theorem 11.26

The continuous random variable X has the lack-of-memory property if and only if X is exponentially distributed.

Proof. The backward implication is routine to prove. Suppose X is positive and continuous and the lack-of-memory property holds. Then let

$$G(u) = P(X > u).$$

We have

$$\frac{G(u+v)}{G(u)} = G(v)$$

by the definition of conditional probability on the left hand side. So

$$G(u+v) = G(u)G(v).$$

This is one of the standard functional equation with solution

$$G(u) = e^{-\lambda u}$$

under the assumptions we have made.

Proof sketch of theorem 11.23. We have

$$P(X_1 > u) = P(N_u = 0) = e^{-\lambda u}.$$

So

$$X_1 \sim \text{Exp}(\lambda).$$

Now by condition (4), we have X_2 is independent of X_1. This step is somewhat incomplete, since X_1 being a random variable itself somewhat complicates the matter. Now by condition (5), arrivals occur homogeneously in time, since the probability of an arrival in $(t, t+h)$ depends only on h and not t. Consequently, X_2 is independent and identically distributed to X_1. This can be extended for all of the X_i, proving the theorem.

Why does X_1 being a random variable matter? Because X_1 is not chosen independently of the process itself, as required by condition (4), meaning it cannot apply without more analysis. Here is a more concrete example. Let L be the time of the last arrival before time 1, assuming one exists. Then the arrivals in $(0, L]$ are clearly not independent of the arrivals in time $(L, 1)$. This is not exactly the same as X_1, as it may require knowledge of the process beyond time L, but in the sense that L is a random variable defined by a property of the process, it is somewhat analogous.

We have shown above that, if N is a Poisson process with parameter λ, the times $X_1, X_2, \ldots$ between arrivals in the process are independent and identically distributed with the exponential distribution with parameter λ. This conclusion characterises the Poisson process, in the sense that Poisson processes are the only "arrival processes" with this property.

Furthermore, if we let $X_1^*, X_2^*, \ldots$ be independent identically distributed random variables with the exponential distribution with parameter λ, and suppose that the telephone is now programmed to ring at the times

$$T_1^* = X_1^*, \qquad T_2^* = X_1^* + X_2^*, \qquad T_3^* = X_1^* + X_2^* + X_3^*, \qquad \ldots$$

Then the process

$$N^* = (N_t^* : t \geq 0)$$

is a Poisson process with rate λ, and behaves exactly the same as the old model. The telephone operator would not be able to distinguish them.

Example 11.29

Suppose that buses for downtown arrive at the bus stop on the corner in the manner of a Poisson process. Knowing this, David expects to wait an exponentially distributed period of time before a bus will pick him up. If he arrives at the bus stop and Doris tells him that she has been

waiting 50 minutes already, then this is neither good nor bad news for him, since the exponential distribution has the lack-of-memory property. Similarly, if he arrives just in time to see a bus departing, then he need not worry that his wait will be longer than usual. These properties are characteristics of the Poisson process.

We may use the ideas we have developed so far to develop a model for population growth in continuous time. For instance, consider an organism which multiplies in the following way. At time $t = 0$, there are a number I of organisms in a pond. These multiply over time by dividing into two identical copies. This is somewhat reminiscent of a branching process, but cell divisions occur at random times, like the telephone calls in our last model.

We make these two assumptions:

- each cell in the pond at time t has a chance of dividing in the interval $(t, t + h]$. There is a constant λ, the birth rate, such that the probability that a cell:
  - divides once in the interval is $\lambda h + o(h)$.
  - does not divide in the interval is $1 - \lambda h + o(h)$.
- each division of a cell occurs independently of both its past history and of all the activities of all other amoebas present at time t.

This looks very much like a Poisson process. How is it different? In a Poisson process, the overall rate of arrivals is a single fixed constant λ, no matter how many arrivals have already occurred. Here, it's each individual cell that divides at rate λ, but the number of amoebas is itself growing over time. This is described in the assumptions. While the expressions look similar, the Poisson process makes statements directly about the count N_t, while the same statement is made over each cell in this example.

Let M_t be the number of cells at time t.

If $M_t = k$, then

$$P(M_{t+h} = k \mid M_t = k) = 1 - \lambda kh + o(h).$$

And

$$P(M_{t+h} = k + 1 \mid M_t = k) = \lambda kh + o(h).$$

Again, comparing this to (5) for Poisson processes, we have similar expressions, but here we have a factor of k in the right-hand expressions. The rate at which M increases is proportional to M itself, and we call the process M a simple linear birth process or a pure birth process.

Theorem 11.36

If $M_0 = I$ and $t > 0$, then

$$P(M_t = k) = \binom{k-1}{I-1} e^{-I\lambda t}(1 - e^{-\lambda t})^{k-1}.$$

The proof follows similarly to the corresponding theorem for Poisson processes.

Note that M_t has a negative binomial distribution with parameters I and

$$p = e^{-\lambda t}.$$

We have

$$E(M_t) = Ie^{\lambda t}$$

and

$$\mathrm{Var}(M_t) = Ie^{2\lambda t}(1 - e^{-\lambda t}).$$

Additionally, note that in the case $I = 1$, a single initial cell, we have

$$P(M_t = k) = e^{-\lambda t}(1 - e^{-\lambda t})^{k-1},$$

so that M_t follows a geometric distribution with probability $p = e^{-\lambda t}$. This can be understood without the standard generating function differential equation method.

Each cell has an $\mathrm{Exp}(\lambda)$ time until it next divides. This follows similarly to our derivations of exponential distributions in Poisson processes. Now the minimum of n independent $\mathrm{Exp}(\lambda)$ random variables is itself $\mathrm{Exp}(n\lambda)$. So while the population is at size n, the time until the next division follows $\mathrm{Exp}(n\lambda)$. Consequently, the waiting times $\tau_1, \tau_2, \ldots$ follow

$$\tau_i \sim \mathrm{Exp}(i\lambda).$$

Define $X_1, \ldots, X_{k-1} \sim \mathrm{Exp}(\lambda)$ independently and with identical distribution. Then another fact of exponential distributions is that

$$\max(X_1, \ldots, X_{k-1}) = T_k$$

in distribution, where

$$T_k = \tau_1 + \tau_2 + \cdots + \tau_{k-1}$$

for $\tau_i \sim \mathrm{Exp}(i\lambda)$. So

$$P(T_k \leq t) = P(\max(X_1, \ldots, X_{k-1}) \leq t) = (1 - e^{-\lambda t})^{k-1}$$

by independence. We then obtain the result for the $I = 1$ case.

We now introduce birth and death processes, in which the division postulate of the last section is replaced by two postulates concerning divisions and deaths respectively. Our cells will now satisfy:

- division rate; each cell present at time t has a chance of dividing in the short time interval $(t, t + h]$. There is a constant λ, the birth rate, such that the probability that any cell divides once during this interval equals $\lambda h + o(h)$ and the probability that it divides more than once is $o(h)$.
- death rate: each cell present at time t has a chance of dying in the short time interval $(t, t + h]$. There is a constant μ, the death rate, such that the probability that any cell dies during this interval equals $\mu h + o(h)$.

We assume deaths occur independently of other deaths and of all births.

Several things can now happen to a cell present at time t in the short time interval $(t, t + h]$. We have:

- death, with probability $\mu h + o(h)$.
- a single division, with probability $\lambda h + o(h)$.
- no change of state, with probability $1 - (\mu + \lambda)h + o(h)$.
- some other combination with probability $o(h)$, which may be disregarded.

We write L_t for the number of cells alive at time t. We have the equations:

$$\begin{aligned} P(L_{t+h} = k \mid L_t = k) &= 1 - (\mu + \lambda)h + o(h), \\ P(L_{t+h} = k+1 \mid L_t = k) &= 1 - \lambda h + o(h), \\ P(L_{t+h} = k-1 \mid L_t = k) &= 1 - \mu h + o(h), \\ P(L_{t+h} < k-1 \text{ or } L_{t+h} > k+1 \mid L_t = k) &= o(h). \end{aligned}$$

As before, we have that the rate of birth in the population is a linear function of k, but now we also have a linear function of k as a rate of death.

This birth-death process now has an absorbing state, in the sense that if we ever reach $L_t = 0$, then there will never be living cells subsequently.

We cannot easily write down the mass function of L_t this time. However, we are able to find the probability generating function of L_t using the same differential-difference equation method as before. We have the following theorem.

Theorem 11.53

If $L_0 = I$, then L_t has probability generating function

$$E(s^{L_t}) = \begin{cases} \left(\dfrac{\lambda t(1-s)+s}{\lambda t(1-s)+1}\right)^I, & \mu = \lambda, \\ \left(\dfrac{\mu(1-s)-(\mu-\lambda s)e^{t(\mu-\lambda)}}{\lambda(1-s)-(\mu-\lambda s)e^{t(\mu-\lambda)}}\right)^I, & \mu \neq \lambda. \end{cases}$$

It is possible that the process L will ultimately become extinct in that $L_t = 0$ for some t. Theorem 11.53 allows us to calculate the probability of this event.

Theorem 11.57

Let $L_0 = I$. Then as $t \to \infty$,

$$P(L_t = 0) \to \begin{cases} 1, & \lambda \leq \mu, \\ (\mu/\lambda)^I, & \lambda > \mu. \end{cases}$$

Hence, extinction is certain if and only if the death rate is at least as big as the birth rate.

Proof.

$$P(L_t = 0) = G(0, t).$$

By theorem 11.53,

$$G(0,t) = \left(\frac{\lambda t}{\lambda t + 1}\right)^I \quad \text{for } \mu = \lambda.$$

$$G(0,t) = \left(\frac{\mu - \mu e^{t(\mu-\lambda)}}{\lambda - \mu e^{t(\mu-\lambda)}}\right)^I \quad \text{for } \mu \neq \lambda.$$

The result follows on taking $t \to \infty$.

We now consider models of queues. We may categorize queues by the ways in which

- the people arrive in the system.
- the people are stored in the system prior to being served.
- the people are served.

The first and third of these can be described with probability theory.

We shall consider a simple model of a queue. We say there is one shop assistant. Customers arrive in the shop, wait for their turn to be served, and leave after their service has been completed. We suppose the following:

- customers arrive in the manner of a Poisson process with rate λ.
- if N_t is the number of customers who have arrived by time t, then $N = (N_t : t \geq 0)$ is a Poisson process with rate λ.
- the service time of each customer is a random variable with the exponential distribution with parameter μ.
- the service times of different customers are independent.
- Service times are independent of arrival times.

We shall also suppose that the customers form a single line and the customer at the head of this line is served first. This queue discipline is called first come, first served.

The arrivals assumption is equivalent to demanding that the times between successive arrivals are independent, exponentially distributed random variables with parameter λ.

Consequently, both inter-arrival times and service times have the exponential distribution, which is crucial for this example.

By memorylessness, the times of the next arrival and the next departure do not depend on the times of the last arrival and the last departure.

Thus, for example, for $h > 0$, the probability of a single arrival during the short time interval $(t, t + h]$ is $\lambda h + o(h)$, and the probability of no arrival is $1 - \lambda h + o(h)$.

Similarly, if the shop assistant is serving someone at time t, then the probability that this service is completed during the short time interval is $\mu h + o(h)$.

Let Q_t be the number of people in the queue, including the person being served, at time t. Suppose $Q_0 = 0$. The process

$$Q = (Q_t : t \geq 0)$$

is something analogous to a birth-death process. We have the probabilities

$$\begin{aligned} P(Q_{t+h} = k \mid Q_t = k) &= 1 - (\mu + \lambda)h + o(h), \\ P(Q_{t+h} = k + 1 \mid Q_t = k) &= \mu h + o(h), \\ P(Q_{t+h} = k - 1 \mid Q_t = k) &= \lambda h + o(h), \\ P(Q_{t+h} > k + 1 \text{ or } Q_{t+h} < k - 1 \mid Q_t = k) &= o(h). \end{aligned}$$

This look very similar to the corresponding equations for the birth-death process, except arrivals and departures here occur at rates which do not depend upon the current queue size. This is somewhat reminiscent of the relation between a Poisson process and our simple model for population with cell division.

We may again invoke the method of differential-difference equations to find the probability mass function $p_k = P(Q_t = k)$, though in this case we end up with non-elementary integrals (in particular, using modified Bessel functions).

It is more interesting to study the long-term behaviour of the queue. If service times are long in comparison with inter-arrival times, the queue will tend to grow, so that after a long period of time it will be very large. On the other hand, if service times are relatively short, it is reasonable to expect that the queue length will settle down into some "steady state".

The asymptotic behaviour of the queue length Q_t as $t \to \infty$ is described by the sequence

$$\pi_k = \lim_{t\to\infty} P(Q_t = k), \qquad k = 0, 1, 2, \ldots.$$

$$\pi_k = \lim_{t\to\infty} p_k(t).$$

Differentiating both sides — this requires the limit to exist — results in

$$0 = \lim_{t\to\infty} p_k'(t).$$

We will now return to the differential-difference equations which we omitted previously. From the four equations describing this model, we obtain using the standard method (see theorem 11.6 for an example):

$$p_k'(t) = \lambda p_{k-1}(t) - (\lambda + \mu)p_k(t) + \mu p_{k+1}(t) \quad \text{for } k \in \mathbb{Z}_+.$$

$$p_0'(t) = -\lambda p_0(t) + \mu p_1(t).$$

Now letting $t \to \infty$ in these equations, we obtain

$$0 = \lambda \pi_{k-1} - (\lambda + \mu)\pi_k + \mu \pi_{k+1} \quad \text{for } k \in \mathbb{Z}_+.$$

$$0 = -\lambda \pi_0 + \mu \pi_1.$$

Now, call a non-negative sequence $\{\pi_k\}$ a steady-state distribution of the queue if it satisfies the above equations together with

$$\sum_{k=0}^{\infty} \pi_k = 1.$$

Theorem 11.74

If $\lambda < \mu$, the queue has a unique steady-state distribution given by

$$\pi_k = \left(1 - \frac{\lambda}{\mu}\right)\left(\frac{\lambda}{\mu}\right)^k, \qquad k = 0, 1, 2, \ldots.$$

If $\lambda \geq \mu$, there is no steady-state distribution.

Proof. We will solve the equations below.

$$0 = \lambda \pi_{k-1} - (\lambda + \mu)\pi_k + \mu \pi_{k+1} \quad \text{for } k \in \mathbb{Z}_+.$$

$$0 = -\lambda \pi_0 + \mu \pi_1.$$

We define the ratio

$$\rho = \frac{\lambda}{\mu}$$

and obtain

$$\pi_1 = \rho\pi_0.$$

Then

$$0 = \rho\pi_0 - (\rho + 1)\pi_1 + \pi_2,$$

so

$$\pi_2 = (\rho^2 + \rho - \rho)\pi_0,$$
$$\pi_2 = \rho^2\pi_0.$$

We can then prove either by induction or otherwise the obvious guess

$$\pi_k = \rho^k\pi_0.$$

Now

$$\sum_{k=0}^{\infty} \pi_k = \pi_0 \sum_{k=1}^{\infty} \rho^k.$$

Then we require $\rho < 1$ and $\pi_0 = 1 - \rho$. We have no restrictions on π_0, and so we deduce that $\rho < 1$ implies the existence of a steady-state solution which necessarily has $\pi_0 = 1 - \rho$.

We call ρ the traffic intensity of the queue. We therefore have:

- if $\rho < 1$, the queue length Q_t settles down as $t \to \infty$ into a steady-state distribution, for which the probability of k customers being present is $(1 - \rho)\rho^k$.
- if $\rho \geq 1$, there is no steady-state distribution, and the queue length either grows beyond bound or has fluctuations of large order.

12 Markov Chains

Due to the length of this chapter, we provide a summary of the results contained therein below.

We will begin with an introduction to discrete-time Markov chains and the use of matrix products and linear algebra in their study. We will introduce the concepts of recurrence and transience and prove a necessary and sufficient criterion for recurrence. We will then derive Polya's theorem, show how to calculate hitting probabilities and hitting times, introduce stopping times, and present the strong Markov property. We will discuss invariant distributions. We will prove the convergence theorem for positive recurrent chains. We will provide a criterion for time reversibility and apply this in the case of a random walk on a finite graph.

A stochastic process is said to have the *Markov property* if, conditional on its present value, its future is independent of its past. This is restrictive enough to allow for study while also being very applicable to processes we find in the real world.

Let S be a countable set called the *state space* and let $X = (X_n : n \geq 0)$ be a sequence of random variables taking values in S.

The X_n are functions on some common probability space which we shall not be specific about.

Informally, we have that the sequence X is a Markov chain if, conditional on X_n, the future $(X_r : r > n)$ is independent of the past $(X_m : m < n)$.

Definition 12.1

The sequence X is called a Markov chain if it satisfies the following property, the Markov property,

$$P(X_{n+1} = i_{n+1} \mid X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) = P(X_{n+1} = i_{n+1} \mid X_n = i_n)$$

for all $n \geq 0$ and all $i_0, i_1, \ldots, i_{n+1} \in S$.

We call the Markov chain *homogenous* if for all $i, j \in S$ the conditional probability $P(X_{n+1} = j \mid X_n = i)$ does not depend on n.

Here are some examples of Markov chains.

- The branching processes described in Chapter 9. Note that clearly the population size of the $(n+1)$th generation is dependent on the size of the nth generation; this is fine, as it is not dependent on the mth generations for $m < n$.
- The random walks described in Chapter 10. At each time, the size of the jump is independent of the previous jumps such that the position is only dependent on the previous position.
- The continuous-time random processes described in Chapter 11. Recall these were the Poisson process, a pure birth process, a linear birth-death process, and the simple queue model. These are all quite clearly Markov chains by recalling their definitions, though to be explicit, we had:
  - The Poisson process had jumps of $+1$ at a constant rate λ. The simple queue model had jumps of $+1$ or -1 at constant rates λ or μ (given $Q_t > 0$).
  - The birth process had jumps of $+1$ at a rate λN_t dependent only on the current state. The birth-death process had jumps of $+1$ or -1 at rates λL_t or μL_t dependent only on the current state. Note that for these birth processes, we defined them in Chapter 11 using the rates of birth (and death) for each individual in the population, and obtained the rates given here as a consequence of these. In the study of Markov chains, it would be more common to define a birth process using the defining property $P(N_{t+h} = n+1 \mid N_t = n) = \lambda_n h + o(h)$ for a non-negative sequence λ_n, with some supplemental condition to ensure the per-individual behaviour.

A final example is "Markov chain Monte Carlo". This is an important example of the use of Markov chains in statistics; in Bayesian statistics, we often find a distribution of an unknown parameter.

These "posterior" distributions can be easier to work with by the following method: construct a Markov chain with the posterior π as its *invariant distribution*, a concept we will later discuss.

Then, simulate this chain for a long time such that the outcome is "nearly" distributed as π - by this, we mean the probabilities $P(X_n = i_1), P(X_n = i_2), \ldots$ approach the distribution π such that sampling the result of the chain simulated for a long time is essentially the same as sampling from the posterior distribution.

In this chapter, we will only study homogenous Markov chains. Recall we define a Markov chain homogenous if $P(X_{n+1} = j \mid X_n = i)$ does not depend on n.

This is perhaps easy to confuse with the properties of the birth and birth-death processes of the previous chapter, where we have the size of jumps dependent on the size of the population, however, these do not break this sense of homogeneity as we have the distribution of, say, $(X_2 \mid X_1 = k)$ identical to that of $(X_{10} \mid X_9 = k)$ as an example.

A Poisson process whose rate varies with time, say phone calls arriving faster during the day than at night, is an example of a non-homogenous Markov chain.

I have included many tangents in the paragraphs above for my own understanding, so we shall recall that our primary focus for now is to study discrete homogenous Markov chains. Let us return to this. In order to calculate properties associated with these chains, we need to know two quantities:

- the transition matrix $P = (p_{i,j} : i, j \in S)$ given by $p_{i,j} = P(X_1 = j \mid X_0 = i)$.
  - By the assumption of homogeneity, we have $p_{i,j} = P(X_{n+1} = j \mid X_n = i)$ also.
- the initial distribution $\lambda = (\lambda_i : i \in S)$ given by $\lambda_i = P(X_0 = i)$.
  - Note that so far we have mostly seen random processes with trivial initial distributions with $P(X_0 = a) = 1$ for some starting value a.

The pair (λ, P) is characterised as follows.

Proposition 12.3

- The vector λ is a distribution in that $\lambda_i \geq 0$ for $i \in S$ and $\sum_{i \in S} \lambda_i = 1$.
- The matrix $P = (p_{i,j})$ is a stochastic matrix in that:
  - $p_{i,j} \geq 0$ for $i, j \in S$.
  - $\sum_{j \in S} p_{i,j} = 1$ for $i \in S$.

The second condition for a stochastic matrix results in P having row sums 1.

Proof.

- λ_i is a probability, so it is non-negative and $\sum_{i \in S} \lambda_i = P(X_0 \in S) = 1$.
- $p_{i,j}$ is a probability, so it is non-negative.

$$\sum_{j \in S} p_{i,j} = \sum_{j \in S} P(X_1 = j \mid X_0 = i).$$

$$\sum_{j \in S} p_{i,j} = P(X_1 \in S \mid X_0 = i).$$

$$\sum_{j \in S} p_{i,j} = 1.$$

Theorem 12.4

Let λ be a distribution and P a stochastic matrix. The random sequence $X = (X_n : n \geq 0)$ is a Markov chain with initial distribution λ and transition matrix P if and only if

$$P(X_0 = i_0, X_1 = i_1, \ldots, X_n = i_n) = \lambda_{i_0} p_{i_0,i_1} \cdots p_{i_{n-1},i_n}$$

for all $n \geq 0$ and $i_0, i_1, \ldots, i_n \in S$.

Proof.

Write $A_k = \{X_k = i_k\}$ such that we have the left hand side $P(A_0 \cap A_1 \cap \cdots \cap A_n)$.

Now the forward implication is quite easy to prove by induction.

Suppose conversely that we allow our condition to hold for all n and sequences $\{i_m\}$.

Setting $n = 0$, we have $P(X_0 = i_0) = \lambda_{i_0}$, so the initial distribution is λ.

Now

$$P(A_{n+1} \mid A_0 \cap A_1 \cap \cdots \cap A_n) = \frac{P(A_0 \cap A_1 \cap \cdots \cap A_{n+1})}{P(A_0 \cap A_1 \cap \cdots \cap A_n)}.$$

So $P(A_{n+1} \mid A_0 \cap A_1 \cap \cdots \cap A_n) = p_{i_n,i_{n+1}}$ by our assumption.

Now we may do some more work to show that $P(A_{n+1} \mid A_n)$, which is what we need to show is dependent only on i_n, is equal to this left hand side.

Essentially, we write $P(A_n \cap A_{n+1})$ as the sum of the probabilities of each possible case of the sequence up to the $(n-1)$th state such that A_n, A_{n+1} occur. We rewrite this using the partition theorem as a sum over cases of the sequence up to the $(n-1)$th state of $P(A_{n+1} \mid$ a case of the sequence up to the nth state$)P($that case$)$, which we use the previous result to rewrite as $p_{i_n,i_{n+1}} P(A_n)$. Now $P(A_n \cap A_{n+1}) = p_{i_n,i_{n+1}} P(A_n)$ gives the desired result upon division by $P(A_n)$.

Since $P(A_{n+1} \mid A_n)$ only depends on i_n, we have that X is a homogenous Markov chain with transitional matrix P.

The Markov property can be interpreted as "the past affects the future only via the present". This is made formal in the next theorem, in which X_n is the present value, F is a future event, and H is a historical event.

Theorem 12.8: extended Markov property

Let X be a Markov chain. For $n \geq 0$, for any event H in terms of the past $X_0, \ldots, X_{n-1}$ and any event F given in terms of the future $X_{n+1}, \ldots$, we have

$$P(F \mid X_n = i, H) = P(F \mid X_n = i)$$

for $i \in S$.

Proof idea.

We apply theorem 12.4 upon rewriting $P(F \mid X_n = i, H)$ using the definition of conditional probability.

We write this as a sum over all possible sequences with the events H, $X_n = i$, F divided by the sum over all possible sequences with the events H, $X_n = i$.

We simplify this to obtain the result.

Let X be a Markov chain with transition matrix P. The elements $p_{i,j}$ are called the one-step transition probabilities. More generally, the n-step transition probabilities are given by

$$p_{i,j}(n) = P(X_n = j \mid X_0 = i).$$

The n-step transition probabilities form a matrix called the n-step transition matrix $P(n) = (p_{i,j}(n) : i, j \in S)$. The matrices $P(n)$ satisfy the equations below.

Theorem 12.13: Chapman-Kolmogorov equations

$$p_{i,j}(m+n) = \sum_{k \in S} p_{i,k}(m) p_{k,j}(n)$$

for $i, j \in S$ and $m, n \geq 0$.

Thus, $P(m+n) = P(m)P(n)$ for the matrices P.

Proof.

$$p_{i,j}(m+n) = P(X_{m+n} = j \mid X_0 = i).$$

$$p_{i,j}(m+n) = \sum_{k \in S} P(X_{m+n} = j \mid X_m = k, X_0 = i) P(X_m = k \mid X_0 = i)$$

by the partition theorem.

Note that $P(A \mid B, C) = P(A \mid B \cap C) = P_c(A \mid B)$ for P_c a probability measure with $P_c(E) = P(E \mid C)$ for a rigorous proof of the partition theorem here, if this has not already been covered in a prior chapter.

$$p_{i,j}(m+n) = \sum_{k \in S} P(X_{m+n} = j \mid X_m = k) P(X_m = k \mid X_0 = i)$$

by the extended Markov property.

We may be getting a bit lost in notation by now, so let's review. Theorem 12.13 says the probability of moving from state i to state j after $m+n$ steps is the sum over all states k of the product of the probability of moving from i to k in m steps and the probability of moving from k to j in n steps.

By the Chapman-Kolmogorov equations, we have $P(n) = P(1)^n$. Thus, the n-step transition probabilities may be found by finding the nth power of the matrix of 1-step transition probabilities. This can be done with standard diagonalisation techniques applied to P.

Example 12.15: a two-state Markov chain

Suppose $S = \{1, 2\}$ and

$$P = \begin{pmatrix} 1-a & a \\ b & 1-b \end{pmatrix}$$

with $a, b \in (0, 1)$. Find the n-step transition probabilities.

As a reminder, we have the probability of moving from state 1 to state 2 being a, etc.

Solution.

We shall diagonalise P. The eigenvalues κ of P are the roots of the equation $\det(P - \kappa I) = 0$.

We obtain $\kappa_1 = 1$ and $\kappa_2 = 1 - a - b$.

Thus,

$$P = U^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 1-a-b \end{pmatrix} U$$

for some invertible matrix U.

We have

$$P^n = U^{-1} \begin{pmatrix} 1 & 0 \\ 0 & (1-a-b)^n \end{pmatrix} U$$

by standard matrix theory.

Thus, $p_{1,1}(n) = A + B(1-a-b)^n$ for some constants A, B.

We obtain the constants A, B by using our known value for $p_{1,1}(1)$ and $p_{1,1}(0) = 1$.

We have $p_{1,2}(n) = 1 - p_{1,1}(n)$ and $p_{2,2}(n)$ may be found by interchanging a and b.

We can therefore find a closed form for P^n, which is not very nice and hence omitted here.

However, we do have

$$P^n \to \frac{1}{a+b} \begin{pmatrix} b & a \\ b & a \end{pmatrix} \qquad \text{as } n \to \infty,$$

which is to say that:

$$p_{i,1}(n) \to \frac{b}{a+b} \qquad \text{and} \qquad p_{i,2}(n) \to \frac{a}{a+b}.$$

These results are crucially independent of i, which hints at a general property of Markov chains. The matrix P^n will approach, except for special cases, to a matrix with identical rows.

We note that this problem may also be solved with difference equations, which is the quicker approach for small Markov chains which we might have encountered in STEP not necessarily presented as Markov chains. This approach breaks down for larger Markov chains.

We summarise the matrix method. Suppose that the state space is finite so that $|S| = N$. Then P is an $N \times N$ matrix.

It is a general result for stochastic matrices that $\kappa_1 = 1$ is an eigenvalue of P, and no other eigenvalue has larger absolute value.

We write $\kappa_1, \kappa_2, \ldots, \kappa_N$ for the (possibly complex) eigenvalues of P, arranged in decreasing order of absolute value. We assume for simplicity that the κ_i are distinct.

There exists an invertible matrix U such that $P = U^{-1}KU$ where K is the diagonal matrix with entries $\kappa_1, \kappa_2, \ldots, \kappa_N$.

Then

$$P^n = U^{-1} \begin{pmatrix} \kappa_1^n & 0 & \cdots & 0 \\ 0 & \kappa_2^n & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \kappa_N^n \end{pmatrix} U,$$

and the individual probabilities $p_{i,j}(n)$ may thus be found.

The situation is simplified further, and leads to a generalisation of the result in example 12.15, if the chain has two further properties, *irreducibility* and *aperiodicity*. We will encounter these conditions in due time.

Under these conditions, by the *Perron-Frobenius theorem*, we have $\kappa_1 = 1$ as the unique eigenvalue with absolute value 1, so that $\kappa_k^n \to 0$ as $n \to \infty$ for $k \geq 2$.

Consequently, the matrix P^n approaches

$$U^{-1} \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix} U$$

in the limit as $n \to \infty$.

Now the rows of U are the normalised left eigenvectors of P. One can verify that for any stochastic matrix P, $P\mathbf{1}' = \mathbf{1}'$ where $\mathbf{1}'$ is the column vector of ones. Therefore, this is an eigenvector and the first column of U^{-1} is $\mathbf{1}'$.

Note that in linear algebra this decomposition is more common as $P = VDV^{-1}$ rather than having the inverse first, but this is just notation.

Hence, by examining the product of our limiting result for P^n, we find that $p_{i,j}(n) \to \pi_j$ for some vector $\pi = (\pi_j : j \in S)$ that does not depend on the initial state i.

This is essentially the result of example 12.15, where we found P^n tends to a matrix with identical rows so that the initial state does not matter for where the Markov chain ends up after a very long time.

Note that for this result, we made three big assumptions:

- irreducibility
- aperiodicity
- a finite state space

Remark 12.19

Much of the theory of Markov chains involves the manipulation of vectors and matrices. For instance, if X_0 has distribution λ interpreted as a row vector $(\lambda_i : i \in S)$, then

$$P(X_1 = j) = \sum_{i \in S} \lambda_i p_{i,j} \qquad \text{for } j \in S.$$

So that the distribution of X_1 is the row vector λP. We also have X_2 having distribution λP^2, etc. This of course makes sense for P^n being equivalent to the n-step transition matrix, which we derived from the Chapman-Kolmogorov equations.

Thus, we adopt the convention that probability distributions are by default row vectors acting on the left side of matrices.

We define the column vector λ' as the transpose of λ.

An important element in the theory of Markov chains is the interaction between the state space S and the transition mechanism P.

Let X be a homogenous Markov chain with state space S and transition matrix P. For $i, j \in S$ we say that i leads to j, written $i \to j$, if $p_{i,j}(n) > 0$ for some $n \geq 0$. By setting $n = 0$, we have that $i \to i$ for all $i \in S$. We write $i \leftrightarrow j$ if $i \to j$ and $j \to i$. In this case, we say that i and j communicate.

Proposition 12.23

The relation $\leftrightarrow$ is an equivalence relation.

Proof.

We must show that the relation $\leftrightarrow$ is reflexive, symmetric, and transitive.

- We have that $\leftrightarrow$ is reflexive as $i \leftrightarrow i$ for all i.
- We have that $\leftrightarrow$ is symmetric as $i \leftrightarrow j$ if and only if $j \leftrightarrow i$.

Now suppose $i \leftrightarrow j$ and $j \leftrightarrow k$. Since $i \to j$ and $j \to k$, there exist $m, n \geq 0$ such that $p_{i,j}(m) > 0$ and $p_{j,k}(n) > 0$. By the Chapman-Kolmogorov equations,

$$p_{i,k}(m+n) = \sum_{l \in S} p_{i,l}(m) p_{l,k}(n).$$

Now the right hand side is positive as it includes the term $p_{i,j}(m)p_{j,k}(n)$ which is known to be positive.

So $i \to k$ and we can similarly show $k \to i$. So $i \leftrightarrow k$, meaning $\leftrightarrow$ is transitive.

So $\leftrightarrow$ is an equivalence relation.

Therefore, $\leftrightarrow$ has equivalence classes (which are either identical or disjoint, and thus partition the state space). These are the subsets of S of the form $C_i = \{j \in S : i \leftrightarrow j\}$. These classes are called communicating classes. We now arrive at our definition of irreducibility which was one of our assumptions in a previous theorem: the chain X is called irreducible if there is a single communicating class, so that $i \leftrightarrow j$ for all $i, j \in S$.

A subset C of S is called closed if

$$i \in C, \quad i \to j \quad \Rightarrow \quad j \in C.$$

If the chain ever hits a closed set C, then it remains in C forever afterwards. If a singleton set $\{i\}$ is closed, we call i an *absorbing state*.

Why does this definition result in the chain remaining in C? Suppose the chain is at some state $i \in C$. Then for any j not in C, it must be the case that $i \to j$ must be false, so the chain cannot reach j from i, and we are stuck in C.

To provide an example, suppose we have the three state chain with

$$P = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0 & 0.4 & 0.6 \\ 0 & 0.7 & 0.3 \end{pmatrix}.$$

Then we have the communicating classes $\{1\}$ and $\{2, 3\}$. We have the closed sets $\{1, 2, 3\}$ and $\{2, 3\}$. So $\{2, 3\}$ is both a closed set and a communicating class, $\{1\}$ is only a communicating class, and $\{1, 2, 3\}$ is only a closed set, where "only" here refers to a categorisation into closed sets and communicating classes.

Proposition 12.25

A subset C of states is closed if and only if $p_{i,j} = 0$ for $i \in C$, $j \notin C$.

We will omit the proof as intuitively this is quite obvious.

We may illustrate a Markov chain by adding points to represent states and directed lines between these states. We can then identify closed sets as regions without any arrows pointing out of them. The communicating classes are somewhat more difficult to spot, but still not difficult for a small chain.

Let X be a homogenous Markov chain with state space S and transition matrix P. For brevity, we write henceforth $P_i(A)$ for $P(A \mid X_0 = i)$ and $E_i(Z)$ for $E(Z \mid X_0 = i)$.

The *first-passage time* to state j is defined as $T_j = \min\{n \geq 1 : X_n = j\}$.

The *first-passage probabilities* to state j are given by $f_{i,j}(n) = P_i(T_j = n)$ so that $f_{i,j}(n)$ may be interpreted as the probability that state j is first reached at time n.

Definition 12.30

A state i is called recurrent if $P_i(T_i < \infty) = 1$. A state is called transient if it is not recurrent.

The word persistent is sometimes used instead of recurrent.

Theorem 12.31

The state i is recurrent if and only if

$$\sum_{n=0}^{\infty} p_{i,i}(n) = \infty.$$

We proved theorem 10.12 regarding conditions for the recurrence of the simple random walk using generating functions. This method may be extended to prove theorem 12.31, but we will provide an arguably simpler proof here.

Proof.

Define V as the number of times that $X_n = i$ with $n \geq 0$. Let $r = P_i(T_i < \infty)$ be the probability of ever returning.

Since V is the sum of the indicator functions $\mathbf{1}_{\{X_n=i\}}$, we have

$$E_i(V) = \sum_{n=0}^{\infty} P_i(X_n = i).$$

Or

$$E_i(V) = \sum_{n=0}^{\infty} p_{i,i}(n).$$

Now we claim $P_i(V \geq k+1) = r^k$.

$$P_i(V \geq k+1) = \sum_{n \geq 1} P_i(T_i = n \text{ and there are at least } k \text{ further visits to } i \text{ after time } n).$$

By the extended Markov property, we have the right hand side factoring, and we get:

$$P_i(V \geq k+1) = \sum_{n \geq 1} P_i(T_i = n) P_i(V \geq k).$$

$$P_i(V \geq k+1) = rP_i(V \geq k).$$

$$P_i(V \geq k+1) = r^k,$$

as we have $P_i(V \geq 1) = 1$ for all i.

Now we prove the forward implication. We assume $r = 1$.

Then $P_i(V \geq k+1) = 1$, so $V = \infty$ almost surely. It can be made rigorous that this implies $E_i(V) = \infty$.

Intuitively, this direction is quite clear. A recurrent state returns with certainty, and each return restarts the chain from i, so it returns again with certainty, and so on forever. Infinitely many visits means an infinite expected number of visits.

Now we prove the converse. We have $r < 1$ implying $E_i(V) = \sum_{k \geq 0} r^k$, which converges given our assumption. The middle step is a standard rewriting of the mean for integer random variables.

Taking the contrapositive, we have $E_i(V) = \infty \Rightarrow r = 1$, which is our definition for recurrence.

Theorem 12.37

Let C be a communicating class. We have:

1. either every state in C is recurrent or every state is transient.
2. if C contains a recurrent state, then C is closed.

Proof.

We prove part (a) first.

Let $i \leftrightarrow j$ and $i \neq j$. By theorem 12.31, it suffices to show that $\sum_{n=0}^{\infty} p_{i,i}(n) = \infty$ if and only if $\sum_{n=0}^{\infty} p_{j,j}(n) = \infty$.

Since $i \leftrightarrow j$, there exist $m, n \geq 1$ such that $p_{i,j}(m)p_{j,i}(n) > 0$.

By Chapman-Kolmogorov, we have

$$p_{i,i}(m+r+n) \geq p_{i,j}(m)p_{j,j}(r)p_{j,i}(n).$$

So

$$\sum_{r=0}^{\infty} p_{i,i}(m+r+n) \geq a \sum_{r=0}^{\infty} p_{j,j}(r)$$

summing over r and for some constant.

The condition follows.

Now proving part (b).

Suppose $i \in C$ is recurrent and C is not closed.

By proposition 12.25, there exists $j \in C$, $k \notin C$ such that $p_{j,k} > 0$.

But since C is a communicating class, we have that $k \not\to j$.

By part (a), we have that j is recurrent.

Now,

$$P_j(X_n \neq j \text{ for all } n \geq 1) \geq P_j(X_1 = k) = p_{j,k} > 0.$$

This contradicts the recurrence of j.

Therefore we have that if $i \in C$ is recurrent, C must be closed.

What is the motivation for this contradiction? This seems a bit out-of-nowhere, but the key idea is that if $j \to k$ but $k \not\to j$, then $P_j(T_j = \infty) > 0$, so j is transient, which is quite intuitive. So if j is recurrent and $j \to k$, then $k \to j$, so k is in the communicating class of j for any state k which we initially assumed to be outside of this communicating class. Hence, the communicating class is closed.

For the following theorem, recall a chain is irreducible if it has a single communicating class.

Proposition 12.40

Let $i, j \in S$. If j is transient, then $p_{i,j}(n) \to 0$ as $n \to \infty$.

Proof sketch.

Using a similar idea to our approach to theorem 12.31, we may show that the chain almost surely visits j only finitely many times if j is transient.

Then

$$P_i(X_m = j \text{ for some } m \geq n) \to 0$$

as $n \to \infty$.

Now $p_{i,j}(n) \leq P_i(X_m = j \text{ for some } m \geq n)$, so $p_{i,j}(n) \to 0$ also.

Theorem 12.39

Suppose that the state space S is finite. Then:

1. There exists at least one recurrent state.
2. If the chain is irreducible, all states are recurrent.

Proof.

We prove part (a) first.

Suppose $|S| < \infty$. Then $\sum_{j \in S} p_{i,j}(n) = 1$.

Now if every state is transient, then by proposition 12.40, we have $p_{i,j}(n) \to 0$ as $n \to \infty$, contradicting the above line.

Now proving part (b).

Suppose the chain is irreducible. By theorem 12.37, either every state is recurrent or every state is transient. By part (a), we must have every state recurrent.

Part (b) of theorem 12.39 is quite useful in problem solving.

We explored one-dimensional random walks in Chapter 10. We will use the theory of Markov chains to study random walks in up to three dimensions.

The graphs in these sections are d-dimensional lattices. Let $\mathbb{Z}$ be the set of integers and $\mathbb{Z}^d$ the set of all d-vectors of integers written $x = (x_1, x_2, \ldots, x_d)$ with $x_i \in \mathbb{Z}$. The set $\mathbb{Z}^d$ may be interpreted as a graph with vertex set $\mathbb{Z}^d$ and with edges joining any two vectors x, y separated by Euclidean distance 1. Two such vertices are declared adjacent and said to be neighbours. We denote the resulting graph by $\mathbb{Z}^d$ also and note that each vertex has exactly $2d$ neighbours.

Note that $\mathbb{Z}^d$ is connected in that any pair of vertices is joined by a path of edges.

Let $d \geq 1$. The symmetric random walk on $\mathbb{Z}^d$ is the Markov chain on the state space $\mathbb{Z}^d$ which, at each step, jumps to a uniformly chosen neighbour. The transition matrix is given by $p_{x,y} = 1/(2d)$ if y is a neighbour of x, and $p_{x,y} = 0$ otherwise.

Since the graph is connected, the chain is irreducible. By theorem 12.37, either every state is recurrent or every state is transient.

Theorem 12.44: Polya's theorem

The symmetric random walk on $\mathbb{Z}^d$ is recurrent if $d = 1, 2$ and transient if $d \geq 3$.

Proof.

Let $d = 1$ and $X_0 = 0$.

We have

$$p_{0,0}(2n) = \left(\frac{1}{2}\right)^{2n} \binom{2n}{n},$$

which we obtained in Chapter 10.

By Stirling's formula,

$$p_{0,0}(2n) \sim \frac{1}{\sqrt{\pi n}}.$$

Thus,

$$\sum_{n=0}^{\infty} p_{0,0}(2n) = \infty.$$

By theorem 12.31, the state 0 is recurrent. By theorem 12.37, all states are recurrent.

Let $d = 2$ and $X_0 = \mathbf{0} := (0, 0)$.

We have

$$p_{\mathbf{0},\mathbf{0}}(2n) = \left(\frac{1}{4}\right)^{2n} \sum_{m=0}^{n} \frac{(2n)!}{\left(m!(n-m)!\right)^2}$$

using a combinatorial argument for how equal numbers of upward and downward steps may be taken.

$$p_{\mathbf{0},\mathbf{0}}(2n) = \left(\frac{1}{4}\right)^{2n} \binom{2n}{n} \sum_{m=0}^{n} \binom{n}{m}\binom{n}{n-m}.$$

$$p_{\mathbf{0},\mathbf{0}}(2n) = \left(\frac{1}{2}\right)^{4n} \binom{2n}{n}^2.$$

Notice that this is the square of the corresponding result for one dimension. So we have

$$p_{\mathbf{0},\mathbf{0}}(2n) \sim \frac{1}{\pi n}.$$

Thus,

$$\sum_{n=0}^{\infty} p_{\mathbf{0},\mathbf{0}}(2n) = \infty.$$

By theorem 12.31, the state 0 is recurrent. By theorem 12.37, all states are recurrent.

Let $d = 3$ and $X_0 = \mathbf{0} := (0, 0, 0)$.

Similarly,

$$p_{\mathbf{0},\mathbf{0}}(2n) = \left(\frac{1}{6}\right)^{2n} \sum_{i+j+k=n} \frac{(2n)!}{(i!j!k!)^2}.$$

$$p_{\mathbf{0},\mathbf{0}}(2n) = \left(\frac{1}{2}\right)^{2n} \binom{2n}{n} \sum_{i+j+k=n} \left(\frac{n!}{3^n i!j!k!}\right)^2.$$

$$p_{\mathbf{0},\mathbf{0}}(2n) \leq \left(\frac{1}{2}\right)^{2n} \binom{2n}{n} M \sum_{i+j+k=n} \frac{n!}{3^n i!j!k!}$$

where

$$M = \max\left\{\frac{n!}{3^n i!j!k!} : i, j, k \geq 0 \text{ and } i + j + k = n\right\}.$$

Now the maximum is attained where i, j, k are all closest to $n/3$, so that

$$M \leq \frac{n!}{3^n(\lfloor n/3 \rfloor !)^3}.$$

$$M \leq \frac{3\sqrt{3}}{2\pi} \cdot \frac{1}{n}$$

by Stirling's formula.

Thus, applying Stirling's formula again, $p_{\mathbf{0},\mathbf{0}}(2n) \leq Cn^{-3/2}$ for some constant C.

Summing over n provides a convergent series, implying that the origin $\mathbf{0}$ is transient by theorem 12.31.

By theorem 12.37, all states are transient.

Are there heuristics for why this theorem should hold? One heuristic is that after n steps the walk is spread over a region of radius $\sqrt{n}$, which contains about $n^{d/2}$ states. The probability of being at a particular state is therefore about one over that volume. Recall the p-series test and apply the results to theorem 12.31. This is of course far from a proof.

Some interesting ways of presenting Polya's theorem. There is a direct analogy to an electrical network with a 1-ohm resistor placed on every edge of $\mathbb{Z}^d$. I do not understand it, but it sounds interesting. We also have the following very serious result attributed to Shizuo Kakutani: "a drunk man will find his way home, but a drunk bird may get lost".

Let A be a subset of S. The hitting time of the subset A is the earliest epoch n of time at which $X_n \in A$ so that

$$H^A = \inf\{n \geq 0 : X_n \in A\}.$$

The infimum of an empty set is taken by convention to be ∞, so that H^A takes values in $\mathbb{N} \cup \{\infty\}$.

The hitting time is essentially the first-passage time with two distinctions:

- the definition admits sets of states.
- we have $H^A = 0$ if $X_0 \in A$; the initial state is considered in calculating H^A.

In problems, one can sometimes use either. However, hitting times are more common when dealing with reachability or absorption, while first-passage times are more common when analysing recurrence and transience.

We study the hitting probability $h_i^A = P_i(H^A < \infty)$ of ever hitting A starting from i.

We also study the mean value of H^A.

If A is closed, then h_i^A is called an absorption probability.

Theorem 12.54

The vector $h^A = (h_i^A : i \in S)$ is the minimal non-negative solution to the set of equations (an equation for each i):

$$h_i^A = 1 \qquad \text{for } i \in A.$$

$$h_i^A = \sum_{j \in S} p_{i,j} h_j^A \qquad \text{for } i \notin A.$$

The latter line may be interpreted as "the probability we ever reach A starting from i is the sum of the probability we ever reach A starting from j multiplied by the probability we reach j from i", which makes the statement a bit more clear.

We take h^A to be a column vector, as it multiplies $p_{i,j}$ on the right.

Proof of theorem 12.54.

The statement for $i \in A$ is obvious.

For the second part of the piecewise expression, we write

$$h_i^A = \sum_{j \in S} p_{i,j} P_i(H^A < \infty \mid X_1 = j) = \sum_{j \in S} p_{i,j} h_j^A.$$

Some more steps may be included to make the application of the partition theorem and Markov property clearer.

We must show the h_i^A are minimal. Let $x = (x_i : i \in S)$ be a non-negative solution to the set of equations.

Let $i \notin A$. We have:

$$x_i = \sum_{j \in S} p_{i,j} x_j = \sum_{j \in A} p_{i,j} x_j + \sum_{j \notin A} p_{i,j} x_j.$$

Since $x_j = 1$ for $j \in A$, and x is non-negative,

$$x_i = \sum_{j \in A} p_{i,j} + \sum_{j \notin A} p_{i,j} x_j.$$

$$x_i = P_i(X_1 \in A) + \sum_{j \notin A} p_{i,j} x_j.$$

We now expand the x_j in the second sum using a similar decomposition.

$$x_i = P_i(X_1 \in A) + \sum_{j \notin A} p_{i,j} \left(\sum_{k \in A} p_{j,k} x_k + \sum_{k \notin A} p_{j,k} x_k \right).$$

Hence $x_i = P_i(X_1 \in A) + P_i(X_1 \notin A, X_2 \in A)$ + another sum.

Or, $x_i \geq P_i(X_1 \in A) + P_i(X_1 \notin A, X_2 \in A)$ as the other sum has positive terms.

Iterating, we obtain

$$\begin{aligned} x_i \geq\ & P_i(X_1 \in A) \\ & + P_i(X_1 \notin A, X_2 \in A) \\ & + P_i(X_1 \notin A, X_2 \notin A, X_3 \in A) + \cdots . \end{aligned}$$

Or, $x_i \geq P_i(H^A \leq n)$ for all $n \geq 0$.

Taking the limit as $n \to \infty$, we deduce $x_i \geq P_i(H^A < \infty) = h_i^A$ by definition.

We provide an example of the calculation of h^A by theorem 12.54.

Consider the three state Markov chain with

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 0 & 1 \end{pmatrix}$$

so that state 3 is an absorbing state.

Let $A = \{3\}$. We have $h_3^A = 1$, $h_1^A = h_2^A$, $h_2^A = 1$ upon simplifying the equations from theorem 12.54.

So $h^A = (1, 1, 1)$. Hence, from any starting state, the chain will eventually reach state 3 with probability 1.

Similarly, we may obtain $h^B = (1, \frac{1}{2}, 0)$ for $B = \{1\}$ and $h^C = (1, 1, 0)$ for $C = \{2\}$.

We now turn to the mean hitting times. Write $k_i^A = E_i(H^A)$. Note that $k_i^A = \infty$ if $P_i(H^A = \infty) > 0$.

Theorem 12.57

The vector $k^A = (k_i^A : i \in S)$ is the minimal non-negative solution to the set of equations (an equation for each i):

$$k_i^A = 0 \qquad \text{for } i \in A.$$

$$k_i^A = 1 + \sum_{j \in S} p_{i,j} k_j^A \qquad \text{for } i \notin A.$$

Proof sketch.

The equation is obtained by conditioning on the first step of the chain.

Showing that the k_i^A are minimal follows very similarly to the proof of theorem 12.54, with a recursive splitting of sums establishing a bound on any solution y_i.

Example 12.59: Gambler's ruin

Let $S = \mathbb{Z}_+$ and $p \in (0,1)$. A random walk on S moves one unit right with probability p and one unit left with probability $q = 1 - p$. There is an absorbing barrier at 0. Find the probability of absorption from a given starting point.

Solution.

Let h_i be the probability of absorption starting at i. We have (h_i) as the minimal non-negative solution to the equations $h_0 = 1$, $h_i = ph_{i+1} + qh_{i-1}$.

The difference equation has general solution

$$h_i = A + B(q/p)^i.$$

If $p < q$, we must have $B = 0$ for h_i to remain bounded. Then $h_0 = 1 \Rightarrow A = 1$, so $h_i = 1$ for all $i \geq 0$.

If $p > q$, we have $A + B = 1$ from $h_0 = 1$, and we then use the minimality of the h_i to obtain $A = 0$ upon rewriting h_i with B eliminated. We obtain $h_i = (q/p)^i$.

If $p = q$, the difference equation has solution $h_i = A + Bi$, and the above arguments yield $(A, B) = (1, 0)$ so that $h_i = 1$.

The Markov property of Definition 12.1 requires that, conditional on the value of the chain at a given time n, the future evolution of the chain is independent of its past. We frequently require an extension of this property to a random time n.

The Markov property cannot be true for all random times, and it turns out that the random times for which it holds are those satisfying the following definition of a *stopping time*. Random times that fail to be stopping times are the ones that break it.

Definition 12.63

A random variable T, taking on values in the extended non-negative integers, is called a stopping time for the chain X if, for all $n \geq 0$, the event $\{T = n\}$ is given in terms of $X_0, X_1, \ldots, X_n$ only.

That is to say, a random time T is a stopping time if you can tell whether it equals any given time by examining only the present and past of the chain.

Whereas if T "looks into the future", conditioning on $\{T = n\}$ tells you something about $X_{n+1}, X_{n+2}, \ldots$, so the future after T is no longer a fresh copy of the chain.

The definition of a stopping time removes this problem. The event $\{T = n\}$ depends only on $X_0, \ldots, X_n$, so conditioning on it uses only the past and present. Thus, the Markov property carries over to T. A rigorous proof will be provided in the following theorem.

The principal examples of stopping times are the hitting times of Section 12.6. Let A be a subset of S, and consider the hitting time H^A. We have that

$$\{H^A = n\} = \{X_n \in A\} \cap \left(\bigcap_{0 \leq m < n} \{X_m \notin A\} \right).$$

Thus, one can tell whether or not $H^A = n$ by examining $X_0, X_1, \ldots, X_n$ only; the distribution of H^A is dependent on $X_0, X_1, \ldots, X_n$ only.

For some more examples, we have $T = H^A + 1$ is a stopping time, but $T = H^A - 1$ is not a stopping time, as if $\{H^A - 1 = n\}$ occurs, we are given information about X_{n+1} and thus the future is not independent of the past up to time $T = H^A - 1$.

Theorem 12.64: strong Markov property

Let X be a Markov chain with transition matrix P and let T be a stopping time. Given that $T < \infty$ and $X_T = i$, the sequence $Y = (Y_k : k \geq 0)$, given by $Y_k = X_{T+k}$, is a Markov chain with transition matrix P and initial state $Y_0 = i$. Furthermore, Y is independent of $X_0, X_1, \ldots, X_{T-1}$.

Proof idea.

We use many of the same techniques as we did in the introduction to Markov chains to show chains obeyed the Markov property. We must remember to show independence of $X_0, X_1, \ldots, X_{T-1}$ also.

Why is the final part about independence there? For many non-stopping random times T, we have that Y is not even a Markov sequence. For instance, if $T = H^A - 1$, then $Y_1 = A$ with certainty (letting A be a singleton). Sometimes though, Y still may be Markov, but with dependence on $X_0, X_1, \ldots, X_{T-1}$ based on how T was selected. An example is given below.

Example

Let $X_0, X_1, X_2, \ldots$ be independent fair bits.

This is a Markov chain with transition matrix

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

and uniform initial distribution.

Define $T = 1$ if $X_3 = X_0$ and $T = 2$ if $X_3 \neq X_0$.

T is not a stopping time, as X_3 is in the future relative to both times 1 and 2, so the event $\{T = n\}$ is never given in terms of $X_0, X_1, \ldots, X_n$ only.

In the first case, $Y = (X_1, X_2, X_3, \ldots)$ is a Markov chain as each time is independent of the past and future of Y. However, $X_3 = X_0$ and so Y is not independent of $X_0, X_1, \ldots, X_{T-1}$.

Looking only at Y, we know nothing about X_3 from either the past or future of Y, and so it is still Markov.

A similar idea applies in the second case. Hence, the additional independence property is indeed unique to stopping times.

The point is that we constructed a T such that values in the future of T, and hence in Y, are dependent on the past, but only the past values of X which are not included in Y itself, keeping Y as Markov.

Example 12.66: Gambler's ruin

Let $S = \mathbb{Z}_+$ and $p \in (0, 1)$. A random walk on S moves one unit right with probability p and one unit left with probability $q = 1 - p$. There is an absorbing barrier at 0. Let H be the time until absorption at 0. Find the distribution and mean of H given $X_0 = 1$.

How is this different from example 12.59? There, we ask about the probability of absorption ever. It was in the general case for any starting state, but we would be finding the total mass $P(H < \infty)$. Here, we find the distribution of H so that not only we know $P(H < \infty)$ but in fact $P(H = k)$ for any k.

Solution.

The mean may in fact be calculated using techniques from the previous chapter, specifically theorem 12.57. It could also be obtained from our method below by standard generating function theory.

We have $H = \min\{n \geq 0 : X_n = 0 \mid X_0 = 1\}$. Define $G(s) = \sum_{n=0}^{\infty} s^n P(H = n)$ with $X_0 = 1$. Here, $H = \infty$ contributes 0 to the sum $G(s)$, so H can be infinite if and only if $G(1) < 1$.

Lemma

If the walk starts at i, then $H_i = \min\{n \geq 0 : X_n = 0 \mid X_0 = i\}$ has probability generating function $G(s)^i$.

The walk starts at i. Let T_i be the first time it reaches $i - 1$, T_2 the first time it reaches $i - 2$, and so on up to T_i.

Define the durations of the stages $D_k = T_k - T_{k-1}$ and $D_1 = T_1$ so that $H = D_1 + D_2 + \cdots + D_i$.

Now the T_i are hitting times so they are also stopping times.

By the strong Markov property, we have that the chain onward from time T_k is itself a Markov chain with the same transition matrix as X. We repeatedly apply this property to split our process into identical independent Markov chains Y_k with components $X_{T_k}, X_{T_k+1}, \ldots, X_{T_{k+1}-1}$.

Now the D_k are each determined by the path of one of these chains, so they are independent. By identical distribution, we have $D_k = D_1 = H$ in distribution.

Hence, H_i is a sum of i independent copies of H, and thus the probability generating function of H_i is $G(s)^i$.

End of lemma.

We return to the solution.

With probability q, $X_1 = 0$ so $H = 1$. With probability p, $X_1 = 2$ so $H = 1 + H_2$.

Recall H_2 has probability generating function $G(s)^2$ by our lemma.

So

$$G(s) = qs + psG(s)^2.$$

Hence,

$$G(s) = \frac{1 \pm \sqrt{1 - 4pqs^2}}{2ps}.$$

Since $G(0) \leq 1$ while the positive case diverges for $s \to 0^+$, we take the negative root.

$$G(s) = \frac{1 - \sqrt{1 - 4pqs^2}}{2ps}.$$

From here, we use standard algebra to obtain

$$P(H = 2k - 1) = \frac{(2k-2)!}{k!(k-1)!} \frac{(pq)^k}{p},$$

and $P(H = 2k) = 0$.

Note that in this case, our lemma was more general than it needed to be, as we only needed the probability generating function of H_2. However, it is quite a typical approach for problems of this type.

We have seen already that a state i is recurrent if, starting from i, the chain returns to i with probability 1. The state is transient if it is not recurrent.

It follows that if the starting state i is recurrent, the chain is bound to return to it infinitely often.

Theorem 12.71

Suppose $X_0 = i$ and let $V_i = |\{n \geq 1 : X_n = i\}|$ be the number of subsequent visits by the Markov chain to i. Then V_i has the geometric distribution

$$P_i(V_i = r) = (1 - f)f^r$$

where $f = f_{i,i}$ is the return probability $P_i(X_n = i \text{ for some } n \geq 1)$. In particular:

1. $P_i(V_i = \infty) = 1$ if i is recurrent. We will investigate the rate of this divergence in theorem 12.105.
2. $P_i(V_i < \infty) = 1$ if i is transient.

Proof.

The proof makes use of $T_i = \inf\{n \geq 1 : X_n = i\}$, the first-passage time to i. If $X_0 = i$, then T_i is the recurrence time of i.

Let $f_{i,i} = P_i(T_i < \infty)$ so that i is recurrent if $f_{i,i} = 1$ and transient if $f_{i,i} < 1$.

Let T_i^r be the time of the rth visit to i, with $T_i^r = \infty$ if $V_i < r$.

Since the T_i^r are increasing,

$$P_i(V_i \geq r) = P_i(T_i^r < \infty).$$

$$P_i(V_i \geq r) = P_i(T_i^r < \infty \mid T_i^{r-1} < \infty) P_i(T_i^{r-1} < \infty).$$

$$P_i(V_i \geq r) = P_i(T_i < \infty) P_i(T_i^{r-1} < \infty)$$

by the strong Markov property.

$$P_i(V_i \geq r) = f_{i,i} P_i(T_i^{r-1} < \infty)$$

by the definition of $f_{i,i}$.

$$P_i(V_i \geq r) = f_{i,i}^r$$

by recursion since $P_i(V_i \geq 0) = 1$.

It is a routine computation to find $P_i(V_i = r)$ now.

And taking $r \to \infty$, we obtain results (a) and (b) in the theorem.

Definition 12.74

1. The mean recurrence time μ_i of the state i is defined by $\mu_i = E_i(T_i) = \sum_{n=1}^{\infty} n f_{i,i}(n)$ if i is recurrent and $\mu_i = \infty$ if i is transient. This makes sense as recall transient states have non-zero $P_i(T_i = \infty)$.
2. If i is recurrent, we call it null if $\mu_i = \infty$ and positive or non-null if $\mu_i < \infty$.
3. The period d_i of the state i is given by $d_i = \gcd\{n : p_{i,i}(n) > 0\}$. The state i is called aperiodic if $d_i = 1$ and periodic if $d_i > 1$.
4. State i is called ergodic if it is aperiodic and positive recurrent.

Examples of the definitions in 12.74.

- Take two states, 0 and 1, with $p_{0,1} = a$ and $p_{1,0} = b$. It is easy to verify we have positive recurrent state 0.
- If $a = b = 1$, we have the chain $0, 1, 0, 1, \ldots$. Then $p_{0,0}(n) > 0$ only for even n, so $d_0 = 2$ and the state 0 is periodic (and positive recurrent).
- If $a < 1$ then $p_{0,0}(1) = 1 - a > 0$ so $d_0 = 1$ and we have the state 0 being aperiodic and positive recurrent, and hence ergodic.

By theorem 12.37, we have that recurrence is a property of a communicating class. We extend this in the following theorem.

Theorem 12.74

If $i \leftrightarrow j$ then:

1. i and j have the same period.
2. i is recurrent if and only if j is recurrent.
3. i is positive recurrent if and only if j is positive recurrent.
4. i is ergodic if and only if j is ergodic.

Proof.

We assume $i \neq j$.

1. Since $i \leftrightarrow j$, there exist $m, n \geq 1$ such that $\alpha := p_{i,j}(m)p_{j,i}(n) > 0$.

   Then by the Chapman-Kolmogorov equations,

   $$p_{i,i}(m + r + n) \geq \alpha p_{j,j}(r)$$

   for all $r \geq 0$.

   We have $p_{i,i}(m + n) \geq \alpha$ also by Chapman-Kolmogorov or by $r = 0$ in the above.

   Hence, d_i divides $(m + n)$.

Take any $r \geq 1$ and suppose d_i does not divide r. Then d_i does not divide $(m + r + n)$ by the above.

Hence $p_{i,i}(m + r + n) = 0$ and so $p_{j,j}(r) = 0$.

We have shown d_i does not divide $r \Rightarrow p_{j,j}(r) = 0$. Take the contrapositive. $p_{j,j}(r) > 0 \Rightarrow d_i$ divides r.

Now consider the set $\{r : p_{j,j}(r) > 0\}$. Then d_i divides each r, so d_i divides $\gcd\{r : p_{j,j}(r) > 0\} = d_j$.

We have $d_i \mid d_j$. By reversing the argument, we have $d_j \mid d_i$, and hence $d_i = d_j$.

2. This was proven in theorem 12.37.

3. Suppose that i is positive recurrent, and let C be the communicating class containing i.

   By theorem 12.37, C is closed, so if $X_0 \in C$ then $X_n \in C$ for all n.

   Consequently, the chain restricted to the state space C is a Markov chain, and it is irreducible (even if the chain on the entire state space need not be).

   By part (a) of theorem 12.83, which we shall see in due time, it possesses an invariant distribution.

   By part (b) of theorem 12.83, every state of C is positive recurrent.

   If $i \leftrightarrow j$ then $j \in C$, so j is positive recurrent.

4. This follows from (a), (b), and (c).

Proposition 12.77

If the chain is irreducible and $j \in S$ is recurrent, then

$$P(X_n = j \text{ for some } n \geq 1) = 1$$

regardless of the initial distribution of the chain.

Proof.

Let i, j be distinct states and recall $f_{i,j} = P_i(X_n = j \text{ for some } n \geq 1)$. We wish to show $f_{i,j} = 1$ for all i.

Now we will consider $p_{j,i}$. We have information about the chain starting from j due to the assumption of recurrence, so this is somewhat natural to consider.

Since the chain is irreducible - recall this means $i \leftrightarrow j$ for all states - there exists a least integer m such that $p_{j,i}(m) > 0$.

Since m is least, $p_{j,i}(m) = P_j(X_m = i, X_r \neq i \text{ for } 1 \leq r < m)$.

Consider the event $\{X_0 = j, X_m = i, \text{ and no return to } j \text{ takes place after time } m\}$. Then by the above line, no return to j ever takes place. It follows by the Markov property at time m that

$$p_{j,i}(m)(1 - f_{i,j}) \leq 1 - f_{j,j}$$

where the left hand side is the event we are considering and the right hand side is the event that no return to j ever takes place, of which the left hand side is a subset and so the probability is an upper bound of the probability of the left hand side.

If j is recurrent, then $f_{j,j} = 1$ so that $f_{i,j} = 1$ for all $i \in S$. Then we must have $1 - f_{i,j} = 0$.

So $f_{i,j} = 1$. We never said anything about the state i, so this holds for all i.

The proposition is now clearly true, but if we want to be rigorous, given an initial distribution λ we have

$$P(T_j < \infty) = \sum_{i \in S} \lambda_i f_{i,j} = 1.$$

We now ask what can be said about X_n in the limit as $n \to \infty$. We will see that the distribution of X_n settles into an equilibrium.

Definition 12.82

Let X be a Markov chain with transitional matrix P. The vector $\pi = (\pi_i : i \in S)$ is called an invariant distribution of the chain if:

- $\pi_i \geq 0$ for all $i \in S$ and $\sum_{i \in S} \pi_i = 1$.
- $\pi = \pi P$.

An invariant distribution is also called a stationary, equilibrium, or steady-state distribution.

An invariant distribution is invariant under the passage of time. If X_0 has distribution π, then X_n has distribution $\pi P^n = \pi$, so that every X_n has distribution π.

For example, if we have the three-state Markov chain with $S = \{1, 2, 3\}$ and

$$P = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0.2 & 0.3 & 0.5 \\ 0.1 & 0.1 & 0.8 \end{pmatrix},$$

then we have $\pi = (\frac{9}{44}, \frac{10}{44}, \frac{25}{44})$. If the Markov chain has distribution π, then these are the probabilities that it will be in state 1, 2, or 3.

Most of our attention will be spent on proving the following theorem.

Theorem 12.83

Consider an irreducible Markov chain. We have:

1. There exists an invariant distribution π if and only if some state is positive recurrent.
2. If there exists an invariant distribution π, then every state is positive recurrent, and $\pi_i = 1/\mu_i$ for $i \in S$, where μ_i is the mean recurrence time of state i. In particular, π is the unique invariant distribution.

Recall a positive recurrent state is a recurrent state with its mean recurrence time being finite.

We shall prove theorem 12.83 by exhibiting an explicit solution of the vector equation $\rho = \rho P$. We will fix a state $k \in S$ and start the chain from this state.

Let W_i be the number of subsequent visits to state i before the first return to the initial state k. Then we have:

$$W_i = \sum_{m=1}^{\infty} \mathbf{1}_{\{X_m = i,\, T_k \geq m\}}.$$

$$W_i = \sum_{m=1}^{T_k} \mathbf{1}_{\{X_m = i\}},$$

where $\mathbf{1}_A$ is the indicator function of A and $T_k = \inf\{n \geq 1 : X_n = k\}$ is the standard first return time to k.

Note that $W_k = 1$ if $T_k < \infty$.

We consider the vector ρ given by $\rho_i = E_k(W_i)$ for $i \in S$. Recall that $E_k(Z)$ denotes the mean of Z given that $X_0 = k$. The following proposition gives some results about ρ.

Proposition 12.86

For an irreducible, recurrent chain (this means that every state is recurrent) and any given $k \in S$, which we will start the chain from, the vector $\rho = (\rho_i : i \in S)$ satisfies:

1. $\rho_k = 1$.
2. $\sum_{i \in S} \rho_i = \mu_k$ whether or not μ_k is finite.
3. $\rho = \rho P$.
4. $0 < \rho_i < \infty$ for $i \in S$.

Here is a useful consequence of theorem 12.83 and proposition 12.86.

Consider an irreducible, positive recurrent Markov chain and fix a state k.

Since the chain is positive, we have $\mu_k < \infty$.

By theorem 12.83, there exists a unique invariant distribution π with $\pi_k = 1/\mu_k$.

By proposition 12.86, specifically (b) and (c), $v = \pi_k \rho$ satisfies $v = vP$ and $\sum_{i \in S} v_i = 1$.

By the uniqueness of the invariant distribution, $\pi = v$.

Therefore, $\rho_i = v_i/\pi_k = \pi_i/\pi_k$. We state this as a corollary.

Corollary 12.87

Let $i, k \in S$ be distinct states of an irreducible, positive recurrent Markov chain with invariant distribution π. The mean number of visits to state i between two consecutive visits to k is π_i/π_k.

Proof of proposition 12.86.

1. Since we assume recurrence, $P_k(T_k < \infty) = 1$. Then $W_k = 1$ so that $\rho_k = E_k(W_k) = E_k(1) = 1$.
2. We have that $T_k = \sum_{i \in S} W_i$, which makes sense intuitively.

   With an unjustified interchange of expectation and summation, we have $\mu_k = \sum_{i \in S} \rho_i$.
3. We have

$$W_j = \sum_{m=1}^{\infty} \mathbf{1}_{\{X_m = j, T_k \geq m\}}.$$

Taking expectations,

$$\rho_j = \sum_{m=1}^{\infty} P_k(X_m = j, T_k \geq m).$$

The event $\{T_k \geq m\}$ depends only on $X_0, X_1, \ldots, X_{m-1}$.

$$\begin{aligned} &P_k(X_m = j, T_k \geq m) \\ &= \sum_{i \in S} P_k(X_{m-1} = i, X_m = j, T_k \geq m), \end{aligned}$$

this is just splitting into cases.

$$\begin{aligned} &P_k(X_m = j, T_k \geq m) \\ &= \sum_{i \in S} P_k(X_m = j \mid X_{m-1} = i, T_k \geq m) \\ &\quad \cdot P_k(X_{m-1} = i, T_k \geq m). \end{aligned}$$

$$\begin{aligned} &P_k(X_m = j, T_k \geq m) \\ &= \sum_{i \in S} P_k(X_m = j \mid X_{m-1} = i) \\ &\quad \cdot P_k(X_{m-1} = i, T_k \geq m) \end{aligned}$$

by the extended Markov property, as the event $\{T_k \geq m\}$ depends only on the past and present in terms of time $(m-1)$. Since we have specified $X_{m-1} = i$, the additional conditional only impacts the past $X_0, X_1, \ldots, X_{m-2}$ and hence we can apply extended Markov to get rid of it. However, it seems there is some extra detail around the case $X_{m-1} = k$, which I am not totally sure how to resolve, but for now we shall take this at face value.

$$P_k(X_m = j, T_k \geq m) = \sum_{i \in S} p_{i,j} P_k(X_{m-1} = i, T_k \geq m).$$

Then

$$\rho_j = \sum_{i \in S} \sum_{m=1}^{\infty} p_{i,j} P_k(X_{m-1} = i, T_k \geq m).$$

Rewriting with $r = m - 1$, we obtain

$$\rho_j = \sum_{i \in S} p_{i,j} \sum_{r=0}^{\infty} P_k(X_r = i, T_k \geq r + 1).$$

$\rho_j = \sum_{i \in S} p_{i,j} \rho_i$ by considering $i = k$ and $i \neq k$ separately.

Then $\rho = \rho P$.

4. We already have $\rho_k = 1$. Since the chain is irreducible, there exist $m, n \geq 0$ such that $p_{i,k}(m)$ and $p_{k,i}(n)$ are positive.

   Since $\rho = \rho P$ and hence $\rho = \rho P^r$, we have that:

   - $\rho_k = \sum_i \rho_i p_{i,k}(m) \geq \rho_i p_{i,k}(m)$ for any i by reading off the kth coordinate.
   - $\rho_i = \sum_k \rho_k p_{k,i}(n) \geq \rho_k p_{k,i}(n)$ for any k by reading off the ith coordinate.

Since $\rho_k = 1$, we obtain

$$p_{k,i}(n) \leq \rho_i \leq \frac{1}{p_{i,k}(m)}.$$

The result follows.

Proof of theorem 12.83.

We prove part (a) first.

Suppose $k \in S$ is positive recurrent, so that $\mu_k < \infty$. Since the chain is irreducible and k is recurrent, by theorem 12.37 every state is recurrent. So proposition 12.86 applies. Let ρ be given by $\rho_i = E_k(W_i)$. By proposition 12.86, $\sum_{i \in S} \rho_i = \mu_k < \infty$. Hence $\pi := (1/\mu_k)\rho$ satisfies $\pi_i \geq 0$ and $\sum_{i \in S} \pi_i = 1$. By proposition 12.86(c), $\rho = \rho P$. Dividing by the constant μ_k, we have $\pi = \pi P$. So π is an invariant distribution.

Now proving part (b).

Suppose that π is an invariant distribution of the chain.

Step 1: $\pi_i > 0$ for all $i \in S$.

Since $\sum_{i \in S} \pi_i = 1$, there exists $k \in S$ with $\pi_k > 0$.

Let $i \in S$. Since the chain is irreducible, there exists $m \geq 1$ such that $p_{k,i}(m) > 0$.

We have $\pi = \pi P$ and hence $\pi = \pi P^m$. Reading off the ith coordinate,

$$\pi_i = \sum_{j \in S} \pi_j p_{j,i}(m) \geq \pi_k p_{k,i}(m) > 0,$$

since all terms in the sum are non-negative.

Step 2: all states are recurrent.

Since the chain is irreducible, by theorem 12.37 either all states are transient or all are recurrent.

Suppose all states are transient. Then by proposition 12.40, $p_{i,j}(n) \to 0$ as $n \to \infty$ for all $i, j \in S$.

Since $\pi = \pi P^n$, we have for any $j \in S$

$$\pi_j = \sum_{i \in S} \pi_i p_{i,j}(n) \to 0$$

as $n \to \infty$.

This contradicts $\pi_j > 0$. Hence all states are recurrent.

Step 3: $\pi_i \mu_i = 1$ for all $i \in S$.

Suppose X_0 has distribution π. Since π is invariant, $(X_1, X_2, \ldots, X_{n-1})$ and $(X_0, X_1, \ldots, X_{n-2})$ have the same joint distributions.

Since the chain is recurrent, $T_i < \infty$ almost surely, and we have

$$\mu_i = E_i(T_i) = \sum_{n=1}^{\infty} P_i(T_i \geq n),$$

which is the standard rewriting of the mean of a non-negative integer valued random variable. Then

$$\pi_i \mu_i = P(X_0 = i) \sum_{n=1}^{\infty} P_i(T_i \geq n)$$

$$= \sum_{n=1}^{\infty} P(T_i \geq n, X_0 = i),$$

since $P(X_0 = i)P_i(A) = P(A, X_0 = i)$ by definition of conditional probability.

For $n = 1$, we have $P(T_i \geq 1, X_0 = i) = P(X_0 = i)$, since $T_i \geq 1$ always.

For $n \geq 2$, we have

$$\{T_i \geq n\} = \{X_m \neq i \text{ for } 1 \leq m \leq n-1\},$$

so

$$P(T_i \geq n, X_0 = i) = P(X_0 = i, X_m \neq i \text{ for } 1 \leq m \leq n-1).$$

The event $\{X_m \neq i \text{ for } 1 \leq m \leq n-1\}$ is the disjoint union of the same event intersected with $\{X_0 = i\}$ and with $\{X_0 \neq i\}$. Rearranging, we obtain

$$\begin{aligned} &P(X_0 = i, X_m \neq i \text{ for } 1 \leq m \leq n-1) \\ &= P(X_m \neq i \text{ for } 1 \leq m \leq n-1) \\ &\quad - P(X_m \neq i \text{ for } 0 \leq m \leq n-1). \end{aligned}$$

The first term is the same as if we shift all times back by one by our joint distribution argument, so

$$\begin{aligned} &P(X_m \neq i \text{ for } 1 \leq m \leq n-1) \\ &= P(X_m \neq i \text{ for } 0 \leq m \leq n-2). \end{aligned}$$

Define $a_r = P(X_m \neq i \text{ for } 0 \leq m \leq r)$. Then for $n \geq 2$,

$$P(T_i \geq n, X_0 = i) = a_{n-2} - a_{n-1}.$$

Summing over n, the sum telescopes:

$$\begin{aligned} &\sum_{n=1}^{N} P(T_i \geq n, X_0 = i) \\ &= P(X_0 = i) + \sum_{n=2}^{N} (a_{n-2} - a_{n-1}) \\ &= P(X_0 = i) + a_0 - a_{N-1}. \end{aligned}$$

Letting $N \to \infty$, we obtain

$$\pi_i \mu_i = P(X_0 = i) + a_0 - \lim_{n \to \infty} a_n.$$

Now $a_0 = P(X_0 \neq i)$, so $P(X_0 = i) + a_0 = 1$, and hence

$$\pi_i \mu_i = 1 - \lim_{n \to \infty} a_n.$$

The events $\{X_m \neq i \text{ for } 0 \leq m \leq n\}$ are decreasing in n, so by continuity of probability

$$a_n \to P(X_m \neq i \text{ for all } m \geq 0)$$

as $n \to \infty$.

Since i is recurrent and the chain is irreducible, by proposition 12.77 we have $P(X_m = i \text{ for some } m \geq 1) = 1$ regardless of the initial distribution. So $P(X_m \neq i \text{ for all } m \geq 0) = 0$, and $a_n \to 0$.

Hence $\pi_i \mu_i = 1$.

A heuristic. We are probably totally lost in notation by now, so it is nice to return to a heuristic argument on occasion. We have that $\pi_i \mu_i = 1$ says that the proportion of time spent in i, which is π_i, is the reciprocal of the mean time between visits to i. If the chain returns to i every μ_i steps on average, then it is in i about one step out of every μ_i steps.

We now give the principal result for discrete time Markov chains.

Theorem 12.98: convergence theorem for Markov chains

Consider a Markov chain that is aperiodic, irreducible, and positive recurrent. For $i, j \in S$,

$$p_{i,j}(n) \to \pi_j \qquad \text{as } n \to \infty,$$

where π is the unique invariant distribution of the chain.

Writing out this result in words, we have "the probability that, starting at i, we are at state j after n steps converges to π_j for $n \to \infty$."

Proof.

The proof uses a technique known as coupling.

Construct an ordered pair $Z = (X, Y)$ of independent aperiodic, irreducible, positive recurrent Markov chains $X = (X_n : n \geq 0)$ and $Y = (Y_n : n \geq 0)$, each of which has state space S and transition matrix P.

Then $Z = (Z_n : n \geq 0)$ is given by $Z_n = (X_n, Y_n)$. Z is a Markov chain (this requires a verification we omit) with state space $S \times S$ and transition probability $p_{ij,kl} = p_{i,k} p_{j,l}$ by independence.

We have the result that for an aperiodic state i of a Markov chain, there exists $N_1 \geq 1$ such that $p_{i,i}(n) > 0$ for all $n \geq N_1$. This is because the set of possible return times to i had gcd 1 by definition, and if m, n are in this set then so is $(m + n)$. Such a set must contain all sufficiently large integers.

Then if the chain is irreducible, we have $p_{i,j}(m) > 0$ for some m.

Set $N_2 = N_1 + m$ and let $n \geq N_2$.

By Chapman-Kolmogorov, we have $p_{i,j}(n) \geq p_{i,i}(n - m) p_{i,j}(m)$ and thus $p_{i,j}(n) > 0$ for all $n \geq N_2$.

Since X is irreducible and aperiodic, for states i, j, k, l there exists $N = N(i, j, k, l)$ such that $p_{i,k}(n) p_{j,l}(n) > 0$ for all $n \geq N$.

Therefore, Z is irreducible. Only here is aperiodicity used.

Now we use the positive recurrence of X. By theorem 12.83, X has a unique invariant distribution π, and it follows that Z has the invariant distribution v, with $v_{i,j} = \pi_i \pi_j$. Thus, Z is also positive recurrent by theorem 12.83.

Now let $X_0 = i$ and $Y_0 = j$.

Fix $s \in S$ and let $T = \min\{n \geq 1 : Z_n = (s, s)\}$ be the first-passage time of Z to (s, s). By the recurrence of Z and proposition 12.77,

$$P_{ij}(Z_n = (s, s) \text{ for some } n \geq 1) = 1,$$

or

$$P_{ij}(T < \infty) = 1.$$

Now since T is a stopping time, by the strong Markov property, X_n and Y_n have the same conditional distributions given the event $\{T \leq n\}$. This is because X_n and Y_n are both at state s here.

We now use this fact to show that the limiting distributions of X and Y are independent of their starting points.

We have $p_{i,k}(n) = P_{ij}(X_n = k)$. These notations look very similar so we will recall $p_{i,k}(n)$ means the probability that starting from i, we end at k after n steps, while the right hand side is the probability given $Z_0 = (i, j)$, we end at $X_n = k$ after n steps. These are equal by independence.

Then

$$\begin{aligned} p_{i,k}(n) = P_{ij}(X_n = k, T \leq n) \\ + P_{ij}(X_n = k, T > n). \end{aligned}$$

$$\begin{aligned} p_{i,k}(n) = P_{ij}(Y_n = k, T \leq n) \\ + P_{ij}(X_n = k, T > n) \end{aligned}$$

since given $T \leq n$, X_n and Y_n are identically distributed. This is the application of strong Markov, given our discussion above.

$$p_{i,k}(n) \leq P_{ij}(Y_n = k) + P_{ij}(T > n).$$

$$p_{i,k}(n) \leq p_{j,k}(n) + P_{ij}(T > n).$$

Now we can obtain the same inequality with i and j interchanged and thus

$$|p_{i,k}(n) - p_{j,k}(n)| \leq P_{ij}(T > n) \to 0 \qquad \text{as } n \to \infty,$$

as we have shown that T is finite with probability 1.

Thus, $p_{i,k}(n) - p_{j,k}(n) \to 0$, and hence if the limit of $p_{i,k}$ exists, then it does not depend on the choice of i.

Now

$$\pi_k - p_{i,k}(n) = \sum_{l \in S} \pi_l (p_{l,k}(n) - p_{i,k}(n)) \to 0$$

as $n \to \infty$ - this can be shown quite easily - and so by the result we have for the limit of the right hand side, $\pi_k - p_{i,k}(n) \to 0$, and so $p_{i,k}(n) \to \pi_k$ as $n \to \infty$.

Example 12.102

Here is an elementary example which shows why we need aperiodicity. Let X be a Markov chain with state space $S = \{1, 2\}$ and transition matrix

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then X alternates between the two states, and thus the limit of $p_{i,j}(n)$ does not exist for $i, j \in S$.

Example 12.103: coupling game

A pack of playing cards is shuffled and dealt one by one. A friend is asked to select some card, secretly, from amongst the first six or seven cards, say. If the face value of this card is m, the next $m-1$ cards are allowed to pass, and your friend is asked to note the face value of the mth card. Continuing according to this rule, there arrives a last card in this sequence, with face value X and with fewer than X cards remaining. We call X your friend's score.

With high probability, you are able to guess accurately your friend's score, as follows. You follow the same rules as the friend, starting for simplicity at the first card. You obtain thereby a score Y. There is a high probability that $X = Y$.

Suppose your friend picks the m_1th card, m_2th card, and so on, and you pick the n_1th, n_2th, ... If $m_i = n_j$ for some i, j, the two of you are "stuck together" forever after. When this occurs first, we say that "coupling" has occurred. Prior to coupling, each time you read the value of a card, there is a positive probability that you will arrive at the next stage on exactly the same card as the other person. If the pack of cards were infinitely large, then coupling would take place sooner or later. It turns out that there is a reasonable chance that coupling takes place before the last card of a regular pack of 52 cards has been dealt.

This chapter is yet to be fully completed (October 2026).